Banks, Sentiments, and Business Cycles*

Tyler J. Pike

University of Maryland

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Abstract

This paper measures the animal spirits of U.S. banks and asks whether they are an important determinant of credit conditions and source of business cycle fluctuations. I first construct a novel semi-structural measure of bank-level sentiment, revealing heterogeneous animal spirits across banks and common dynamics marked by surges in pessimism during crises and excessive optimism during periods of elevated asset prices. Using a structural BVAR that identifies bank sentiment alongside standard macroeconomic shocks, I find that bank sentiment shocks account for 24% of fluctuations in output growth and 20% in the policy rate. However, effects are time-varying and most acute in recessions when sentiment becomes a dominant source of volatility in real activity and financial conditions. These effects arise as pessimistic sentiment shocks act like a negative credit supply shock propagating primarily through the business lending channel, raising borrowing costs and tightening lending standards independent of realized risk. Monetary policy can only partially offset these effects.

Keywords Sentiment, Financial Intermediaries, Credit Supply Shocks, Business Cycles
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Department of Economics, University of Maryland; E-Mail: tjpike@umd.edu; Web: https://tylerjpike.github.io/

1 Introduction

Economic sentiments, or animal spirits in the language of Keynes, have long been hypothesized to shape investors' willingness to supply credit and play a central role in driving business cycle fluctuations. This is because the potential effects of *animal spirits*, an economic agent's irrational optimism or pessimism about the current and future state of the world, are quite clear.[1] For example, when banks become excessively pessimistic, they forecast greater losses in their loan portfolios and, in turn, increase loan rates to compensate for the expected losses. As loan rates rise, firms and households are priced out of debt markets, leading to a subsequent decline in economic activity that reinforces the precipitating fear, making the credit crunch self-enforcing. A similar mechanism may also explain how banks' excessive optimism can lead to a credit boom.[2]

However, animal spirits are unobservable and difficult to measure, which has limited empirical evidence of their impacts on credit conditions and the economy more broadly. Attempts have been made to measure sentiments, although they suffer a common limitation. Measures based on asset prices provide revealed-preference proxies for beliefs but cannot disentangle whether those beliefs originate from the demand or supply side of credit markets.[3] Survey-based measures, while agent-specific, primarily capture the expectations of households or firms and therefore reflect demand-side sentiments.[4] As a result, neither approach directly identifies the beliefs of financial intermediaries or allows for a clean test of how animal spirits operate through the supply of credit.

This paper measures the animal spirits of U.S. banks and evaluates their effects on the supply of credit and business cycle dynamics. Specifically, I measure *bank risk sentiment* (BRS), defined as a bank's expectation of future loan losses in excess of its rational expectations benchmark. I then document three main findings. First, bank sentiment is broadly acyclical but exhibits sharp increases in pessimism during recessions. Second, pessimistic sentiment shocks act like negative credit supply shocks, tightening lending conditions and affecting real activity primarily through the business lending channel, with effects that are only partially offset by conventional monetary policy. Third, these shocks are a quantitatively important driver of macroeconomic fluctuations, accounting for a substantial share of variation in credit conditions and real activity during crises.

The key challenge is that neither a bank's expectations nor the corresponding rational expectations

[1] I will use the terms sentiments and animal spirits interchangeably. This particular definition of animal spirits may be micro-founded as shocks in a noisy signal and dispersed information problem, as in Angeletos and La'o 2013.

[2] These dynamics have long been discussed in textbook settings in various forms, see for example Kindleberger 1978, Akerlof and Shiller 2010, and Shiller 2020.

[3] See López-Salido et al. 2017 for a recent example and review of the literature of this approach.

[4] See Lagerborg et al. 2023 for a recent example and review of the literature of this approach.

benchmark are directly observable. I overcome this hurdle by taking inspiration from the semi-structural residual-based measurement strategies widespread in the productivity measurement literature (see Blackwood et al. 2021 for a recent detailed discussion on measuring productivity), and the fact that banks incorporate their expectations of future loan losses into the loan rates they charge borrowers. The semi-structural estimation proceeds in two steps.

I start by developing a structural model of the U.S. banking sector. The theoretical framework embeds animal spirits shocks à la Angeletos and La'o 2013 into a heterogeneous banking model similar in spirit to Jamilov and Monacelli 2025. The resulting model is rich enough to take to the data but tractable enough to yield a closed-form solution to a bank's loan rate schedule. In this setting, a bank's loan rate equation is shown to be a function of the bank's market power, capital costs, regulatory costs, and expected loan default rate. Importantly, a bank's risk sentiment, as the difference between the bank's forecast of its loan losses and its rational expectations forecast of these losses, is therefore present in, and identifiable through, the structural loan rate equation.

I then take this structure to the data by estimating the log-linearized loan pricing equation using bank-level panel data. The resulting loan rate schedule can be used to decompose loan rates into observable pricing factors and unobservable bank risk sentiments. Using this approach, I estimate bank-level risk sentiments at a quarterly frequency for the universe of U.S. commercial banks from 1986 through 2024 using regulatory Call Reports.

Common aggregate movements in bank sentiments are broadly acyclical, but characterized by sharp increases in times of financial stress and uncertainty in the United States. For example, BRS increases during the Savings and Loans (S&L) Crisis, Dot-com bust, Global Financial Crisis (GFC), and COVID-19 pandemic. Conversely, banks appear to be excessively optimistic during the mid-2000s housing boom, among other commonly labeled asset bubbles. It is important to clarify that while uncertainty and financial stress may lead to a rational increase in forecasted loan losses, the rise in bank sentiment indicates that banks become even more pessimistic about future states of the economy than is necessarily implied by a rational expectations view of the world. That is, BRS indicates banks become *excessively* pessimistic in times of uncertainty and financial stress.

However, focusing on the common movements in banking sector sentiments obscures a larg e degree of heterogeneity at the bank-level. There is significant dispersion in bank-level sentiments, although the extent of the dispersion varies over time. Disagreement in sentiments is both countercyclical and in secular decline. These differences in bank-level sentiments may occur because banks are either hit with different sentiment shocks or they have different biases and persistence

in sentiments. I show it is a combination of both, although banks are on average unbiased and experience relatively small, short-lived, sentiment shocks.

Before using the estimated bank risk sentiments to proxy for sentiment shocks in the macroeconomic analysis, I verify that they can be interpreted as animal spirits and consider their behavioral origins with three tests. First, bank-level sentiments do not contain systematically useful information for forecasting future loan defaults, implying they do not enter a rational expectations forecast of loan losses. Second, sentiments are not predictable using observable bank characteristics, consistent with being unanticipated, noisy signals. Third, periods of optimistic sentiment are followed by higher realized defaults, indicating that banks' loan pricing behavior is consistent with overreaction and subsequent correction dynamics documented in the behavioral finance literature.

With a measure of bank sentiment in hand, I turn to evaluating the relative importance of sentiment shocks in driving the supply of credit and business cycle fluctuations. To do so, I study the economy through the lens of a structural Bayesian vector-autoregression (BVAR) in which sentiment shocks enter as an external instrument exogenously shifting loan rates.

Bank sentiment shocks are an important source of business cycle fluctuations. In a decomposition of average business cycle variation, bank sentiment accounts for approximately 12 percent of variation in credit conditions, and is a dominant force driving short-term fluctuations in GDP growth and monetary policy, accounting for 24 and 20 percent of the variation in either outcome respectively. Bank sentiment shocks are comparable to household sentiment and aggregate demand shocks as a source of variation in real activity across the cycle, and financial risk or aggregate supply shocks in terms of driving policy actions.

However, the economic effects of bank sentiment are time-varying. A historical decomposition of macroeconomic outcomes into contributions by exogenous shocks shows that bank sentiment has a neutral effect on the economy during "normal" times, but plays a dual role during recessions. Across the past four U.S. recessions, bank sentiment was excessively optimistic prior to each downturn, boosting activity, before deteriorating rapidly and dampening GDP growth by approximately 1.5 percentage points on average, and by up to four percentage points during the GFC.

I lastly examine the transmission of sentiment shocks through loan markets to the broader economy by tracing impulse responses to a pessimistic shock in an expanded BVAR.

A pessimistic bank sentiment shock acts as a negative credit supply shock, but with asymmet-

ric effects across lending markets. As banks become more pessimistic, mortgage rates rise and originations slow, consumer lending remains largely unchanged, and business lending deteriorates along both pricing and constraint margins. More precisely, businesses' borrowing costs increase and lending volumes decline, while banks simultaneously tighten lending standards and loan covenants, restricting access to new credit and reducing borrowing capacity on existing relationships. This asymmetry indicates that sentiment shocks operate most strongly in segments of the credit market that rely on banks' subjective risk assessments.

These changes in credit conditions propagate to the real economy primarily through the business lending channel. Higher borrowing costs and tighter constraints reduce investment, production, employment, and trade, consistent with a contraction in firms' access to external finance. In contrast, housing activity and prices respond only weakly, suggesting limited pass-through to household borrowing conditions. At the same time, the deterioration in bank lending spills over into broader financial markets, raising corporate bond rates and lowering equity valuations.

The limited transmission to housing markets in part reflects the role of monetary policy. While lower policy rates partially offset increases in borrowing costs, they do not reverse tighter lending standards or contractual constraints. As a result, real activity declines even under accommodative policy. By contrast, mortgage lending is less affected by such constraints, allowing monetary policy to more effectively offset increases in mortgage rates and dampen the transmission of sentiment shocks through the household sector.

These findings imply that the macroeconomic consequences of sentiment shocks depend on whose sentiment is changing. For example, household sentiment shocks resemble aggregate demand disturbances, characterized by declines in consumption that can be offset with conventional monetary policy (Lagerborg et al. 2023). In contrast, bank sentiment shocks resemble credit supply disturbances that primarily affect credit conditions, do not impact consumption, and are only partially offset by monetary policy.

Contributions and related literature. This paper contributes to the literatures on economic sentiments, credit supply, and business cycle fluctuations. Its first contribution is a novel measure of bank-level animal spirits that bridges the two dominant empirical approaches to measuring sentiment while addressing their respective limitations. The first approach uses surveys of households and firms to elicit expectations about the economic outlook (e.g., Barsky and Sims 2012, Mian et al. 2015, Lagerborg et al. 2023, Faccini and Melosi 2022, Bianchi et al. 2022). These measures offer agent-specific information and can track beliefs across horizons and over time, which is valuable

for controlling for individual biases and institutional features. However, they rely on self-reported expectations and are therefore subject to measurement error and survey biases (Stantcheva 2023). The second approach infers sentiment from risk premia in financial markets, including corporate bonds (Gilchrist and Zakrajšek 2012, Leiva-Leon et al. 2022, Boeck and Zörner 2023), equities (Baron and Xiong 2017, Pflueger et al. 2020), and syndicated loans (Saunders et al. 2021, Kwak 2022). While these measures benefit from a revealed-preference foundation, they cannot cleanly identify whose beliefs drive observed prices and are thus limited in their ability to account for institutional features of the underlying agents.[5]

I address these limitations by constructing a bank-level measure of sentiment from loan pricing behavior. Because banks embed their expectations of future loan losses into the rates they charge, a decomposition of loan rates allows sentiment to be recovered at the level of the lending institution. This approach combines the agent specificity of survey-based measures with the quantitative discipline and revealed-preference foundation of asset-price-based measures, while directly targeting sentiments originating from the supply of credit.

The paper's second contribution is to provide an empirical assessment of sentiment shocks as a source of credit conditions and business cycle fluctuations. In other words, a modern assessment of the Keynesian hypothesis that animal spirits work through credit markets and drive business cycles. A growing literature studies the macroeconomic effects of sentiment-driven fluctuations, from early accounts (e.g., Minsky 1977, Kindleberger 1978) to modern quantitative frameworks based on diagnostic expectations (e.g., Bordalo et al. 2018, Bianchi et al. 2022, Maxted 2023, Krishnamurthy and Li 2025). However, this work does not isolate sentiment shocks originating in banks' supply of credit. This paper provides the first empirical estimates of such shocks and their macroeconomic effects. In doing so, it complements studies of sentiment in corporate bond markets (e.g., López-Salido et al. 2017, Leiva-Leon et al. 2022) and among households (e.g., Lagerborg et al. 2023), showing that the macroeconomic transmission of sentiment depends on which agents' beliefs are distorted. For example, I find that bank sentiment shocks drive credit conditions during crises but do not change household consumption behavior, while household sentiment shocks do not affect credit conditions but do strongly drive consumption behavior (Lagerborg et al. 2023).

The paper's third contribution is a tractable behavioral banking model that embeds sentiment shocks into a heterogeneous intermediary framework building on Gertler and Karadi 2011. Rel-

[5] Saunders et al. 2021 and Kwak 2022 both construct excess loan return-based sentiment measures using syndicated loan data. However, because non-bank lenders dominate this market (Fleckenstein et al. 2020), these measures are best interpreted as syndicated loan market sentiment rather than commercial bank sentiment.

ative to existing models of heterogeneous banks (e.g., Corbae and D'Erasmo 2021, Jamilov and Monacelli 2025, Coimbra and Rey 2024), the framework introduces sentiment-driven heterogeneity and yields a closed-form loan pricing equation that can be directly estimated using standard linear methods. This tractability allows the model to serve as a bridge between theory and measurement, linking bank beliefs to observed lending behavior.

Roadmap. The remainder of the paper is organized as follows: Section 2 details the measurement strategy and estimation, Section 3 introduces and describes the empirical measure of bank risk sentiment, Section 4 evaluates the macroeconomic importance of sentiments, Section 5 traces the transmission of bank sentiment shocks across the broader economy, and Section 6 concludes.

2 Measuring bank risk sentiment

I begin by constructing a measure of banks' animal spirits. This section first describes the theoretical framework necessary to define bank risk sentiment and guide the empirical estimation. The remaining subsections detail the resulting measurement strategy, required data, and estimated measurement equations in turn.

2.1 Theoretical framework: A model of sentiments in bank lending markets

The theoretical framework presents a partial equilibrium bank lending market, building on the financial blocks of the canonical Gertler and Kiyotaki 2010 model as a foundation. Risk neutral banks raise capital each period to form one-period loan portfolios, face (indirect) net worth constraints, and satisfy a generic CES loan demand. However, this model will diverge from the canonical macro-banking model in two key aspects.

First, banks are subject to animal spirit shocks, which act as a wedge between the rational expectations forecast of loan losses and the bank's actual forecast of loan losses, following Angeletos and La'o 2013.[6] I choose this particular approach to defining sentiments because it can be easily and flexibly mapped to a measurement equation that can be taken to the data.

Second, banks operate in a (Cournot) competitive credit market. The oligopolistic market structure has two distinct advantages over the monopolistically competitive environment common in

[6] An alternative approach would have been to define a sentiment formation process explicitly, for example with diagnostic expectations, which have been used to describe banks in structural settings before (see specifically Maxted 2023 or Krishnamurthy and Li 2025). However, this approach is restrictive when taken to the data because it only allows for one type of sentiment to emerge. I take a more structurally agnostic approach.

the macro-banking literature. First, oligopolistic competition allows for a tractable representation of both time- and firm-varying endogenous markups, which have been shown to play an important role in loan spread dynamics (see for example Corbae and D'Erasmo 2025). Second, oligopolistic competition falls more in line with the competitive market structure studied in the industrial organization literature of the U.S. banking system, such as in the canonical Monti-Klein banking model or more recently Wang et al. 2022, and incorporates demand factors and competitive pressures on loan pricing into the model.

2.1.1 Demand and supply for loans

Time is infinite and discrete. Banks issue one period, uncollateralized, risky loans. Loans are risky because borrowers will default with an exogenous time-varying and bank-specific probability. When loans default, they yield a gross return of zero and the principal is lost.

Demand. A continuum of entrepreneurs require bank loans to fund projects. These entrepreneurs behave as if a representative borrower demands loans to maximize a CES utility preference over bank-specific credit products.[7] Thus, loan demand for bank i at time t is standard and given as:

$$L_{i,t} = \psi_{i,t}\left(\frac{R_t}{R_{i,t}}\right)^{\theta_t} L_t \tag{1}$$

where $L_{i,t}$ is bank i's loans at time t, L_t are total banking sector loans, $R_{i,t}$ is bank i's portfolio average loan rate, R_t is the sector-wide loan rate, $\psi_{i,t}$ is the bank-specific appeal (i.e. the CES preference weight), and $\theta_t \geq 1$ is the time-varying elasticity of intermediate loan substitution. The bank's product appeal $\psi_{i,t}$ may subsume aggregate and bank-specific demand shocks, as well as account for non-pecuniary differences that create persistent differences in bank-specific demand (such as the strictness of loan terms and branch locations).[8]

Supply. Banks operate in a Cournot competitive loan market with the goal of maximizing their expected present discounted flow of future profits (which is equivalent to their net worth in this

[7] To be concrete, take the representative borrower as having CES preferences as: $U_t = \left(\int_{i\in\mathscr{B}} \psi_{i,t} L_{i,t}^{\frac{\theta_t-1}{\theta_t}} d_i\right)^{\frac{\theta_t}{\theta_t-1}}$, where the banking sector is the set of banks $\mathscr{B}$, which implies the corresponding aggregate loan rate: $R_t = \left(\int_{i\in\mathscr{B}} \psi_{i,t}^{\theta_t} R_{i,t}^{1-\theta_t} d_i\right)^{\frac{1}{1-\theta_t}}$ solving the standard cost minimization problem then yields the demand schedule for bank-specific loans.

[8] See Hottman et al. 2016 or Eslava et al. 2024 for a more thorough description of firm-specific appeal in a similar oligopolistic competition and CES demand setting, but in a non-financial firm application.

setting). Bank i's problem is:

$$\max_{L_{i,t}} \mathbf{V}(N_{i,t}, C_{i,t}; L_t, R_t) = \mathbb{E}\Pi_{i,t} + \beta_i \mathbb{E}\mathbf{V}(N_{i,t+1}, C_{i,t+1}; L_{t+1}, R_{t+1}) \quad \text{s.t.} \tag{2}$$

$$\Pi_{i,t} = R^*_{i,t} L_{i,t} - (L_{i,t} - N_{i,t}) C_{i,t} - \Gamma\left(\sigma^R_{i,t} \frac{L_{i,t}}{N_{i,t}}\right) \qquad \text{(Profit function)}$$

$$R^*_{i,t} = (1 - \lambda_{i,t}) R_{i,t} \qquad \text{(Realized gross return)}$$

$$N_{i,t} = N_{i,t-1} + \Pi_{i,t-1} \qquad \text{(Net worth)}$$

where the bank maximizes profits, and in turn net worth, by issuing $L_{i,t}$ loans. The gross realized portfolio return on loans, $R^*_{i,t}$, is realized at the beginning of period $t+1$ after $\lambda_{i,t}$ percent of borrowers default at the end of period t. Thus, profits $\Pi_{i,t}$ are known at the beginning of period $t+1$. The bank's net worth in period t is denoted $N_{i,t}$ and is simply the previous period's net worth plus realized gains or losses from the current period's loan portfolio. Future net worth is discounted at rate $\beta_i < 1$. I will make the simplifying assumption that banks are sufficiently well funded to cover loan losses so that I may abstract from the possibility of bank failures.[9]

Banks can use their net worth, $N_{i,t}$, to fund loans and can source deposits or other funding from an inter-bank funding market at the marginal gross cost $C_{i,t} = 1 + c_{i,t}$. The assumption of bank-specific capital costs is motivated by classic work on deposit market competition (see Freixas and Rochet 2008 for a textbook discussion of the canonical Monti-Klein model) and recent work on banking market power in deposit markets, such as Drechsler et al. 2017, or deliberate spatial sorting into markets with cheap deposits, such as Oberfield et al. 2024.

Banks also pay a regulatory cost based on their risk-adjusted leverage ratio, $\sigma^R_{i,t} \cdot \frac{L_{i,t}}{N_{i,t}}$, where $\sigma^R_{i,t}$ is the time- and bank-varying variance of loan portfolio returns.[10] The regulatory cost function, $\Gamma(\cdot)$, will be kept general for the remainder of the presentation of the analytical model, and is assumed to be increasing and weakly convex. More formally, I assume $\Gamma'(X) \geq 0$ and $\Gamma''(X) \geq 0$ for all $X \in \mathbb{R}$. The convex regulatory costs acknowledge the real presence of such costs borne by banks, as well as establishes a connection between a bank's net worth, $N_{i,t}$, and ability to make loans. Moreover, by explicitly incorporating the portfolio return risk, the regulatory cost term captures a reduced-form

[9] Relaxing this assumption would not qualitatively change the subsequent analysis, but would require a richer description of the Households or Government who would ultimately have to foot the bankruptcy bill.

[10] Various authors take up a similar object of interest when formulating regulatory costs and constraints. For example, Gabaix and Maggiori 2015 focus on a liquidity ratio while Coimbra and Rey 2024 employ a leverage ratio. I model regulation as a convex cost rather than a hard constraint, as in Gabaix and Maggiori 2015, to preserve differentiability and to allow regulatory tightness to map continuously into loan pricing.

risk-averse asset pricing dynamics, mirroring those highlighted by the intermediary asset pricing literature (such as He and Krishnamurthy 2013 and Brunnermeier and Sannikov 2014, or more recently Akinci et al. 2022). For concreteness of exposition, I will use $\Gamma(\sigma^R_{i,t} \cdot \frac{L_{i,t}}{N_{i,t}}) = \zeta \cdot \sigma^R_{i,t} \cdot \frac{L_{i,t}}{N_{i,t}}$ as the regulatory cost function in the following estimation, where ζ is the regulatory cost coefficient.

2.1.2 Loan Rates

Bank i solves its loan quantity problem and in turn charges the equilibrium loan rate:

$$R_{i,t} = \frac{1}{\beta_i} \cdot \underbrace{\frac{\theta_t}{\theta_t - 1} \frac{1}{1 - s_{i,t}}}_{\text{markups}} \cdot \underbrace{\frac{1}{1 - E\lambda_{i,t}}}_{\text{risk compensation}} \cdot \underbrace{[C_{i,t} + \Gamma'\left(\sigma^R_{i,t} \frac{L_{i,t}}{N_{i,t}}\right)]}_{\text{marginal cost}} \tag{3}$$

so that as the expected default rate, loan market share $s_{i,t}$, market frictions θ_t, cost of capital, or marginal regulatory cost increases, so does the loan rate. Conversely, as the size of the bank increases, $N_{i,t}$, the loan rate decreases and the quantity supplied increases. The markup term is partly exogenous, arising from market frictions that make it harder to substitute across loan suppliers (e.g. search and matching frictions), and partly endogenous, arising from oligopolistic rents a bank can extract due to its market power. See Appendix B.3 for the formal proof.

One may further express the loan rate as:

$$R_{i,t} = \frac{1}{\beta_i} \cdot \frac{\theta_t}{\theta_t - 1} \frac{1}{1 - s_{i,t}} \cdot \frac{C_{i,t}}{1 - E\lambda_{i,t}} \cdot \underbrace{\frac{\sigma^R_{i,t}}{1 - E\lambda_{i,t}}}_{\text{quantity of risk}} \cdot \underbrace{\frac{\zeta}{N_{i,t}}}_{\text{price of risk}} \tag{4}$$

where two new terms emerge from combining the regulatory cost and compensation for expected defaults, a pseudo-price and quantity of risk. The former is the variance of portfolio returns divided by the expected loan default rate (which determines losses), while the latter is the investor's wealth (in this case the net worth of the bank) times a risk aversion parameter (an alternative interpretation of the regulatory cost coefficient). The emergence of these two pricing factors highlights the potential for banks to behave *as if* risk averse investors pricing risky assets, and its consideration in the measurement of the bank's risk sentiments.

2.1.3 Bank Risk Sentiments

I define a bank's risk sentiment as a wedge differentiating the bank's rational expectations benchmark and actual forecast of loan losses. Therefore, I must postulate a law of motion for loan losses

that will provide an analytical forecast to benchmark a bank's expectations against.[11]

In the spirit of Jamilov and Monacelli 2025, I will assume that a bank's loan losses are a function of idiosyncratic risk (reflecting a bank's innate ability to manage and perceive risk) and aggregate risk (reflecting uninsurable shocks to the entire economy). Additionally, in keeping with evidence presented in Falato and Xiao 2023, the law of motion for loan losses will take on an AR(1) process. I will postulate that $\lambda_{i,t}$ follows a stochastic process with an idiosyncratic and aggregate component:

$$\lambda_{i,t} = \gamma_i + \rho_1 \lambda_{i,t-1} + \rho_2 \lambda_{t-1} + \omega_{i,t} \tag{5}$$

where $\lambda_{i,t}$ is a bank's loan default rate in time t, λ_t is the aggregate default rate, and $\omega_{i,t}$ is a mean zero, idiosyncratic, and exogenous shock.

The rational expectations forecast of loan default rates is then:

$$E_{RE}(\lambda_{i,t}) = \gamma_i + \rho_1 \lambda_{i,t-1} + \rho_2 \lambda_{t-1} \tag{6}$$

which implies the following decomposition of a bank's expectations:

$$\begin{aligned} E(\lambda_{i,t}) &= E_{RE}(\lambda_{i,t}) + \varepsilon_{i,t} \\ &= \gamma_i + \rho_1 \lambda_{i,t-1} + \rho_2 \lambda_{t-1} + \varepsilon_{i,t} \end{aligned}$$

where $\varepsilon_{i,t}$ is the bank's risk sentiment, the bank-level deviation from the rational expectations forecast of loan default rates. With the postulated law of motion for default rates, one can further expand the bank's loan pricing equation to explicitly reflect the presence of the bank's rational expectations and risk sentiment:

$$R_{i,t} = \frac{1}{\beta} \cdot \frac{\theta_t}{\theta_t - 1} \frac{1}{1 - s_{i,t}} \cdot \frac{1}{1 - (\gamma_i + \rho_1 \lambda_{i,t-1} + \rho_2 \lambda_{t-1} + \varepsilon_{i,t})} \cdot [C_{i,t} + \Gamma' \left(\sigma_{i,t}^R \frac{L_{i,t}}{N_{i,t}} \right)] \tag{7}$$

which in turn makes the relationship between bank risk sentiment and bank lending clear. An increase in the bank's rational expectations forecast of default rates or the bank's risk sentiment, $\varepsilon_{i,t}$, leads to an increase in the bank loan rate.

[11] One may take a more agnostic approach to estimating a rational expectations forecast by combining machine learning and large datasets, as in Bianchi et al. 2023 or McCarthy and Hillenbrand 2021. However, these approaches threaten to predict the behavioral sentiment of interest in addition to the fundamental risk of interest. Such an over-prediction problem becomes an identification problem when attempting to isolate sentiment shocks. For this reason I do not adopt these agnostic approaches.

Equilibrium and lending outcomes. Having fully specified the banks' loan pricing equation and formalized bank risk sentiment, I leave a description of the competitive equilibrium and a series of predictions for how bank sentiment may impact both micro and macro lending to Appendix B.

2.2 Measurement strategy

The theoretical framework yields a closed-form solution for a bank's loan pricing equation, which I can in turn use to measure bank-level sentiments in observed data.

The bank's log-linearized loan rate schedule is:

$$r_{i,t} = \gamma_i + \rho_1 \lambda_{i,t-1} + \rho_2 \lambda_{t-1} + \varepsilon_{i,t} + \log\left(\frac{1}{\beta_i}\right) + \log\left(\frac{\theta_t}{\theta_t - 1}\frac{1}{1 - s_{i,t}}\right) + c_{i,t} + \Gamma'\left(\sigma^R_{i,t}\frac{L_{i,t}}{N_{i,t}}\right) \tag{8}$$

assuming small enough marginal capital costs, and marginal regulatory costs. The pricing schedule can then be further written as the linear equation:

$$r_{i,t} = a_i + BX_{i,t} + \varepsilon_{i,t} \tag{9}$$

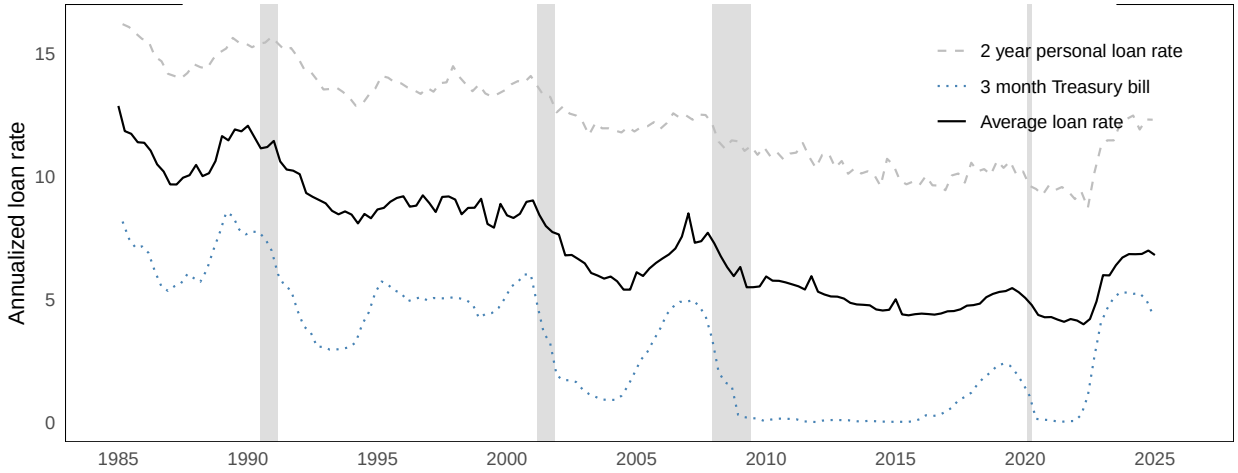
$$a_i = \gamma_i + \log\beta_i^{-1} \qquad X_{i,t} = \begin{bmatrix} \lambda_{i,t-1} \\ \lambda_{t-1} \\ \log\left(\frac{\theta_t}{\theta_t - 1}\frac{1}{1-s_{i,t}}\right) \\ c_{i,t} \\ \sigma^R_{i,t}/N_{i,t} \end{bmatrix} \qquad B' = \begin{bmatrix} \rho_1 \\ \rho_2 \\ 1 \\ 1 \\ \zeta \end{bmatrix}$$

which is easily recognizable as a linear fixed effects panel model that can be estimated with standard econometric techniques given observable proxies for the covariates $X_{i,t}$.

Sentiments are then recovered from the estimated Equation 9. First, bank-level sentiment is isolated as the unobservable component of the net loan rate, and is captured by the residual of the resulting loan rate decomposition. Then, with estimated bank-level sentiments in hand, measuring aggregate sentiments is straightforward: aggregate bank sentiment is the quarter-by-quarter loan-weighted average bank-level sentiment index. This is directly implied by the CES aggregator used by the generic borrower presented in the theoretical framework, since when $\theta = 1$ (as will be approximately true here) the CES aggregator is effectively the activity-weighted linear aggregator.

Relationship with other measurement problems. Conceptually, this approach parallels strate-

Figure 1: Commercial bank loan rates



Notes: This plot compares the sample implied loan-weighted loan rate to published risk free and consumer loan rates. The bank-level sample loan rate is calculated as the reported quarterly loan income divided by total loans, scaled by the inverse one minus net charge-off ratio. The average loan rate is then the loan-weighted average of bank-level loan rates over the cross-section of banks in the given quarter. The risk-free rate is proxied by the US 3-month T-bill. The consumer loan rate is the Finance Rate on Personal Loans at Commercial Banks, 24 Month Loan, reported by the Federal Reserve Board's G.19 Consumer Credit. Data are quarterly from 1985:Q1 through 2025:Q1.

gies in the productivity measurement literature, where unobservables are inferred from structural residuals. In particular, the measurement strategy taken here mirrors the revenue residual approach to estimating firm-level productivity.[12] When measuring productivity, the econometrician derives a firm's revenue function from its theoretical profit maximization problem, yielding a structural equation that decomposes revenues into inputs and productivity. The log-linearized revenue function is then estimated with observable proxies for input quantities and costs, identifying productivity as the residual revenue. My approach to measuring sentiment follows a similar path, but uses the structural loan rate equation to identify sentiments rather than revenue to identify productivity.

2.3 Data

The measurement strategy requires six ingredients: loan rates, markups, regulatory costs, capital costs, bank-level default rates, and aggregate default rates. Several ingredients are easily collected from quarterly U.S. Call Reports, a regulatory filing required of all commercial banks in the United

[12] See Foster et al. 2016 and Blackwood et al. 2021 for detailed discussions of productivity measurement strategies.

States detailing a bank's balance sheet and income statement.[13] Realized returns are measured as the bank-level loan interest income divided by total loans. Marginal cost of capital is proxied by the bank-level interest expenses divided by total loans, that is, the average interest paid to maintain a dollar of the bank's loan portfolio. Realizations of bank-level loan losses are measured by the bank's charge-off ratio, defined as net charge-offs divided by total loans.[14] Aggregate loan losses are the quarterly loan-weighted average of bank-level charge-off ratios.

However, Call Reports do not have information on loan rates, $R_{i,t}$. Instead, one might infer loan rates from the realized returns and charge-offs, which are observable in Call Reports. Note that given the theoretical framework, $R^* = (1-\lambda)R + \lambda \cdot 0$, then, $R = \frac{R^*}{1-\lambda}$.[15] Therefore, with R^* and λ, two observable bank-level characteristics, one can infer the bank-level portfolio average R.

Figure 1 shows the industry-wide (loan-weighted) average loan rate, and compares it to the risk-free rate proxied by three-month Treasury Bill yield, and the Federal Reserve reported two year personal loan rate. Notably, loan rates have been in secular decline since the beginning of the sample. In response, I will augment Equation 9 with a linear time trend to account for the secular decline in the estimation and further take the year-over-year difference in the resulting residualized loan rate to remove the residual trend.

Lastly, loan markups and regulatory costs are neither directly observable nor easily inferred from Call Report data, so I create quarter-by-quarter estimates of these two bank-level objects. However, I leave a full description of their estimation to Appendix A.1.

While constructing the required empirical ingredients, banks are defined at the bank holding company (BHC) level when applicable, or to the standalone bank-level for institutions without a holding company, following standard practices in the micro-banking literature (see for example Corbae and D'Erasmo 2021). I additionally merger-adjust Call Report data using NIC merger records by consolidating acquired banks backward in time, following standard practice in the banking and macro-finance literature (see for example Drechsler et al. 2017 for a detailed discussion).

[13] Standard micro-data cleaning procedures are applied. Bank-level data are winsorized at the 1st and 99th percentiles, observations with negative capital costs, loan rates, and loans are removed.

[14] Charge-offs are measured net of recoverable assets, thus reflect the net losses to the bank due to the default of a given loan. Therefore, where the analytical model may be unrealistic in ignoring the possibility of recoverable collateral or liens, the empirical exercises allow for this realistic possibility.

[15] Recall that λ is the portfolio average loan-level haircut, net recoveries, and so incorporates the average level of recoverable collateral. This implies that in an empirical setting one can assume a loan yields $R = 0$ if $\lambda = 1$, because the actual collateral recovery rate is incorporated into the average non-default yield via λ.

Table 1: Bank-level quarterly summary statistics

Variable	Symbol	Units	Mean	P(5)	P(25)	P(50)	P(75)	P(95)
Net worth	$N_{i,t}$	\$ Bn	18.7	0.117	0.369	0.873	2.29	15.7
Capital costs	$c_{i,t}$	%	1.37	0.122	0.491	1.16	1.95	3.39
Loan rate	$r_{i,t}$	%	2.14	1.20	1.59	2.15	2.63	3.17
Default rate	$\lambda_{i,t}$	%	0.193	0.000	0.007	0.044	0.158	0.833
Regulatory cost	$\Gamma'(\sigma_{i,t}^R \frac{L_{i,t}}{N_{i,t}})$	p.p.	0.351	0.012	0.076	0.195	0.433	1.24
Markups	$\frac{\theta_t}{\theta_t - 1} \frac{1}{1 - s_{i,t}}$		1.00	0.998	1.000	1.00	1.01	1.01

Notes: This table shows the bank-level quarterly summary statistics for the sample used to estimate bank risk sentiments. The column $P(X)$ presents the x-th percentile of observations in the sample. Observations are at the bank-quarter level. The sample includes 781,738 bank-quarter observations with 19,915 unique banks and quarterly data from 1985:Q1 through 2024:Q4.

Table 1 summarizes the sample used to estimate bank risk sentiment. The dataset comprises 781,738 bank-quarter observations from 19,915 unique U.S. banks over the period 1985:Q1 to 2024:Q4 and captures substantial heterogeneity in bank balance sheets and behavior. Median net worth is modest at \$0.873 billion, with a highly skewed distribution reflected in a 95th percentile of \$15.7 billion. Capital costs and loan rates are low on average, at 1.37% and 2.14% respectively, and exhibit relatively tight interquartile ranges, indicating limited variation in bank funding and lending costs across most institutions. Default rates are generally low, with a median of 4.4 basis points but a right tail extending above 83 basis points, consistent with episodic stress among a small subset of banks. The regulatory cost proxy, constructed as the marginal cost of leverage $\Gamma'\left(\sigma_{i,t}^R \frac{L_{i,t}}{N_{i,t}}\right)$, is similarly skewed, with a 95th percentile of 1.24. Lastly, markups are tightly clustered around one, indicating minimal dispersion in price-setting power among banks. The further description and construction details for each variable are presented in Appendix A.1.

2.4 Estimated loan rate decomposition and measurement equation

I estimate the loan pricing schedule, Equation 9, to decompose loan rates between observable pricing factors and unobservable bank sentiments. Table 2 reports the resulting price schedule. Columns (1) through (4) report the estimated single factor loan rate equations, while column (5) reports the estimated baseline loan rate schedule.

Observable bank-level characteristics and fixed effects account for a large portion of loan rate variation, approximately 91 percent. An increase in capital funding costs increases loan rates in a statistically significant manner. However, after controlling for both bank fixed effects and the trend in

Table 2: Estimated bank loan rate pricing schedule

	(1)	(2)	(3)	(4)	(5)
capital cost: $c_{i,t}$	0.515^{***}				0.513^{***}
	(0.019)				(0.019)
regulatory cost: $\Gamma'_{i,t}$		-0.555^{***}			0.078^{***}
		(0.054)			(0.030)
markups: $\log(\frac{\theta_t}{\theta_t-1}\frac{1}{1-s_{i,t}})$			-0.027^{***}		-0.014
			(0.007)		(0.011)
idio. defaults: $\lambda_{i,t-1}$				0.041	0.076
				(0.053)	(0.054)
agg. defaults: λ_{t-1}				-0.180	-0.121
				(0.137)	(0.076)
Loan share weighted	✓	✓	✓	✓	✓
Time trend	✓	✓	✓	✓	✓
Bank FE	✓	✓	✓	✓	✓
Observations	781,738	781,738	781,738	781,738	781,738
R^2	0.910	0.797	0.800	0.796	0.913

Notes: Columns (1)–(4) report auxiliary regressions corresponding to individual components of the pricing equation. Regulatory cost is measured as the log regulatory cost in these specifications. Column (5) presents the full model weighted by quarterly loan shares. All models include a linear time trend. Column (5) additionally includes bank fixed effects. Parentheses report robust standard errors clustered at the bank level. * denotes significance at 10%, ** at 5%, and *** at 1%.

loan rates, the pass-through of time-varying capital costs is less than one-for-one at approximately 51.3 percent. Rising regulatory costs increase loan rates and are statistically significant, capturing the well-documented relationship between the bank's net worth, risk aversion, and willingness to supply credit. Markups are neither economically nor statistically significant contributors to loan rates. Neither bank-level nor aggregate loan defaults enter the pricing equation at any standard statistical confidence level. These last two observations do not mean that there are no markups or compensation for expected defaults in loan rates, but rather that the within-bank variation in markups and defaults is not economically significant pricing factors. For example, slow or common movements in market frictions or market share growth may be reflected in the time trend or the persistent bank characteristics captured by bank-level fixed effects. Meanwhile, the regulatory

cost based on the bank's risk-adjusted leverage ratio may subsume the bank's loan pricing response to portfolio risk, ameliorating the measured conditional response to lagged observable defaults.

Unobservable bank-level sentiments account for a non-trivial portion of loan rates. Table 2 reports that sentiments explain approximately nine percent of loan rate variation. While it is unlikely that sentiments account for the full nine percent of loan rate variation due to measurement error from the imperfect proxies for regulatory costs and the law of motion for defaults, these estimates nonetheless establish a high ceiling for how much loan rate variation might be accounted for by sentiments.

Measurement robustness. I present evidence that risk sentiment does not systematically vary by bank size, and that measuring aggregate bank risk sentiment is robust to 1) the postulated law of motion for bank-level loan losses, 2) portfolio composition effects, including the average loan maturity and underlying borrower risk, 3) estimation under full sample or expanding window information sets, and 4) excluding prominent historical events, in Appendix C.

3 Describing Sentiments

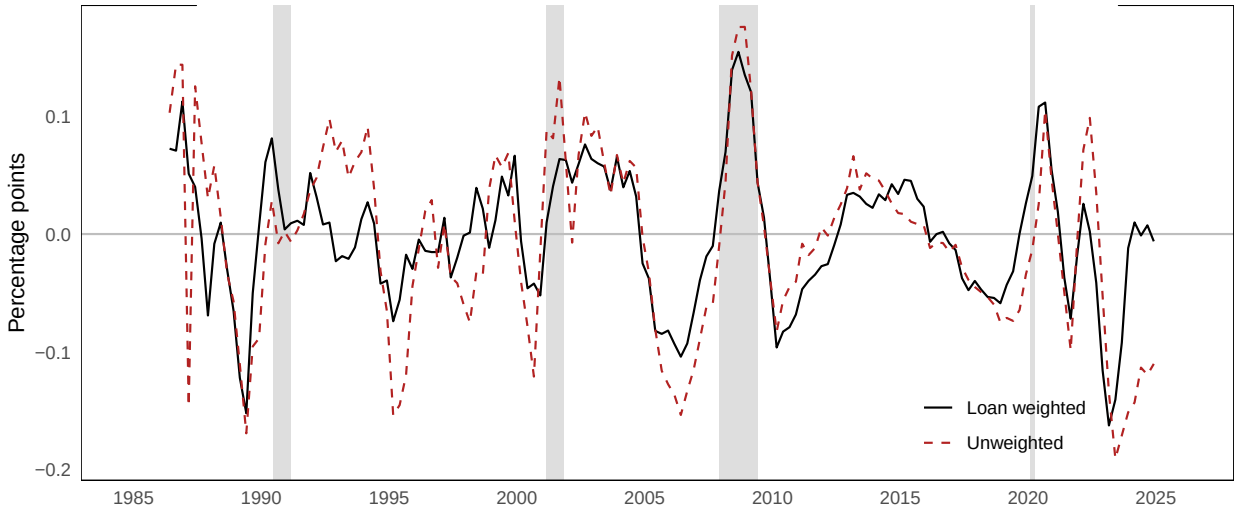
I next describe the estimated bank risk sentiments, starting with common movements across banks before moving to summarize bank-level heterogeneity and verifying that bank risk sentiments may be interpreted as animal spirits shocks.

3.1 Aggregate Sentiments

Common movements in bank sentiment are characterized by sharp increases in times of financial stress and uncertainty in the United States. Figure 2 shows that, for example, BRS spikes during the beginning of the S&L Crisis in the mid-1980s, the Dot-com stock bubble burst and the sudden collapse of Enron and WorldCom in the early 2000s, the GFC, and COVID-19 pandemic. Moreover, BRS also appears to be sensitive to news of foreign crises and uncertainty, with pessimistic sentiment spiking during Europe's "double-dip" recession. Periods of deteriorating sentiments are often precipitated by events elevating uncertainty, such as the September 11 terrorist attacks and the beginning of the Fed's quantitative tightening in 2018 (and subsequent yield curve inversion in 2019). Conversely, marked periods of bank optimism include the S&L crisis recovery, mid-2000s housing boom, and post-COVID-19 recovery.

It is important to clarify that while uncertainty and financial stress may lead to a rational increase in forecasted loan defaults, the rise in bank sentiment indicates that banks become even more pes-

Figure 2: Aggregate bank risk sentiment



Notes: This plot shows the quarterly aggregate bank risk sentiment of U.S. commercial banks. The solid black line presents the loan-weighted sentiment index and the dashed red line presents the unweighted sentiment index. Gray shaded regions mark NBER dated recessions. Data are quarterly from 1986:Q1 through 2024:Q4.

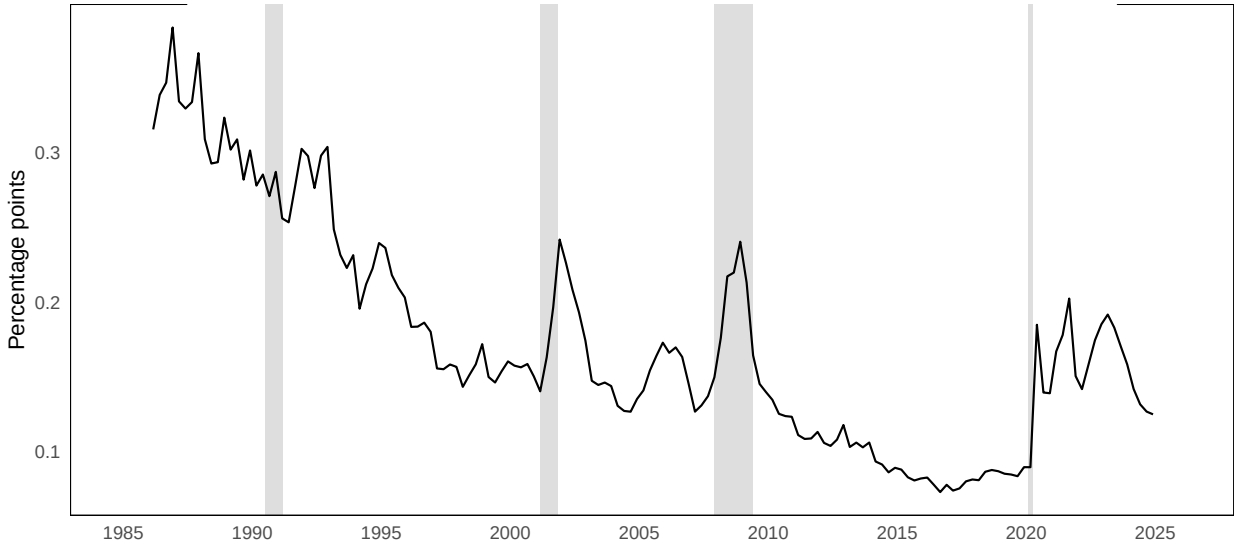
simistic about future states of the economy than is necessarily implied by a rational expectations view of the world. That is, BRS indicates banks become *excessively* pessimistic in times of uncertainty and financial stress. The fact that banks appear to become excessively pessimistic during periods of bad news falls in line with similar behavior documented among households in Lagerborg et al. 2023, which shows that households become pessimistic about economic outcomes when mass shootings are in the news.

3.2 Bank-level Sentiments

However, focusing on the common movements in banking sector sentiments obscures a large degree of heterogeneity at the bank-level.

There is a wide and time-varying distribution of bank-level sentiments. Figure 3 presents the interquartile range of bank sentiments, and shows that the dispersion in bank-level sentiments is both countercyclical and in secular decline. First, the dispersion in bank-level sentiments is generally countercyclical, increasing during the Dot-com recession, GFC, and COVID-19 pandemic. Second, the dispersion of bank-level sentiments has been in secular decline since the beginning of the sample. The interquartile range of bank-level sentiments has fallen from 34.5 basis points in

Figure 3: Dispersion of bank-level sentiments



Notes: This plot shows the dispersion of bank-level sentiments through time. The solid black line presents the annual average inter-quartile range of bank-level sentiments. Gray shaded regions mark NBER dated recessions. Data are quarterly from 1986:Q1 through 2024:Q4.

1986 to approximately 7 basis points in 2018, before rebounding after the COVID-19 pandemic. There may be a number of reasons for the growing coordination in bank sentiments, for example the consolidation in the U.S. banking sector or the gradual increase in the use of hard information to approve loan applications.[16]

The time-varying dispersion in bank-level sentiments may be due to heterogeneity in bank-level sentiment processes, differences in realized sentiment shocks, or a combination of both. That is, if one considers bank-level sentiments to follow an AR(1) process, such as

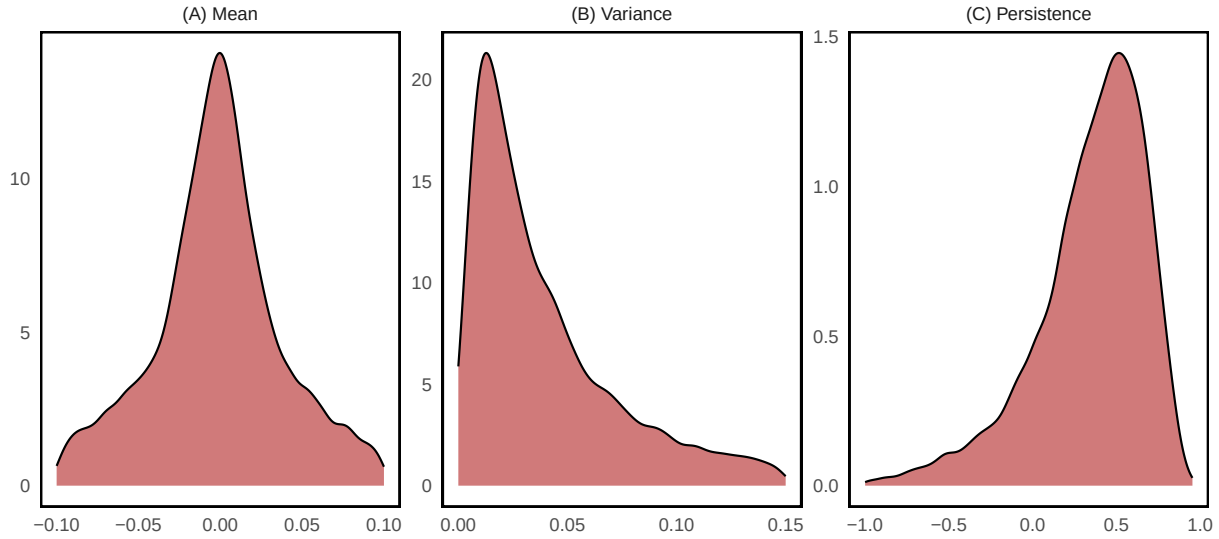
$$\varepsilon_{i,t} = \alpha_i^{\varepsilon} + \rho_i^{\varepsilon} \cdot \varepsilon_{i,t-1} + \eta_{i,t} \tag{10}$$

then differences can arise from differences in the average α_i^{ε}, persistence ρ_i^{ε}, or shocks $\eta_{i,t}$.[17]

[16] *Hard information* is quantifiable and portable, such as accounting information used to evaluate potential loan offers. Banks are gradually relying more on hard information when making decisions about extending loans. Conversely, human loan officers' own sentiments are being gradually removed from the loan application process, which in turn removes them as a source of bank risk sentiment. Fewer sources of variation in sentiments over time may explain the secular convergence in sentiments. See Liberti and Petersen 2019 for further discussion of hard information in banking.

[17] The BIC minimizing lag order for autoregressive representations of the bank-level sentiment process is one in both the pooled sample estimation and when estimating the series bank-by-bank.

Figure 4: Bank-level sentiment processes



Notes: red shaded regions show the empirical density functions of bank-specific (A) mean risk sentiments, (B) variance of risk sentiments, and (C) AR(1) coefficient of risk sentiments. Data are an unbalanced panel of 18,525 banks, quarterly from 1986:Q1 through 2024:Q4.

Differences in bank-level sentiment are driven by differences in both sentiment processes and realized shocks. Figure 4 presents the distributions of bank-level average sentiment, persistence, and variance. Panel (A) shows that the average bank sentiment is centered around zero, but with a large mass of optimistic and pessimistic banks. In fact, there is a small group of banks that on average decrease (increase) loan rates by 10 basis points due to excessive optimism (pessimism). Panel (B) and (C) show that the bank-level variance and persistence in sentiments are also heterogeneous, although most banks have relatively stable sentiments that slowly decay (in fact the mean persistence of approximatley 0.45 suggests the effects of a sentiment shock will take more than a year to decay to less than five percent of its initial impact).

In summary, industry-wide sentiments are generally acyclical, though with pessimism spiking during times of crisis and excessive optimism during commonly labeled asset bubbles. However, these common movements mask a large degree of heterogeneity at the bank-level, which is in secular decline but with cyclical increases during crises.

3.3 Interpreting as animal spirits

Before using my measure of bank risk sentiments to evaluate the macroeconomic importance of animal spirits, I first verify that they may indeed be interpreted as animal spirit shocks. While I

Table 3: (In-sample) Forecasting bank-level loan losses with sentiments

	1-quarter ahead forecast			1-year ahead forecast		
	(1)	(2)	(3)	(4)	(5)	(6)
$(1/1000) \cdot$ sentiment: $\varepsilon_{i,t}$	0.461	0.209	0.111	0.559	0.469	0.434
	(0.315)	(0.158)	(0.121)	(0.502)	(0.303)	(0.284)
Bank characteristics			✓			✓
Bank & Date FE		✓	✓		✓	✓
Observations	687,538	687,538	687,538	637,264	637,264	637,264
R^2	0.001	0.450	0.453	0.001	0.400	0.401

Notes: This table presents the coefficients from in-sample forecasting regressions predicting the bank-level charge-off ratios. Columns (1)-(3) present one-quarter ahead forecasting models, while Columns (4)-(6) present one-year ahead forecasting models. Columns (3) and (6) include observable bank characteristics, including net worth, leverage, and market share. Observations are loan-share weighted across all models. Parentheses report bank-level clustered standard errors, and * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

cannot observe a bank's rational expectations forecast of its future loan losses, I can test that bank risk sentiments 1) do not contain systematically useful information for forecasting loan defaults, and 2) are not forecastable beyond their autoregressive dynamics. The former condition tests that bank risk sentiments are systematically uninformative, thus irrational to include in a bank's forecast of loan defaults. The latter condition tests that bank risk sentiments are unanticipated fluctuations and statistically independent of fundamental lending supply and demand factors.[18]

First, I test the (lack of) information in sentiments with an in-sample forecasting exercise. If sentiments can forecast bank-level loan losses, then they must contain useful information about future states of the world and should necessarily be included in the rational expectations forecast of defaults. If sentiment cannot systematically forecast loan losses, then they represent deviations from the rational expectations forecast and are properly interpreted as animal spirits. The in-sample forecasting model will follow a simple linear form:

$$\lambda_{i,t+h} = \eta_i + \eta_t + \beta_1 \varepsilon_{i,t} + \beta_2 x_{i,t} + e_{i,t+h}$$

which forecasts bank-level default rates $\lambda_{i,t+h}$ either one quarter (h=1) or one year (h=4) ahead with

[18] Aggregate bank risk sentiment is additionally shown to be statistically independent of common aggregate shocks of interest in Appendix E.

Table 4: (In-sample) Forecasting bank-level sentiments

	One-quarter ahead forecast			One-year ahead forecast		
	(1)	(2)	(3)	(4)	(5)	(6)
$(1/1000) \cdot \text{leverage}_{i,t-1}$	0.134		0.134		−0.414	−0.414
	(0.154)		(0.154)		(2.240)	(2.240)
$(1/1000) \cdot \text{appeal}_{i,t-1}$		−0.003	−0.003	−0.001		−0.001
		(0.003)	(0.003)	(0.002)		(0.002)
Bank FE	✓	✓	✓	✓	✓	✓
Observations	637,264	637,264	637,264	591,091	591,091	591,091
R^2	0.455	0.455	0.455	0.095	0.095	0.095

Notes: This table presents coefficients from in-sample forecasting regressions predicting bank-level sentiments. Columns (1)-(3) present one-quarter ahead forecasts, while Columns (4)-(6) present one-year ahead forecasts. Within each horizon, columns are ordered as leverage-only, appeal-only, and both controls. All models include four lags of bank-level sentiment and bank fixed effects. Observations are loan-share weighted. Standard errors clustered at the bank level are reported in parentheses. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

bank-level sentiments $\varepsilon_{i,t}$ and, in some configurations, with time fixed effects η_t (to account for the possible influence of aggregate shocks), bank fixed effects η_i (to account for the bank-specific bias in default rates), and additional bank-level covariates, $x_{i,t}$, including the bank's net worth, leverage ratio, and market share, with forecast errors $e_{i,t+h}$.

Sentiments do not contain systematically useful information for forecasting loan defaults. Table 3 reports the loading on bank-level sentiment (i.e. the regression coefficient) in the estimated forecasting models. Columns (1) through (3) more specifically show that there is no systematically useful information in bank-level sentiments for forecasting one-quarter ahead loan losses in any of the forecasting models tested. That is, bank-level sentiments do not enter any forecasting model with a loading statistically distinguishable from zero at any standard confidence level. Columns (4) through (6) show a similar result when forecasting one-year ahead loan losses and accounting for additional information from fixed effects and bank-level characteristics.

Second, to test if BRS is unforecastable and statistically independent of credit demand and supply shocks, thus reasonably interpreted as an exogenous shock, I turn to another in-sample forecasting exercise. I will forecast sentiment with two bank-level characteristics in particular, bank-specific appeal and leverage ratio. First, a bank's appeal captures bank-specific demand shocks (see

Hottman et al. 2016 for a more detailed discussion of firm-specific appeal).[19] Second, changes in a bank's leverage ratio reflect changes in its willingness to supply credit for idiosyncratic reasons, such as changes in its risk aversion or uncertainty (as in He and Krishnamurthy 2013 and Brunnermeier and Sannikov 2014), or for macroeconomic reasons, such as aggregate shocks deteriorating collateral values (as in Gertler and Kiyotaki 2010). Thus the two bank-level characteristics provide a holistic summary of bank-level and aggregate factors affecting the demand and supply of bank credit. If there are exogenous forces driving bank risk sentiment, then these two characteristics should reasonably subsume these forces and indicate their importance in the forecasting exercise.

The in-sample forecasting model will follow a similar form to the previous exercise:

$$\varepsilon_{i,t+h} = \eta_i + \beta_1 \left(\frac{L_{i,t}}{N_{i,t}} \right) + \beta_2 \psi_{i,t} + \beta_3 x_{i,t} + e_{i,t+h}$$

where one is forecasting bank-level sentiments $\varepsilon_{i,t+h}$ either one quarter (h=1) or one year (h=4) ahead with a vector of four autoregressive lags of sentiment, $x_{i,t}$, the bank's leverage ratio $\frac{L_{i,t}}{N_{i,t}}$, and appeal $\psi_{i,t}$, with forecast errors $e_{i,t+h}$.

Bank risk sentiment is not forecastable with bank-level supply and demand factors. Table 4 reports the loading of the bank leverage ratio and appeal in the estimated forecasting models. The bank characteristics are not informative about future sentiments in any estimated model. However, this is not simply because the relationships are imprecisely estimated, but rather, the loadings are precisely estimated around zero. Moreover, the inability to forecast bank-level sentiments is robust. I additionally tested in-sample forecasting models without loan-weighting observations, in various sub-samples (for example, with and without the COVID-19 pandemic period), with time fixed effects, and with polynomial terms, but without any discernible increase in the ability to forecast bank-level sentiments. These exercises are omitted for brevity.

3.4 Diagnosing the behavioral origins of bank sentiment

Having established that bank risk sentiments behave like animal spirits, I lastly attempt to characterize the behavioral friction or source driving these irrational deviations. The primary hypothesis emerging from the behavioral finance literature is that banks form diagnostic expectations over the realization of loan losses (see for example, Bordalo et al. 2018, Bordalo et al. 2019, Maxted 2023 among others). To test this potential source of sentiments, I estimate the dynamic correlation between bank-level sentiments and realizations of bank-level defaults. This will test whether animal

[19] See Appendix A.1 for details on estimating bank-specific appeal.

Table 5: Sentiment and realized loan default rates

	(1)	(2)	(3)	(4)
$\Delta\lambda_{i,t-1}$	-8.082^{***}	-8.370^{***}	-8.847^{***}	-8.419^{***}
	(1.414)	(1.282)	(2.684)	(2.618)
$\lambda_{i,t-1}$			-1.568	-0.100
			(2.861)	(2.939)
Date FE		✓		✓
Bank FE	✓	✓	✓	✓
Bank Clustered SE	✓	✓	✓	✓
Loan-share weighted	✓	✓	✓	✓
Observations	541,433	541,433	541,433	541,433
R^2	0.041	0.150	0.041	0.150

Notes: This table presents the regression coefficients from a model projecting bank-level sentiments onto realizations of bank-level loan default rates. Columns (1)-(2) only include the change in loan default rates, such that $\Delta\lambda_{i,t-k} = \lambda_{i,t-1} - \lambda_{i,t-k-1}$. Columns (3)-(4) additionally includes the lagged level of default rates. Standard errors clustered at the bank level are reported in parentheses. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

spirits are over-extrapolating realizations of loan defaults, consistent with diagnostic expectations.

Bank sentiments reflect over-reaction, but are inconsistent with diagnostic expectations based on realizations of contemporaneous loan default rates. Table 5 reports the regression coefficients from four models projecting bank-level sentiment onto either the level or changes in loan default rates. Across all specifications, increases in either the level or change of loan defaults are associated with declines in measured bank risk sentiment. Given that higher sentiment corresponds to greater pessimism, this implies that banks become more optimistic as realized defaults increase. In other words: Sentiment is leading realizations and reverting as outcomes materialize. This pattern is inconsistent with diagnostic expectations formed over contemporaneous realizations of loan defaults, which would predict a positive relationship between defaults and pessimism. However, the negative relationship between sentiment and realized defaults is consistent with behavioral models in which agents initially overreact to information and subsequently revise their beliefs too aggressively toward the mean, for example due to overconfidence and self-attribution (Barberis et al. 1998) or representativeness (Daniel et al. 1998). The overreaction and correction dynamics can

also emerge in models of true animal spirits shocks, such as Angeletos and La'o 2013.[20]

4 Evaluating sentiments as a source of business cycles

I next evaluate the effects of bank sentiment shocks on macroeconomic dynamics —fluctuations in prices, activity, monetary policy, and credit conditions— and compare their importance in explaining business cycle fluctuations relative to a host of other macroeconomic shocks.

4.1 Econometric framework

I begin with a parsimonious representation of the economy. More specifically, I summarize the economy through the lens of six endogenous variables: activity, prices, monetary policy rate, credit conditions, consumer sentiment, and the loan rate premium. The first four variables build on a standard parsimonious empirical representation of the economy, see for example Gertler and Karadi 2015 and Caldara and Herbst 2019. The latter two variables are included as observable indicators for household and bank sentiments respectively.

I then postulate that the economy can be well summarized by the joint evolution of these six variables following a standard linear vector-autoregressive law of motion:

$$A_0 Y_t = \sum_{l=1}^{P} A_l Y_{t-l} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0_K, I_K) \tag{11}$$

where Y_t is the vector of $K = 6$ endogenous variables observed at time t, A_0, A_l $\forall l \geq 1$ are conformable coefficient matrices (including a constant for each endogenous variable), $P = 1$ is the lag order of the autoregressive system (following the BIC and AIC criterion), and ε_t is the vector of K white noise structural shocks.[21]

The structural law of motion can be mapped to its reduced form representation by multiplying each side of the equation by A_0^{-1}, yielding:

$$Y_t = \sum_{l=1}^{P} B_l Y_{t-l} + u_t, \quad u_t = S\varepsilon_t \tag{12}$$

[20] The systematic under-pricing of risk during crises may also represent a "gambling for resurrection" type behavior by banks, however Ben-David et al. 2019 find limited evidence of such dynamics in modern banks.

[21] The structural shocks, by definition, satisfy the conditions: $E[\varepsilon_t] = 0, \quad E[\varepsilon_t \varepsilon_t'] = I_K, \quad E[\varepsilon_t \varepsilon_s'] = 0 \ \ \forall s \neq t$, where I_K is the K dimensional identity matrix.

where $B_l = A_0^{-1} C_l$ and $A_0^{-1} = S$. The reduced form and structural shocks therefore mutually satisfy the condition:

$$E[u_t u_t'] = E[SS'] = \Sigma \tag{13}$$

where u_t is the reduced form shock, S is the linear function mapping structural to reduced form shocks, and Σ is the variance-covariance matrix of the reduced form shocks. That is, structural shocks are each a linear combination of the reduced form shocks that preserve the variance-covariance matrix of the reduced form model. The classic macroeconometric question is then how one statistically identifies the columns of S.

Identifying structural shocks. I use a combination of instrumental variables and sign restrictions to jointly identify six structural shocks: bank sentiment, household sentiment, aggregate demand, supply, financial risk, and monetary policy.

The two sentiment shocks will be identified with instrumental variables. First, I follow Lagerborg et al. 2023 and instrument household sentiment shocks with the number of mass shooting events in the United States in a given quarter. I leave further discussion of the data and validity of the relevance and exclusion restrictions to Lagerborg et al. 2023.

Second, I instrument bank sentiment shocks with the impact of bank risk sentiment on the bank loan rate premium. Bank risk sentiment satisfies both the standard relevance condition and exclusion restriction theoretically, following from its definition in Section 2.1, as well as empirically. At the macro-level, I show the first stage F-statistic is 145, indicating a relevant instrument, while bank risk sentiments are statistically independent of the five other aggregate structural shocks that drive endogenous outcomes in the BVAR, satisfying the exclusion restriction. At the micro-level, I additionally show the economically significant, *causal*, effects of bank-level sentiment shocks on loan-level spreads using a Khwaja and Mian 2008 style multi-borrower multi-lender panel approach using DEALSCAN loan data, while Section 3.3 demonstrated that bank-level sentiments do not respond to shifts in bank-level supply or demand factors. However, I leave the full discussion of the BVAR instrument validity tests to Appendix E and the loan-level analysis to Appendix D.

Given bank risk sentiment is a valid instrument for bank sentiment shocks, the procedure for estimating the corresponding column of S is standard and follows Gertler and Karadi 2015 in three steps.[22] 1) Estimate the reduced form VAR and calculate the matrix of reduced form residuals $\mathbf{u}_t$. Let u_t^b be the vector of reduced form residuals from the equation for the loan rate premium (the

[22] I sketch the process for estimating bank sentiment shocks here because it is the focus of this study, although I refer to Mertens and Ravn 2013 and Gertler and Karadi 2015 for a more detailed description of the procedure.

bank sentiment indicator) and let $\mathbf{u}_t^{-b}$ be the reduced form residuals from the equations for all other endogenous variables. Let S^{-b} be the response of $\mathbf{u}_t^{-b}$ to a unit increase in the sentiment shock ε_t^b. The next two steps will then use a standard two stage least squares procedure to estimate the ratio s^{-b}/s^b. 2) In the first stage, project u_t^b onto the instrument vector Z_t to form the fitted value $\widehat{u_t^b}$. The resulting predicted value isolates the variation in the reduced form residual for the loan rate premium that is due to the structural bank sentiment shock. 3) In the second stage, regress u_t^{-b} on $\widehat{u_t^b}$. Given the variation in $\widehat{u_t^b}$ is due only to ε_t^b, the second stage regression yields a consistent estimate of s^{-b}/s^b, so that:

$$u_t^{-b} = \left(\frac{s^{-b}}{s^b}\right)\widehat{u_t^b} + \mathrm{o}_t \tag{14}$$

where $\widehat{u_t^b}$ is orthogonal to the error term o_t, given the exclusion restriction. An estimate for s^b is then derived from the reduced form variance-covariance matrix, Equation 13, and Equation 14.

Table 6: Sign restrictions identifying structural shocks

	GDP growth	Inflation	Policy Rate	Credit Conditions
Demand shock	+	+	+	?
Supply shock	+	-	?	?
Monetary policy shock	-	-	+	?
Risk shock	-	-	-	+

The remainder of the structural shocks are jointly identified by theoretically-motivated sign restrictions following Uhlig 2005, Rubio-Ramirez et al. 2010, and Fry and Pagan 2011, among others. Using GDP growth as activity and inflation as prices the restrictions are as follows:

- A positive demand shock will increase both activity and prices. It then follows from a standard Taylor rule that monetary policy will unambiguously tighten in response to a positive demand shock to prevent the economy from overheating.

- A positive supply shock increases activity and decreases prices. However, the monetary policy response to a supply shock is ambiguous as policymakers need to weigh the cost-benefit trade-off of leaning against the increasing activity at the cost of further pushing inflation below its target rate.

- The tightening monetary policy shock will be identified as an increase in the policy rate, and a decrease in activity and inflation in the economy. See Galí 2015 for a New Keynesian textbook model motivation of these restrictions.

- The financial risk shock is defined by following Christiano et al. 2014, where an increase in financial risk restricts creditors' ability to lend, leading to tightening financial conditions as well as a deterioration in activity and prices, prompting a monetary policy easing.

See Uhlig 2017 for further intuition regarding the standard supply and demand shock identification schemes, as well as a discussion on sign restriction identification more broadly. The impact on sentiment indicators is left ambiguous across shocks.

Data. The macroeconomic time series used in the parsimonious BVAR follow Gertler and Karadi 2015. Activity is represented by the real annual GDP growth rate and prices by core PCE inflation. Monetary policy is proxied by the one year Treasury rate. Credit conditions are proxied by the Gilchrist and Zakrajšek 2012 excess bond premium (EBP), which is interpreted as an indicator of the capacity of intermediaries to extend loans or more generally the overall credit supply conditions in the economy. Household sentiment is represented by the University of Michigan Survey of Consumers' index of consumer sentiment. Bank sentiment is represented by the bank loan rate premium (the loan-weighted average bank loan rate minus the policy rate). Data are quarterly from 1985 through 2024 and are further detailed in Appendix A.2.

Estimation. Model parameters are estimated with standard Minnesota priors following the Giannone et al. 2015 hierarchical model. Not all data are available for the entire sample, so missing observations are handled via Bayesian data augmentation following Tanner and Wong 1987. I additionally include the Lenza and Primiceri 2022 COVID-19 volatility adjustment. Posteriors are sampled with 150 thousand draws and a 50 thousand burn-in. Draws are thinned so that every fifth draw is accepted to reduce autocorrelation in the resulting posterior chain. A structural impact matrix is constructed for each draw following the Cesa-Bianchi and Sokol 2022 procedure for combining IV and sign restriction identifications.

4.2 The role of sentiment over the business cycle

I first examine banks' animal spirits as a source of business cycles. To do so, I estimate the forecast error variance decomposition (FEVD) of the BVAR, which quantifies the contribution of structural shocks to the variation in endogenous variables at different horizons over the business cycle.

Bank sentiment shocks are an important source of business cycle fluctuations. Figure 5 shows that bank sentiment shocks account for 23.65 percent of fluctuations in GDP growth when structural shocks are realized.[23] In fact, short-term fluctuations in GDP growth are dominated by bank and

[23] Table 15 includes longer horizons in Appendix F.

household sentiment shocks, in line with the economically significant effects of sentiment shocks on activity found in Lagerborg et al. 2023. The effects of sentiment on activity fade quickly, with the share of variation due to bank sentiment falling to 18.6 percent one year after shocks are realized. At this point, the amount of variation from sentiment shocks is roughly equivalent to variation from household sentiment and aggregate demand shocks, and dominates aggregate supply, financial risk, and monetary policy shocks. While banks are an economically significant force driving fluctuations in quantities over the cycle, they account for only a small share of fluctuations in prices, only explaining 10.09 percent of inflation three years after shocks are realized.

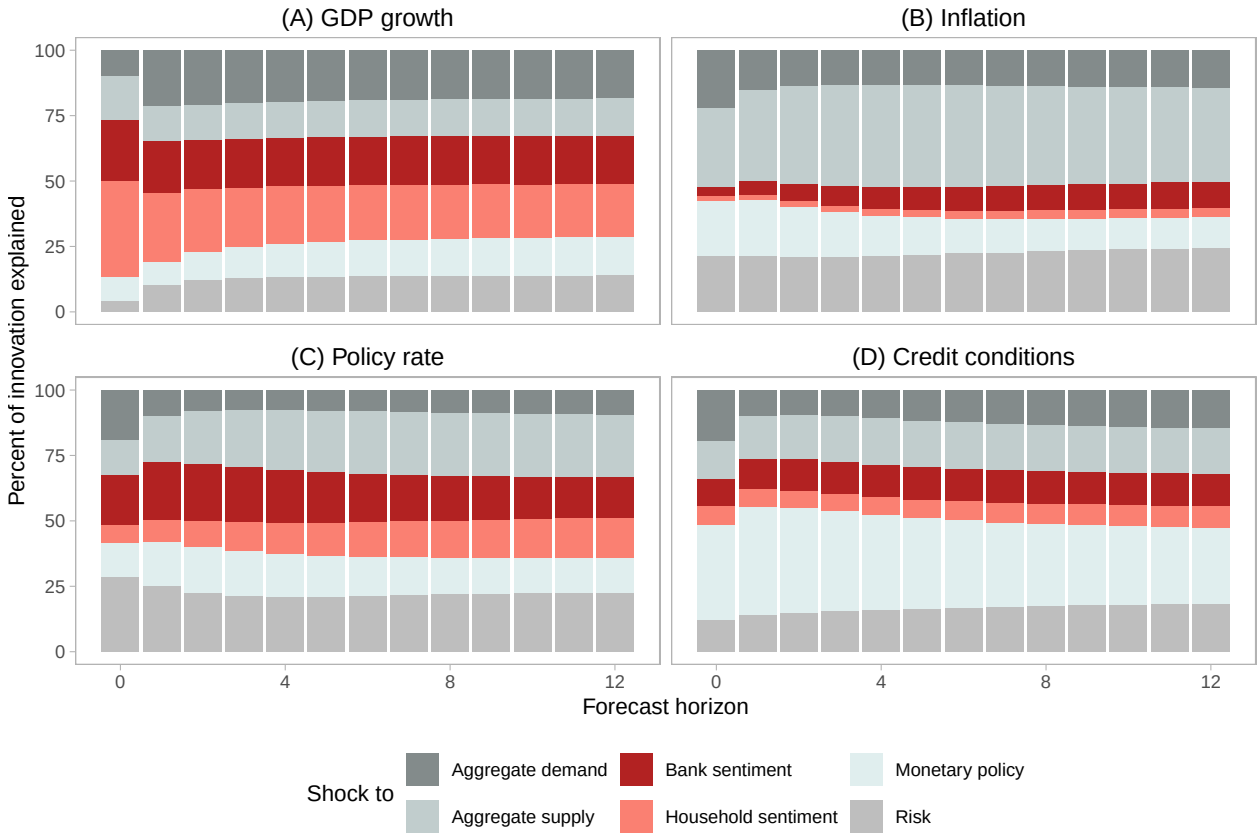
Bank sentiment shocks also have an economically meaningful effect on monetary policy. As bank sentiment shocks induce fluctuations in activity, they spur monetary policy actions. Bank sentiment in turn accounts for 20.09 percent of variation in the policy rate one year after shocks are realized, approximately equivalent to the variation due to aggregate supply and financial risk shocks. These three shocks are the primary source of policy variation over the business cycle. Meanwhile, household sentiment shocks account for less than half the variation in policy than bank sentiment, despite their equivalent roles in driving GDP growth. The difference between bank and household sentiment may be related to their respective effects on broader credit conditions (a primary transmission channel for monetary policy). Although neither shock is a primary driver of broad credit conditions on average, bank sentiment accounts for approximately twice the variation in credit conditions as household sentiments one year after impact.

Historical decomposition. While the variance decomposition suggests that bank sentiment may be an important source of economic fluctuations in activity and policy on average over the business cycle, common movements in bank sentiment are observably acyclical, only becoming acutely pessimistic during times of crisis. This leads me to additionally consider whether the effects of bank sentiment vary over time. To test this, I use the BVAR framework to further decompose the observable endogenous evolution of activity and credit conditions into the causal contributions of structural shocks through time.

The economic effects of bank sentiment are time-varying and asymmetric. Figure 6 shows that the cumulative contribution of bank sentiment to GDP growth and credit conditions is close to zero during expansions. For example, between 1994–1999, 2002–2004, and 2010–2019, sentiment shocks have little net effect on either activity or credit conditions. That is, bank sentiment is largely neutral during "normal" times.

In contrast, bank sentiment plays a dual-role around recession. Figure 7 shows the cumulative con-

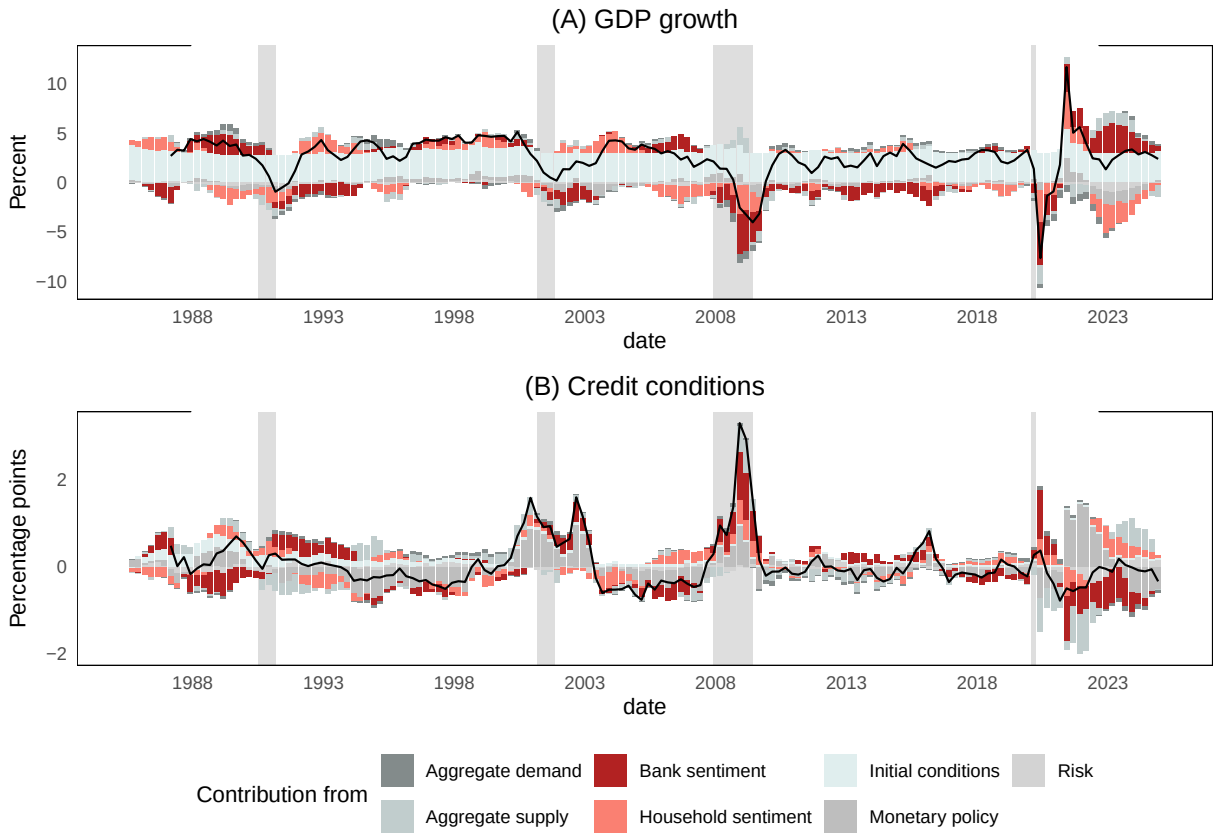
Figure 5: Variance decomposition of policy, activity, prices, and credit conditions



Notes: This plot presents the variance decomposition of business cycle fluctuations into contributions by structural shocks. The contribution of structural shocks account for 100 percent of the horizon-variable innovation. GDP growth is the annualized real quarterly growth rate, inflation is core PCE inflation, financial conditions are proxied by the Gilchrist and Zakrajšek 2012 EBP, and the policy rate is proxied by the one year Treasury rate.

tribution of bank sentiment shocks to GDP growth and credit conditions leading up to and shortly after the four recessions in the estimation sample. First, bank sentiment eases credit conditions directly before recessions, making it easier for borrowers to acquire cheap credit and boosting GDP growth. This behavior is not necessarily the cause of the subsequent recessions. For example, cheap lending in the U.S. did not incubate the COVID-19 pandemic. However, excessive optimism in the banking sector leading to cheap credit running up to the Dot-com bust and GFC is in line with narrative accounts of these asset bubble driven crises (see Brunnermeier and Reis 2023 for a recent textbook review). Second, bank sentiment turns acutely pessimistic during recessions and leads to excessively tight credit conditions during downturns and the early recovery period. In particular, Figure 6 shows that bank sentiment was the dominant force elevating credit conditions

Figure 6: Historical decomposition of activity and credit conditions



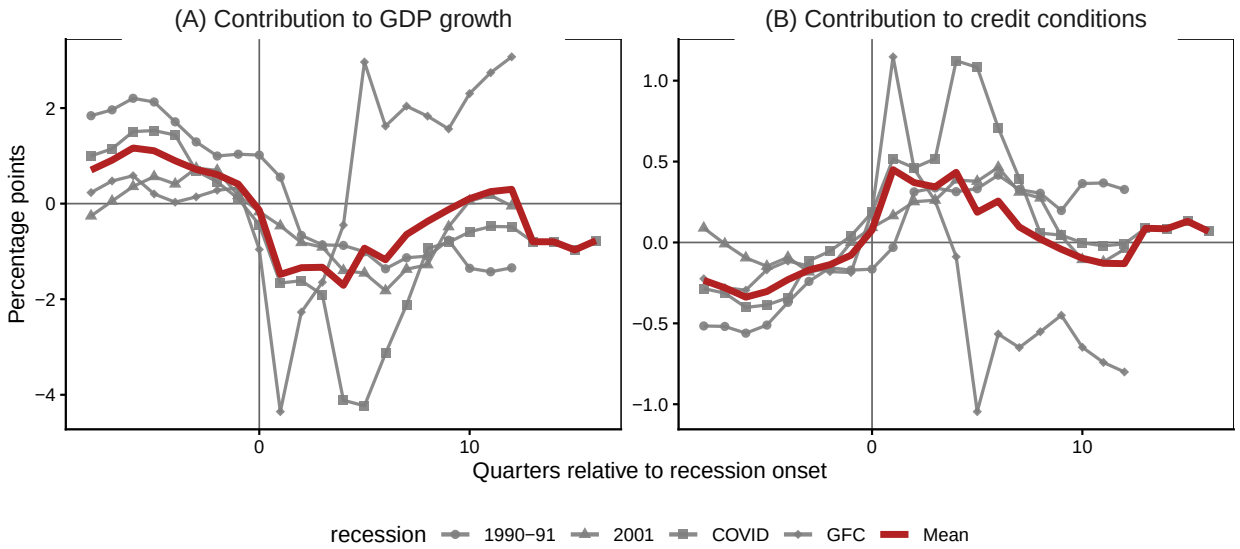
Notes: This plot presents the historical decomposition of business cycles and credit conditions into contributions by structural shocks. GDP growth is the annualized real quarterly growth rate and credit conditions are proxied by the Gilchrist and Zakrajšek 2012 EBP. Data are quarterly from 1986:Q1 through 2024:Q4.

during the GFC and COVID-19 pandemic recessions. The tightening effect on credit in turn spilled over into the real economy, dragging down GDP growth during each U.S. recession, and similarly acting as the primary force dampening GDP growth during the 2001 (Dot-com bust) recession, GFC, and COVID-19 pandemic recession.

4.3 Implications for business cycles and crisis dynamics

These findings suggest that bank sentiment acts as a state-dependent driver of macroeconomic fluctuations, with effects concentrated around turning points in the business cycle. This pattern differs from the standard financial accelerator mechanism (Bernanke et al. 1999, Kiyotaki and Moore 1997, Christiano et al. 2014) in which banks primarily amplify non-financial shocks. Instead, I find that shocks originating within the financial sector itself, through shifts in banks' beliefs about

Figure 7: Contribution of bank sentiment to activity and credit conditions during recessions



Notes: This plot presents the cumulative contribution of bank risk sentiment shocks to activity and credit conditions around NBER dated recessions. GDP growth is the annualized real quarterly growth rate and credit conditions are proxied by the Gilchrist and Zakrajšek 2012 EBP. Each gray line marks the contribution during an individual historical recession, while the red line marks the mean response across all recessions.

default risk, play an independent role in driving business cyclic dynamics. These findings are in line with recent quantitative macro-banking works including Jermann and Quadrini 2012, Ajello 2016, and Coimbra and Rey 2024, or empirical studies such as Ottonello and Song 2025 and Broadbent et al. 2025 that emphasize banks are a distinct source of business cycle fluctuations.

Moreover, these empirical results are suggestive evidence of the hypothesis that animal spirits stoke easy credit conditions before souring and becoming catalysts for severe recessions. Such dynamics have long been discussed in textbook settings, for example Kindleberger 1978, Akerlof and Shiller 2010, and Shiller 2020, although with limited well-identified empirical evidence. Some recent works have explained these dynamics using diagnostic expectations, for example Maxted 2023. However, evidence in Section 3.4 suggests that BRS is capturing an alternative, if not yet structurally specified, behavioral source of loan rate dynamics and in turn the boom-bust dynamics suggested by the historical decompositions.

5 Transmission of sentiment shocks

Having found bank sentiments to be an important source of business cycle fluctuations and crisis dynamics, I lastly turn to understanding the transmission of these shocks from bank lending to the broader economy.

5.1 Expanded econometric framework

I will continue to use the BVAR framework to estimate the effects of bank sentiments. However, as I now turn to focusing on the transmission and impacts of bank sentiment shocks, I will adopt a more comprehensive representation of the economy. To do so, I will make two changes to the previous model. First, I will replace the six variable representation of the economy with a more nuanced 31 variable model of the economy, which includes a set of financial variables summarizing the bank lending market, corporate bond market, stock market, and Treasury yield curve, as well as an expanded set of real variables that give a more detailed view of production, labor market, consumption, and trade. The full set of endogenous variables is presented in Appendix A.2. Second, since the focus of this exercise is to evaluate the impact of bank sentiment shocks, I do not need to meaningfully name the real shocks driving the economy, thus leave the statistical identification of non-bank sentiment shocks to a standard Cholesky decomposition of the BVAR's reduced-form error variance–covariance matrix.

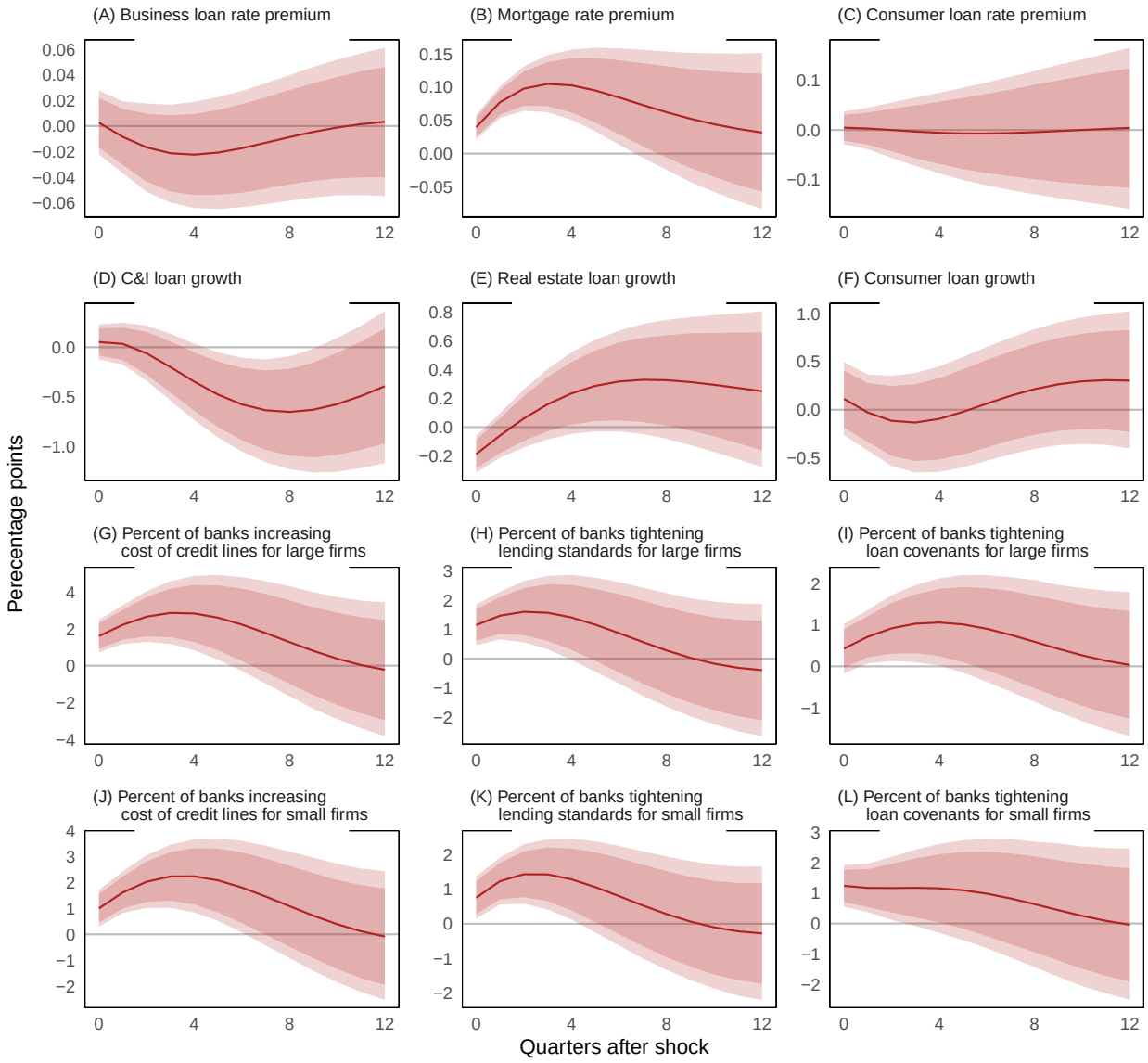
5.2 Effects of a pessimistic bank sentiment shock

I start by examining the effect of a one standard deviation pessimistic bank sentiment shock. More specifically, I decompose bank lending into business loans (for commercial and industrial purposes), mortgages, and consumer loans (such as credit cards and auto-loans), and find the effects of a pessimistic sentiment shock are asymmetric across lending activities.

First, the shock increases the price and decreases the quantity of mortgage lending, analogous to a standard supply shock. Figure 8 panel (B) shows the mortgage rate premium initially increases by five basis points before climbing to a maximum 10 basis point increase one year after impact. The price then recovers slowly, with the point estimate not returning to zero before three years after the shock. Meanwhile, panel (E) shows the growth in mortgage loans declines by 20 basis points. However the effect on quantities is short-lived, becoming statistically indistinguishable from zero one quarter after impact.

Second, the sentiment shock does not affect the price or quantity of consumer loans. Panel (C)

Figure 8: Bank lending market response to a pessimistic bank sentiment shock

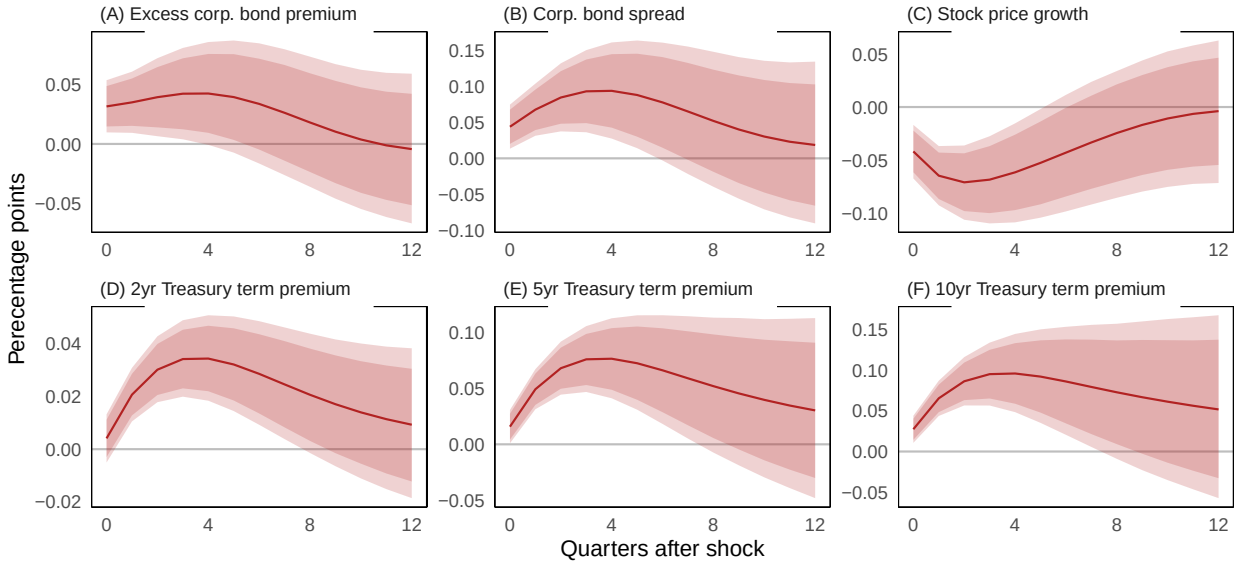


Notes: This plot presents the response of bank loan market conditions to a one standard deviation, pessimistic, bank sentiment shock. The red line marks the mean response. Shaded bands mark the 68 and 90 percent credible sets. All responses are in percentage points.

shows the loan rate premium is unmoved, though imprecisely estimated. In addition, Panel (F) shows a potential decline in consumer lending two to three quarters after the shock, although this and all subsequent changes in the endogenous outcome cannot be statistically distinguished from zero at any standard significance level.

Third, an increase in banks' excessive pessimism tightens both the intensive and extensive margins

Figure 9: Financial market response to a pessimistic bank sentiment shock

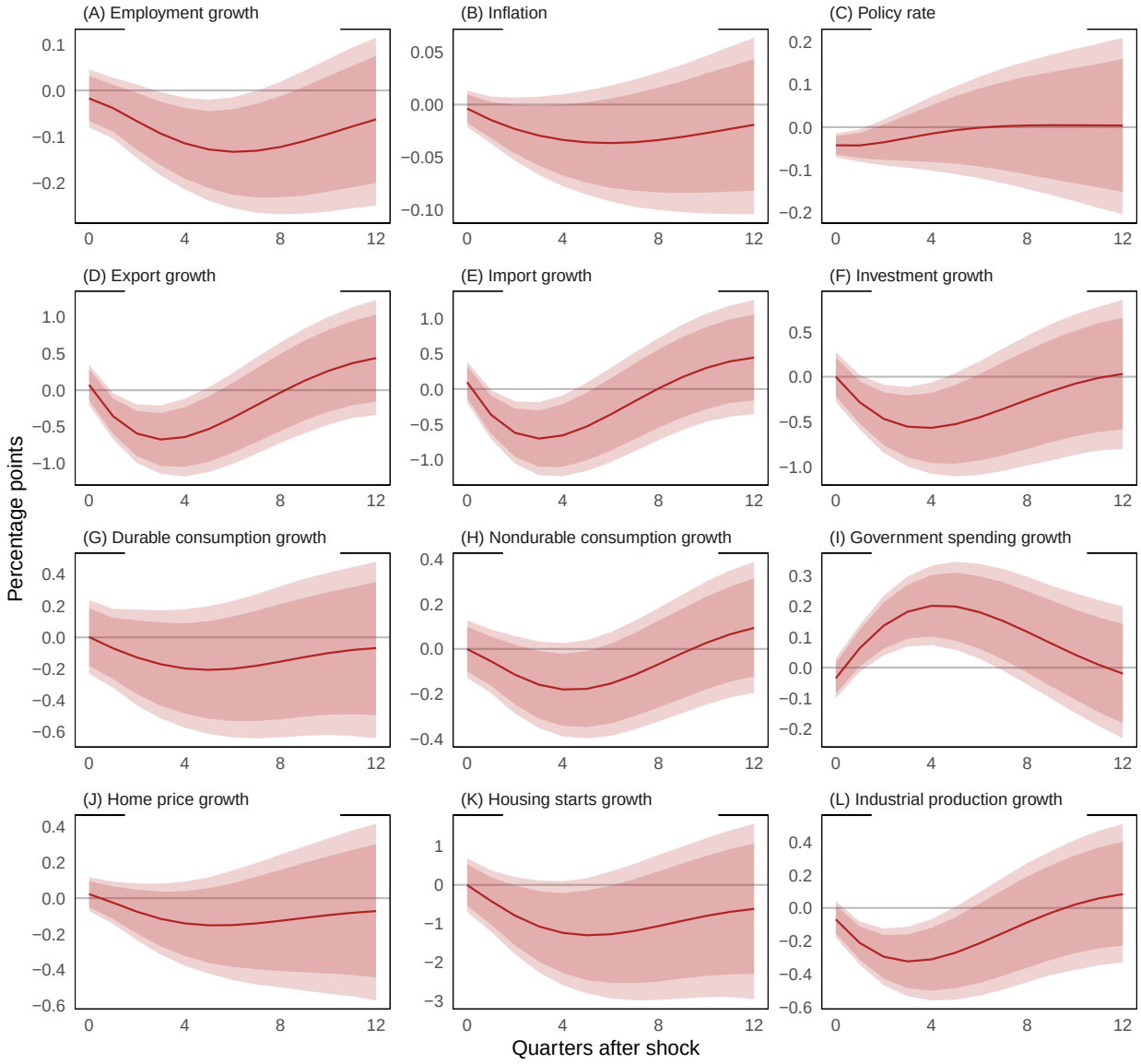


Notes: This plot presents the response of financial markets to a one standard deviation, pessimistic, bank sentiment shock. The red line marks the mean response. Shaded bands mark the 68 and 90 percent credible sets. All responses are in percentage points.

of business lending. While the average loan rate premium on commercial and industrial loans does not increase, the number of banks reporting an increase in the cost of credit lines they offer to businesses increases by approximately two percent for both large and small firms. Moreover, the increased share of banks reporting increases in the cost of credit lines is not offset by banks reporting a decrease in the cost within three years after the shock, indicating that the increase in price is relatively persistent. The cost of credit lines is an important dimension in addition to the overall loan rate on business loans because of how pervasive credit lines are in business' financing mix (for example, Sufi 2009 finds that between 70 and 80 percent of public firms hold direct lines of credit, the most common type of debt instrument in his sample). Panel (D) shows that the increase in the cost of credit lines is then accompanied by a delayed, but economically significant, 78 basis point decline in business lending growth.

While the decline in business lending takes up to two years to fully materialize, other dimensions of business lending are immediately impacted, namely bank lending standards and loan covenants. A bank's *lending standards* are the processes that the bank follows for approving or denying loan applications. Tightening (easing) lending standards indicate an increase (decrease) in the financial health requirements faced by borrowers seeking new loans. Conversely, a bank's *loan covenants* are the specific conditions included in loan contracts, such as collateral requirements and credit

Figure 10: Macroeconomic response to a pessimistic bank sentiment shock



Notes: This plot presents the response of real activity, policy rate, and prices to a one standard deviation, pessimistic, bank sentiment shock. The red line marks the mean response. Shaded bands mark the 68 and 90 percent credible sets. All responses are in percentage points.

limits, and tightening (easing) loan covenants indicate, among other things, more (less) restrictive borrowing constraints faced by borrowers. Therefore, lending standards tend to capture variations in the extensive margin of lending, while terms are more closely related to the intensive margin.

The sentiment shock tightens loan covenants banks offer to both large and small firms. Panels (I) and (L) show that banks tighten loan covenants when they become pessimistic. As a result, banks

may set more stringent restrictions on their borrowers' leverage, for example by lowering the allowable debt to earning ratios borrowers may maintain, or require greater collateral backing for new loans. Moreover, the wave of tightening does not stop after the shock, rather it builds and the number of banks tightening covenants for large firms increases through one year after impact. It is also notable that covenants do not begin to ease, on net, after the tightening. That is, a pessimistic sentiment shock tightens loan covenants, putting stricter borrowing constraints on businesses, but then banks do not ease covenants, thus borrowing constraints, within two years after the pessimistic shock passes.

The sentiment shock tightens lending standards banks offer to both large and small firms. Panels (H) and (K) show that the number of banks reporting that they tightened lending standards to small and large firms, respectively, increases on impact. That is, banks become more pessimistic and then increase the financial health required of borrowers to obtain loans. Similar to loan covenants, the wave of tightening lending standards does not stop after the period of the pessimistic shock, but rather continues through at least a year after impact. Moreover, the lending standards tightening is persistent, making it more difficult for new and existing borrowers alike to obtain loans for at least two years after the pessimistic shock.

Non-bank lending financial markets. The effects of the bank sentiment shock spills over from bank lending markets into financial markets more generally, with a pessimistic shock leading to a broad based deterioration across financial markets. First, there is a rise in Treasury term premium, indicating investors are pricing in greater default risk and/or inflation, suggesting a loss of confidence in the real economy. Second, the corporate bond spread increases, making it more expensive for corporations to borrow via the bond market, in addition to the increase in loan rates directly precipitated by the bank sentiment shock. Third, stock prices decline, further suggesting a decrease in investor confidence in the economy's current and future ability to generate dividends.

Non-financial activity, prices, and policy. Lastly, as the bank sentiment shock leads to a deterioration in credit markets, the effects are transmitted to the real economy.

The bank sentiment shock primarily effects the broader economy through the business lending channel. Despite its effect on mortgage lending, neither the price nor quantity of homes changes as a result, at least not in a statistically significant manner. The point estimates indicate home price growth falling by approximately 14 basis points one year after impact, while the housing starts (i.e. quantity growth) falls by 1.30 percentage points five quarters after impact. In comparison, industrial production growth falls by 32.4 basis points within one year after impact and is statis-

tically significant at the 10 percent confidence level. These two patterns then further emerge in the disaggregated components of GDP. Both non-durable and durable consumption are relatively unaffected by the sentiment shock, in line with its weak effect on consumer lending and the limited mortgage pricing pass-through to the housing market. Simultaneously, private investment (dominated by private business capital investments), along with imports and exports, all decline, in line with the deterioration in business lending conditions.

This broad based deterioration is partly offset by an immediate policy easing. The policy rate falls by approximately five basis points on impact, but quickly recovers as accommodative policy is rescinded. The policy intervention may explain the limited pass-through to inflation, which falls by less than five basis points and is not statistically significant. However, labor markets do slow as the effects of the business slowdown take root. One year after impact, employment growth falls by 13 basis points. The trade-off in stabilizing labor markets and inflation are then in line with the interpretation of the pessimistic bank sentiment shock as a negative supply shock.

Loan-level impacts. The macroeconomic findings that pessimistic sentiment shocks raise loan rates and are more pronounced during crises is mirrored at the bank-loan level. I analyze the loan-level impacts of bank sentiment shocks, using syndicated loan data from DEALSCAN and a multi-borrower multi-lender identification strategy à la Khwaja and Mian 2008. A one percentage point increase in a bank's risk sentiment causes a 14 basis point increase in its syndicated loan rate margins, and this increases to a 19 basis point effect during the GFC (an approximately 35 percent increase relative to normal times). The full exercise is presented in Appendix D.

Robustness. The negative response of macroeconomic conditions to a pessimistic bank sentiment shock is robust to being estimated with the parsimonious model or via local projections. Each robustness exercise is outlined in greater detail in Appendix F.

5.3 Transmission mechanism of bank sentiment shocks

The impulse response functions paint a clear picture of how sentiment shocks propagate across the economy. A pessimistic bank sentiment shock acts like a negative credit supply shock to business and mortgage lending, which in turn precipitates a broader decline in financial market conditions and real activity. However, housing prices and starts are largely unaffected while production declines, suggesting that the bank sentiment shock primarily affects the broader economy through the business lending channel. The sensitivity of real activity to the bank lending channel in particular

may be due to the fact that sentiments affect business lending through two distinct but complimentary margins and how they interact with monetary policy.

First, through the pricing margin, a pessimistic sentiment shock raises the effective cost of external finance by increasing mortgage rates and the cost of business credit lines, while simultaneously decreasing the quantity of loans supplied. Taken together, these responses closely resemble a standard negative credit supply shock, as studied in both the quantitative literature (e.g. Jermann and Quadrini 2012, Gertler and Kiyotaki 2010, Ajello 2016) and empirical work (e.g. Bernanke et al. 1996, Khwaja and Mian 2008, Amiti and Weinstein 2018, Greenstone et al. 2020). However, a key distinguishing feature is the heterogeneity across loan categories: the shock sharply affects business and mortgage lending, while leaving consumer credit largely unchanged.

The asymmetric transmission indicates that sentiment shocks operate through segments of the credit market most exposed to banks' subjective risk assessment. For example, business lending is highly dependent on how loan officers subjectively assess soft information for signals about a borrower's ability to pay back its loans, while consumer loans, such as credit cards and auto-loans are largely dependent on objective hard information like credit scores (see Liberti and Petersen 2019 for a more detailed discussion of hard and soft information in lending practices). This asymmetric behavior is therefore further evidence that the BRS measure captures shifts in bank-level beliefs regarding future default risk.

Second, through the constraint margin, the same increase in pessimism leads banks to tighten both lending standards and loan covenants. Lending standards tighten immediately and persistently, raising the threshold for borrower approval and restricting access to new credit. Simultaneously, loan covenants become more restrictive, tightening borrowing limits conditional on loan origination. This adjustment is consistent with a broad-based tightening of financial constraints, rather than a simple repricing of risk. Moreover, by tightening covenants, sentiment shocks operate by tightening earnings-based borrowing constraints, which are a central feature of modern credit contracts (e.g. Lian and Ma 2021, Drechsel 2023, Caglio et al. 2021). From a macroeconomic perspective, this corresponds to a shock that directly shifts the effective constraint parameters faced by firms, in the spirit of models such as Jermann and Quadrini 2012 or Jones et al. 2022.

As a result of these two business lending channel margins, conventional monetary policy may be ill-suited to fully stabilize the economy in the face of sentiment-driven disruptions to credit supply. In particular, lowering policy rates can partially mitigate increases in borrowing costs, but cannot directly reverse tightening lending standards or contractual constraints that restrict access

to credit. Therefore, even in the presence of accommodative policy, binding borrowing constraints and tightening lending standards may continue to depress real activity. In comparison, mortgages do not face the same type of covenant restrictions that business lending does, thus monetary policy can offset the increase in mortgage rates, insulating the real economy from the mortgage lending channel of sentiment shocks.

5.4 Comparison to other sentiment shocks and policy implications

The estimated impact of sentiment shocks on the macroeconomy creates an important complementarity between this work and the literature estimating the macroeconomic effects of corporate bond market sentiment and household sentiment shocks. First, both López-Salido et al. 2017 and Boeck and Zörner 2023 find that a pessimistic increase in corporate bond market sentiment leads to greater capital costs and in turn decreases macroeconomic activity. I similarly find that an increase in bank sentiment leads to a deterioration in activity, suggesting that pessimism in either credit market is detrimental to financial conditions and the broader economy in turn, despite differences in the borrowers that primarily rely on either market.

Second, Lagerborg et al. 2023 find household sentiment shocks account for a substantial amount of fluctuations in industrial production and household consumption, but do not meaningfully impact credit conditions. In comparison, I estimate that bank sentiment shocks are a primary driver of fluctuations in credit conditions during crises, but do not meaningfully change household consumption behavior. These two papers together show that pessimistic sentiment shocks are bad for economic outcomes, but the source of the pessimism matters for the resulting response of financial conditions and household consumption.

As a result, the policy tools available to combat sentiment shocks may depend on the source of the sentiment. For example, bank sentiment shocks gives rise to a non-trivial stabilization problem for monetary policy. A pessimistic bank sentiment shock translates to a standard aggregate supply shock. This means the monetary authority has to balance the trade-off between accommodating the declining labor market and dampening the rise in prices. In the data, the estimated endogenous policy response to the pessimistic shock answers this policy question by limiting the policy easing such that inflation remains stable while leaving real activity and employment growth to deteriorate. In comparison, a sentiment shock originating among households may instead manifest as a negative demand shock, which can be offset by monetary policy.

6 Conclusion

This paper measures animal spirits in the U.S. banking sector and asks whether they are an important determinant of credit conditions and source of business cycle fluctuations. I find that bank sentiments are characterized by sharp increases in pessimism during periods of uncertainty and crisis, although with substantial heterogeneity at the bank level. Fluctuations in aggregate bank sentiment in turn drive fluctuations in financial conditions, activity, and monetary policy. However the economic consequences of bank sentiments are time-varying, contributing little to business cycle dynamics during "normal" periods but becoming a dominant source of volatility in credit conditions and real activity during crises. Bank sentiments precipitate these crisis dynamics by acting like negative credit supply shocks that particularly effect mortgage and business lending before spilling into other financial markets. The tightening financial conditions lead to a deterioration in real activity, inducing a monetary policy easing. However, the policy easing cannot fully offset the credit tightening because bank sentiment works though both loan rates as well as loan covenants and lending standards which cannot be addressed by conventional monetary policy.

References

Ajello, Andrea (2016). "Financial intermediation, investment dynamics, and business cycle fluctuations". In: *American Economic Review* 106.8, pp. 2256–2303.

Akerlof, George A and Robert J Shiller (2010). *Animal spirits: How human psychology drives the economy, and why it matters for global capitalism*. Princeton university press.

Akinci, Ozge, ebnem Kalemli-Özcan, and Albert Queralto (2022). *Uncertainty Shocks, Capital Flows, and International Risk Spillovers*. Tech. rep. National Bureau of Economic Research.

Amiti, Mary and David E Weinstein (2018). "How much do idiosyncratic bank shocks affect investment? Evidence from matched bank-firm loan data". In: *Journal of Political Economy* 126.2, pp. 525–587.

Angeletos, George-Marios and Jennifer La'o (2013). "Sentiments". In: *Econometrica* 81.2, pp. 739–779.

Barberis, Nicholas, Andrei Shleifer, and Robert Vishny (1998). "A model of investor sentiment". In: *Journal of financial economics* 49.3, pp. 307–343.

Baron, Matthew and Wei Xiong (2017). "Credit expansion and neglected crash risk". In: *The Quarterly Journal of Economics* 132.2, pp. 713–764.

Barsky, Robert B and Eric R Sims (2012). "Information, animal spirits, and the meaning of innovations in consumer confidence". In: *American Economic Review* 102.4, pp. 1343–1377.

Ben-David, Itzhak, Ajay A Palvia, and René M Stulz (2019). *Do distressed banks really gamble for resurrection?* Tech. rep. National Bureau of Economic Research.

Bernanke, Ben, Mark Gertler, and Simon Gilchrist (1996). "The Financial Accelerator and the Flight to Quality". In: *The Review of Economics and Statistics*, pp. 1–15.

Bernanke, Ben S, Mark Gertler, and Simon Gilchrist (1999). "The financial accelerator in a quantitative business cycle framework". In: *Handbook of macroeconomics* 1, pp. 1341–1393.

Bianchi, Francesco, Cosmin Ilut, and Hikaru Saijo (Feb. 2023). "Diagnostic Business Cycles". In: *The Review of Economic Studies*. rdad024. ISSN: 0034-6527. DOI: 10.1093/restud/rdad024.

Bianchi, Francesco, Sydney C Ludvigson, and Sai Ma (2022). "Belief distortions and macroeconomic fluctuations". In: *American Economic Review* 112.7, pp. 2269–2315.

Blackwood, G Jacob et al. (2021). "Macro and micro dynamics of productivity: From devilish details to insights". In: *American Economic Journal: Macroeconomics* 13.3, pp. 142–172.

Boeck, Maximilian and Thomas O Zörner (2023). "The Impact of Credit Market Sentiment Shocks". In: *Journal of Money, Credit and Banking*.

Bordalo, Pedro, Nicola Gennaioli, and Andrei Shleifer (2018). "Diagnostic expectations and credit cycles". In: *The Journal of Finance* 73.1, pp. 199–227.

Bordalo, Pedro et al. (2019). "Diagnostic expectations and stock returns". In: *The Journal of Finance* 74.6, pp. 2839–2874.

Broadbent, Elijah et al. (2025). "Bank Lending Standards and the US Economy". In: *Journal of Economic Dynamics and Control*, p. 105247.

Brunnermeier, Markus K and Ricardo Reis (2023). "A crash course on crises: macroeconomic concepts for run-ups, collapses, and recoveries". In.

Brunnermeier, Markus K and Yuliy Sannikov (2014). "A macroeconomic model with a financial sector". In: *American Economic Review* 104.2, pp. 379–421.

Caglio, Cecilia R, R. Matthew Darst, and ebnem Kalemli-Özcan (Apr. 2021). *Collateral Heterogeneity and Monetary Policy Transmission: Evidence from Loans to SMEs and Large Firms*. Working Paper 28685. National Bureau of Economic Research. DOI: 10.3386/w28685.

Caldara, Dario and Edward Herbst (2019). "Monetary policy, real activity, and credit spreads: Evidence from Bayesian proxy SVARs". In: *American Economic Journal: Macroeconomics* 11.1, pp. 157–192.

Cesa-Bianchi, Ambrogio and Andrej Sokol (2022). "Financial shocks, credit spreads, and the international credit channel". In: *Journal of International Economics* 135, p. 103543.

Christiano, Lawrence J, Roberto Motto, and Massimo Rostagno (2014). "Risk shocks". In: *American Economic Review* 104.1, pp. 27–65.

Coimbra, Nuno and Hélene Rey (2024). "Financial cycles with heterogeneous intermediaries". In: *Review of Economic Studies* 91.2, pp. 817–857.

Corbae, Dean and Pablo D'Erasmo (2021). "Capital buffers in a quantitative model of banking industry dynamics". In: *Econometrica* 89.6, pp. 2975–3023.

– (2025). "A Quantitative Model of Banking Industry Dynamics". In: *Journal of Political Economy Macroeconomics* 3.4, pp. 621–673.

Daniel, Kent, David Hirshleifer, and Avanidhar Subrahmanyam (1998). "Investor psychology and security market under-and overreactions". In: *the Journal of Finance* 53.6, pp. 1839–1885.

Drechsel, Thomas (2023). "Earnings-based borrowing constraints and macroeconomic fluctuations". In: *American Economic Journal: Macroeconomics* 15.2, pp. 1–34.

Drechsler, Itamar, Alexi Savov, and Philipp Schnabl (2017). "The deposits channel of monetary policy". In: *The Quarterly Journal of Economics* 132.4, pp. 1819–1876.

Eslava, Marcela, John Haltiwanger, and Nicolas Urdaneta (2024). "The size and life-cycle growth of plants: The role of productivity, demand, and wedges". In: *Review of Economic Studies* 91.1, pp. 259–300.

Faccini, Renato and Leonardo Melosi (2022). "Pigouvian cycles". In: *American Economic Journal: Macroeconomics* 14.2, pp. 281–318.

Falato, Antonio and Jasmine Xiao (2023). *Expectations and credit slumps*. Tech. rep. Federal Reserve Board of Governors and University of Notre Dame.

Fleckenstein, Quirin et al. (2020). "Nonbank lending and credit cyclicality". In: *NYU Stern School of Business*.

Foster, Lucia et al. (2016). "Firm-level dispersion in productivity: is the devil in the details?" In: *American Economic Review* 106.5, pp. 95–98.

Freixas, Xavier and Jean-Charles Rochet (2008). *Microeconomics of banking*. MIT press.

Fry, Renee and Adrian Pagan (2011). "Sign restrictions in structural vector autoregressions: A critical review". In: *Journal of Economic Literature* 49.4, pp. 938–960.

Gabaix, Xavier and Matteo Maggiori (2015). "International liquidity and exchange rate dynamics". In: *The Quarterly Journal of Economics* 130.3, pp. 1369–1420.

Galí, Jordi (2015). *Monetary policy, inflation, and the business cycle: an introduction to the new Keynesian framework and its applications*. Princeton University Press.

Gertler, Mark and Peter Karadi (2011). "A model of unconventional monetary policy". In: *Journal of Monetary Economics* 58.1, pp. 17–34.

– (2015). "Monetary policy surprises, credit costs, and economic activity". In: *American Economic Journal: Macroeconomics* 7.1, pp. 44–76.

Gertler, Mark and Nobuhiro Kiyotaki (2010). "Financial intermediation and credit policy in business cycle analysis". In: *Handbook of Monetary Economics*. Vol. 3. Elsevier, pp. 547–599.

Giannone, Domenico, Michele Lenza, and Giorgio E Primiceri (2015). "Prior selection for vector autoregressions". In: *Review of Economics and Statistics* 97.2, pp. 436–451.

Gilchrist, Simon and Egon Zakrajšek (2012). "Credit spreads and business cycle fluctuations". In: *American Economic Review* 102.4, pp. 1692–1720.

Greenstone, Michael, Alexandre Mas, and Hoai-Luu Nguyen (2020). "Do credit market shocks affect the real economy? Quasi-experimental evidence from the great recession and" normal" economic times". In: *American Economic Journal: Economic Policy* 12.1, pp. 200–225.

He, Zhiguo and Arvind Krishnamurthy (2013). "Intermediary asset pricing". In: *American Economic Review* 103.2, pp. 732–770.

Hottman, Colin J, Stephen J Redding, and David E Weinstein (2016). "Quantifying the sources of firm heterogeneity". In: *The Quarterly Journal of Economics* 131.3, pp. 1291–1364.

Jamilov, Rustam and Tommaso Monacelli (2025). "Bewley Banks". In: *Review of Economic Studies*.

Jermann, Urban and Vincenzo Quadrini (2012). "Macroeconomic effects of financial shocks". In: *American Economic Review* 102.1, pp. 238–271.

Jones, Callum, Virgiliu Midrigan, and Thomas Philippon (2022). "Household leverage and the recession". In: *Econometrica* 90.5, pp. 2471–2505.

Khwaja, Asim Ijaz and Atif Mian (2008). "Tracing the impact of bank liquidity shocks: Evidence from an emerging market". In: *American Economic Review* 98.4, pp. 1413–1442.

Kindleberger, Charles P (1978). *Manias, panics, and crashes: A history of financial crises*. Wiley.

Kiyotaki, Nobuhiro and John Moore (1997). "Credit cycles". In: *Journal of Political Economy* 105.2, pp. 211–248.

Krishnamurthy, Arvind and Wenhao Li (2025). "Dissecting mechanisms of financial crises: Intermediation and sentiment". In: *Journal of Political Economy* 133.3, pp. 000–000.

Kwak, Seung (2022). "How Does Monetary Policy Affect Prices of Corporate Loans?" In: *FEDS Working Paper*.

Lagerborg, Andresa, Evi Pappa, and Morten O Ravn (2023). "Sentimental business cycles". In: *The Review of Economic Studies* 90.3, pp. 1358–1393.

Leiva-Leon, Danilo et al. (2022). "Introducing the Credit Market Sentiment Index". In: *Richmond Fed Economic Brief* 22.33.

Lenza, Michele and Giorgio E Primiceri (2022). "How to estimate a vector autoregression after March 2020". In: *Journal of Applied Econometrics* 37.4, pp. 688–699.

Lian, Chen and Yueran Ma (2021). "Anatomy of corporate borrowing constraints". In: *The Quarterly Journal of Economics* 136.1, pp. 229–291.

Liberti, José María and Mitchell A Petersen (2019). "Information: Hard and soft". In: *Review of Corporate Finance Studies* 8.1, pp. 1–41.

López-Salido, David, Jeremy C Stein, and Egon Zakrajšek (2017). "Credit-market sentiment and the business cycle". In: *The Quarterly Journal of Economics* 132.3, pp. 1373–1426.

Maxted, Peter (2023). "A macro-finance model with sentiment". In: *Review of Economic Studies*, rdad023.

McCarthy, Odhrain and Sebastian Hillenbrand (2021). "Heterogeneous investors and stock market fluctuations". In: *Available at SSRN 3944887*.

Mertens, Karel and Morten O Ravn (2013). "The dynamic effects of personal and corporate income tax changes in the United States". In: *American Economic Review* 103.4, pp. 1212–1247.

Mian, Atif, Amir Sufi, and Nasim Khoshkhou (2015). *Government economic policy, sentiments, and consumption*. Tech. rep. National Bureau of Economic Research.

Minsky, Hyman P (1977). "The financial instability hypothesis: An interpretation of Keynes and an alternative to "standard" theory". In: *Challenge* 20.1, pp. 20–27.

Oberfield, Ezra et al. (Mar. 2024). *Banks in Space*. Working Paper 32256. National Bureau of Economic Research. DOI: 10.3386/w32256.

Ottonello, Pablo and Wenting Song (2025). "Financial intermediaries and the macroeconomy: Evidence from a high-frequency identification". In: *The Economic Journal*, ueaf119.

Pflueger, Carolin, Emil Siriwardane, and Adi Sunderam (Mar. 2020). "Financial Market Risk Perceptions and the Macroeconomy*". In: *The Quarterly Journal of Economics* 135.3, pp. 1443–1491. ISSN: 0033-5533. DOI: 10.1093/qje/qjaa009.

Rubio-Ramirez, Juan F, Daniel F Waggoner, and Tao Zha (2010). "Structural vector autoregressions: Theory of identification and algorithms for inference". In: *The Review of Economic Studies* 77.2, pp. 665–696.

Saunders, Anthony et al. (2021). "Corporate loan spreads and economic activity". In: *NYU Stern School of Business Forthcoming*.

Shiller, Robert J (2020). "Narrative economics: How stories go viral and drive major economic events". In.

Stantcheva, Stefanie (2023). "How to run surveys: A guide to creating your own identifying variation and revealing the invisible". In: *Annual Review of Economics* 15.1, pp. 205–234.

Sufi, Amir (2009). "Bank lines of credit in corporate finance: An empirical analysis". In: *The Review of Financial Studies* 22.3, pp. 1057–1088.

Tanner, Martin A and Wing Hung Wong (1987). "The calculation of posterior distributions by data augmentation". In: *Journal of the American statistical Association* 82.398, pp. 528–540.

Uhlig, Harald (2005). "What are the effects of monetary policy on output? Results from an agnostic identification procedure". In: *Journal of Monetary Economics* 52.2, pp. 381–419.

– (2017). "Shocks, sign restrictions, and identification". In: *Advances in economics and econometrics* 2, p. 95.

Wang, Yifei et al. (2022). "Bank market power and monetary policy transmission: Evidence from a structural estimation". In: *The Journal of Finance* 77.4, pp. 2093–2141.

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A Data

A.1 Bank data

Bank-level data are from the Consolidated Report of Condition and Income, also known as Call Reports. A form of Call Report data was first collected in 1933 by the Federal Deposit Insurance Corporation (FDIC) as a part of the regulatory reform responding to the Great Depression. However, there was a substantial update to the required reporting frequency, the number of variables collected, and variable definitions in 1984, resulting in a bifurcation between the pre- and post-1984 Call Report sample history. Because of the incongruity between variable definitions across the two sample periods, modern Call Report data effectively begins in 1984.

The bank-level sample is representative of the U.S. banking sector. Figure 14 shows that the sum of deposits and loans reported in the micro-data (approximately) matches the level of deposits and loans reported by the Federal Reserve Board at the national-level. Figure 14 likewise shows that the loan-weighted charge-offs ratio reported in the micro-data matches the national charge-off ratio reported by the Federal Reserve. Banks are defined at the bank holding company (BHC) level when applicable, or to the standalone bank level for institutions without a holding company, following standard practices in the micro-banking literature (see for example Corbae and D'Erasmo 2021). I additionally merger-adjust Call Report data using NIC merger records by consolidating acquired banks backward in time, following standard practice in the banking and macro-finance literature (see for example Drechsler et al. 2017 for a detailed discussion). The sample extends from 1985 through 2025, and includes 19,915 unique banks, although banks enter and exit the sample due to births and failures, so that the maximum number of banks any given period is no more than 12 thousand (see Figure 15 for further description of the number and entry/exit of banks).

Call Reports do not have information on offered loan rates, so they must be inferred from the realized loan returns. Note that given $R^* = (1-\lambda)R + \lambda \cdot 0$, then, $R = \frac{R^*}{1-\lambda}$.[24] Therefore, with R^* and λ, two observable bank-level characteristics, one can infer the bank-level portfolio average R. Figure 12 shows the industry-wide (loan-weighted) average loan rate, and compares it to the risk-free rate proxied by three month Treasury Bill yield, and the Federal Reserve reported two year personal loan rate. Loan rates follow a similar pattern as the realized loan and default rates, falling since the 1980s, while increasing during recessions.

[24] Recall that λ is the portfolio average loan-level haircut, net recoveries, and so incorporates the average level of recoverable collateral. This implies that in an empirical setting one can assume a loan yields $R = 0$ if $\lambda = 1$, because the actual collateral recovery rate is incorporated into the average non-default yield via λ.

Table 7: Bank-level variable details

Variable	Formula	Call Report variables
Loan rates	loan interest income / total loans	RIAD4010, RCON2122
Log inverse net worth	log(1/N)	RCON3210
Loan share	$L_{i,t}/(\sum_{j\in\mathscr{B}_t} L_{j,t})$	RCON2122
Capital funding costs	total interest expense / total loans	RIAD4073, RCON2122
Charge-off ratio	net charge offs / total loans	RIAD4605, RIAD4635, RCON2122

Lastly, Figure 13 shows the quarterly median interest income and costs. The difference between these two loan characteristics is the observable net interest margin, which is used for estimating the regulatory costs and markups described in Section 2.3. Note that similar to the inferred loan rates, both interest income and costs have been in secular decline since the 1980s, but unlike loan rates, have a pro-cyclical component increasing before recessions.

A.1.1 Estimating banks' regulatory costs and markups

The bank-level regulatory cost and industry-wide markups are not observable in the U.S. Call Reports, but can be estimated. First, the regulatory cost function can be consistently estimated from the net realized loan return. To see this, I write the log-linearized realized return equation as follows:

$$r^*_{i,t} = \log\left(\frac{1}{\beta_i}\right) + \log\left(\frac{\theta_t}{\theta_t - 1}\right) + s_{i,t} + c_{i,t} + \Gamma'(\sigma^R_{i,t} L_{i,t}/N_{i,t}) - (\lambda_{i,t} - \mathbb{E}\lambda_{i,t}) \implies$$

$$\underbrace{r^*_{i,t} - s_{i,t} - c_{i,t} - (\lambda_{i,t} - \mathbb{E}\lambda_{i,t})}_{\equiv \chi_{i,t}} = \underbrace{\Gamma'(\sigma^R_{i,t} L_{i,t}/N_{i,t})}_{\equiv\, \zeta\cdot\sigma^R_{i,t}\cdot N^{-1}_{i,t}} + \underbrace{\log\left(\frac{1}{\beta_i}\right) + \log\left(\frac{\theta_t}{\theta_t - 1}\right)}_{\equiv\, \upsilon_{i,t}}$$

where $\chi_{i,t}$ is observable, $\upsilon_{i,t}$ are the unobservable pricing factors, ζ_t is the regulatory cost coefficient from assuming a linear functional form for $\Gamma(\sigma^R_{i,t} L_{i,t}/N_{i,t})$. The regulatory cost coefficient ζ can thus be estimated by projecting observable $\chi_{i,t}$ onto the bank's inverse log net worth, maintaining the identifying assumption that the bank-specific discount factor and industry-wide elasticity of substitution are not correlated with the firms time t net worth.

Second, I then turn to estimating bank-specific markups. For the markups I require a measure of a bank's loan market share —which is easily constructed from Call Report balance sheets— and the elasticity of substitution θ_t. Following the "demand function approach" widely employed by

the industrial organization literature, I estimate θ_t by matching the first moment of the observed realized return distribution with the return equation implied by the banks' FOC conditions.[25] The strategy is to first rewrite the loan pricing equation:

$$\underbrace{r^*_{i,t} - s_{i,t} - c_{i,t} - (\lambda_{i,t} - \mathbb{E}\lambda_{i,t}) - (\zeta_t)N^{-1}_{i,t-1}}_{\equiv \hat{\chi}_{i,t}} = \log\left(\frac{\theta_t}{\theta_t - 1}\right) + \log\left(\frac{1}{\beta_i}\right)$$

and as β_i converges to 1, $\hat{\chi}_{i,t}$ converges to $\log\frac{\theta_t}{(\theta_t-1)}$. Therefore, the ratio $\log\frac{\theta_t}{(\theta_t-1)}$ can be consistently estimated as either the mean of the $\hat{\chi}$ assuming a small enough $|1-\beta_i|$. In the other case, if $\log(\beta_i^{-1})$ is non-zero and confounds the estimated markup, the estimation will be biased by $\mathbb{E}\log(\beta_i^{-1})$. However, this is not a concern for my estimation of sentiments because the bank fixed effect in the loan rate schedule, Equation 9, will subsume the bias.

Figure 11 presents the estimated bank-level regulatory costs, portfolio risk, and industry-wide market frictions. Portfolio risk has been falling over time, but is countercyclical, rising in recessions, such as 2001, 2009, and 2020. In comparison, the regulatory costs have been rising over time. Lastly, estimated markups have been rising over time, with acute deterioration during the S&L crisis in 1986 and the Dot-com bust.

A.1.2 Estimating bank-specific appeal

I use bank-specific appeal as a proxy for aggregate and bank-level demand shocks when testing the forecastability of bank-level risk sentiments. To estimate these measures I go back to the CES preferences of the representative borrower, which yields the aggregate composite loan:

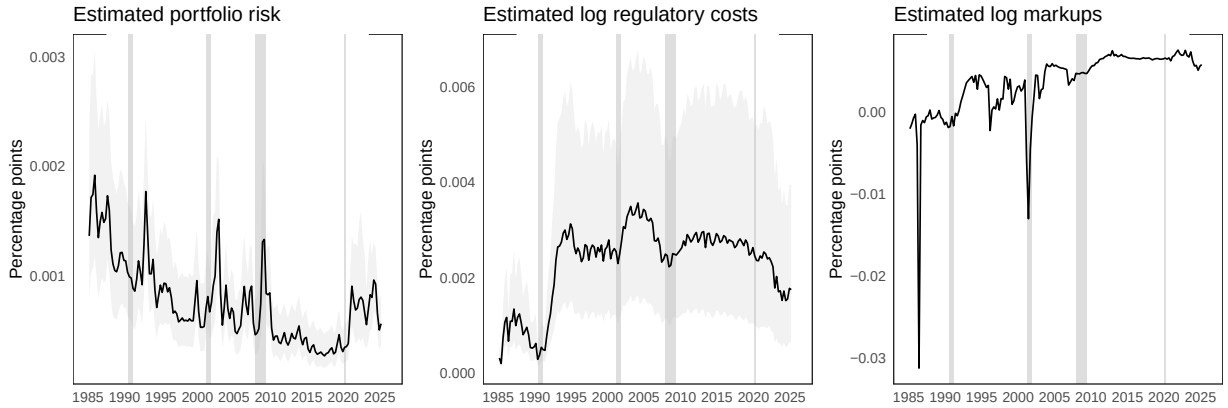
$$L_t = \left(\prod_{i\in\mathscr{B}} \psi_{i,t} L_{i,t}^{\frac{\theta_t-1}{\theta_t}} d_i\right)^{\frac{\theta_t}{\theta_t-1}}$$

where $\psi_{i,t}$ is bank i appeal in time t. Taking the log of the composite loan yields:

$$\left(\frac{\theta_t - 1}{\theta_t}\right) l_t = \sum_{i\in\mathscr{B}} \log\psi_{i,t} + \left(\frac{\theta_t - 1}{\theta_t}\right) l_{i,t}$$

[25] See De Loecker et al. 2020 for a practical overview of the three primary strategies for estimating markups. I take the "demand function approach" in contrast to Corbae and D'Erasmo 2021 or Jamilov and Monacelli 2025 who use the "production function approach" because the former subsumes all unexplained variation in loan rates, over and above the marginal cost, and defines it as a bank's markup, leaving no room for sentiments to play a role.

Figure 11: Estimated bank-level regulatory costs, portfolio risk, and industry-wide market frictions



Notes: This plot shows the estimated bank-level portfolio risk, regulatory costs, and markups. The solid black lines mark the median value of the bank-level distribution and the gray bands mark the inter-quartile range. Markups are common across the industry. Data are quarterly from 1985:Q1 through 2024:Q4.

where $l_{i,t} = \log L_{i,t}$ and $l_t = \log L_t$. I then recast the problem in vector form:

$$l_t = \left(\frac{\theta_t}{\theta_t - 1}\right)\overline{\log \psi_t} \cdot \bar{l}_t$$

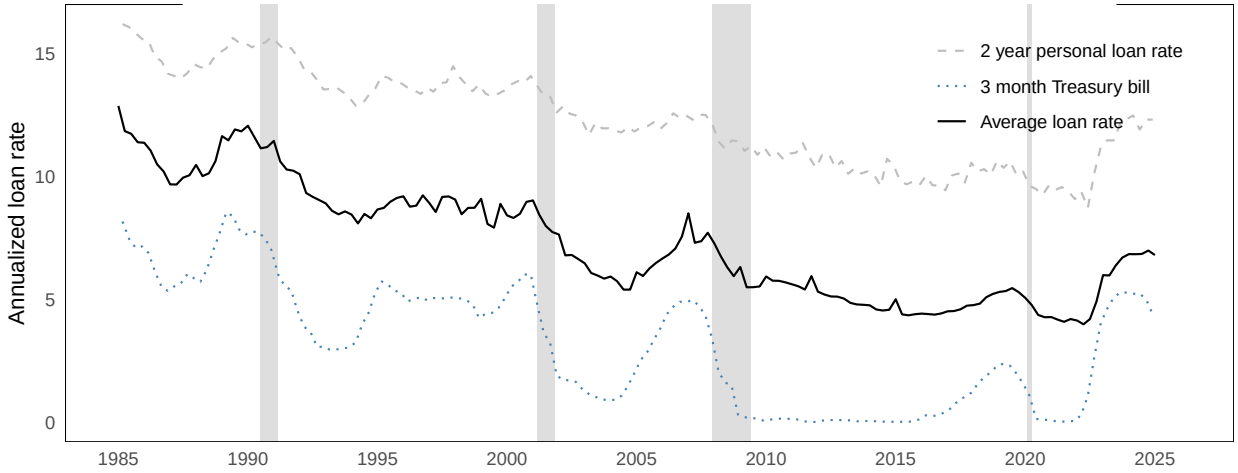
where the overline now marks the vector of bank-level log loans or appeal. Define the vector $\bar{l}_t^{-1} = [l_{1,t}^{-1}\ l_{2,t}^{-1}\ ...l_{n,t}^{-1}]'$ and post-multiple both sides by $\bar{l}_t^{-1}$ so that:

$$l_t \cdot l_t^{-1} = \left(\frac{\theta_t}{\theta_t - 1}\right)\overline{\log \psi_t} \cdot \bar{l}_t \cdot \bar{l}_t^{-1}$$

$$l_t \cdot \bar{l}_t^{-1} = \left(\frac{\theta_t}{\theta_t - 1}\right)\overline{\log \psi_t} \implies \left[\log \psi_{1,t}\ \log \psi_{2,t}\ ...\ \log \psi_{n,t}\right]' = \left(\frac{\theta_t - 1}{\theta_t}\right)\left[\frac{l_t}{l_{1,t}}\ \frac{l_t}{l_{2,t}}\ ...\ \frac{l_t}{l_{n,t}}\right]'$$

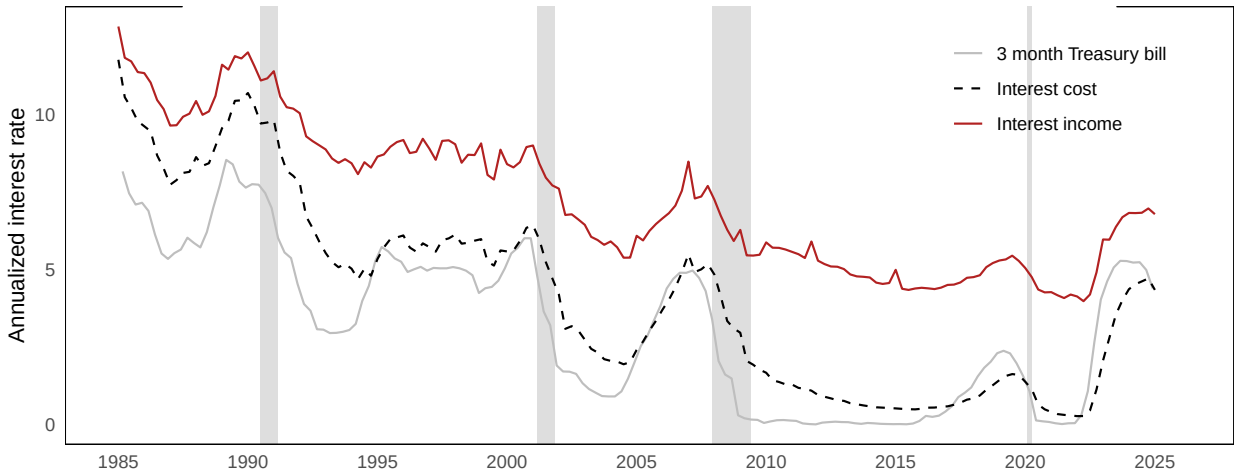
and the elasticity of substitution adjusted ratio of log lending reflects the bank-level preference weight and appeal. Note that with θ_t estimated separately following Section 2.3, calculating log appeal is a simple accounting exercise using observable loans quantities from the U.S. Call Reports.

Figure 12: Commercial bank loan rates



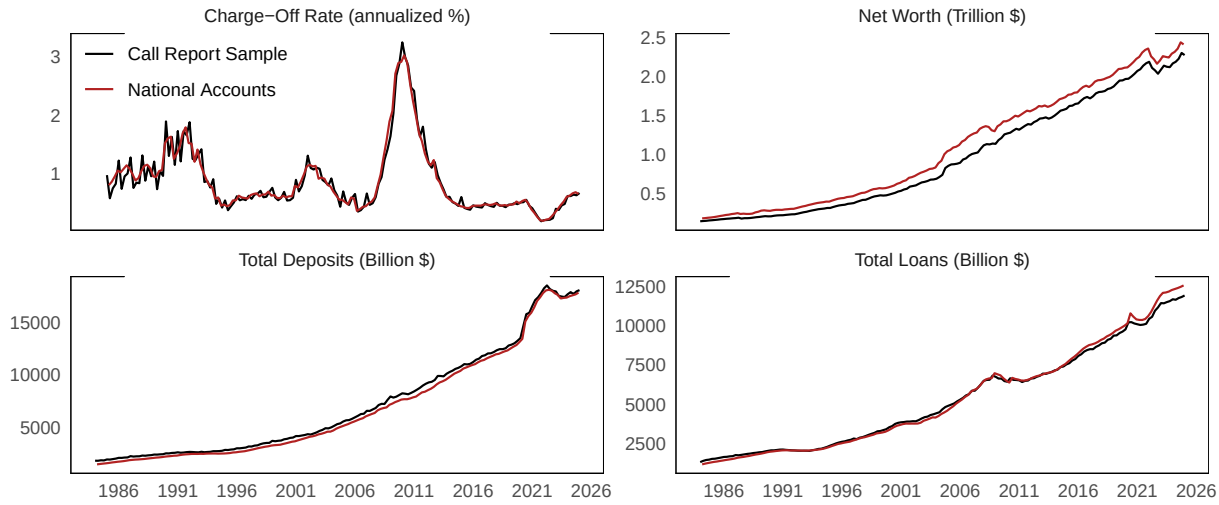
Notes: This plot compares the sample implied loan-weighted loan rate to published risk free and consumer loan rates. The bank-level sample loan rate is calculated as the reported quarterly loan income divided by total loans, scaled by the inverse one minus net charge-off ratio. The average loan rate is then the loan-weighted average of bank-level loan rates over the cross-section of banks in the given quarter. The risk-free rate is proxied by the US 3-month T-bill. The consumer loan rate is the Finance Rate on Personal Loans at Commercial Banks, 24 Month Loan, reported by the Federal Reserve Board's G.19 Consumer Credit. Gray shaded regions mark NBER dated recessions. Data are quarterly from 1984:Q1 through 2025:Q1.

Figure 13: Loan income and costs



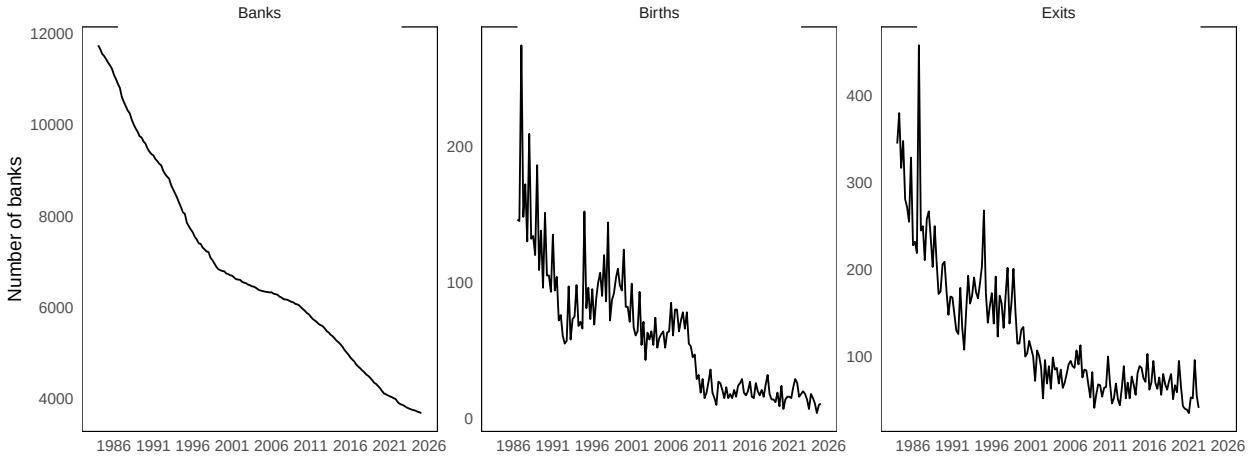
Notes: This plot presents loan interest income, costs, and the 3-month Treasury rate for comparison. Bank loan interest income and costs are loan-weighted. Gray shaded regions mark NBER dated recessions. Data are quarterly from 1985:Q1 through 2025:Q1.

Figure 14: Banking industry size and loan performance



Notes: These plots compare the charge-off ratio, total loans, and banking sector net worth in the sample to those reported by National Accounts. The sample covers the set of U.S. commercial banks that report FFIEC Call Reports. The National Account values are from the H.8 Assets and Liabilities of Commercial Banks in the United States, reported by the Federal Reserve Board. Charge-offs are net of recoveries. Data are quarterly from 1984:Q1 through 2025:Q1.

Figure 15: Banks, entry, and exit

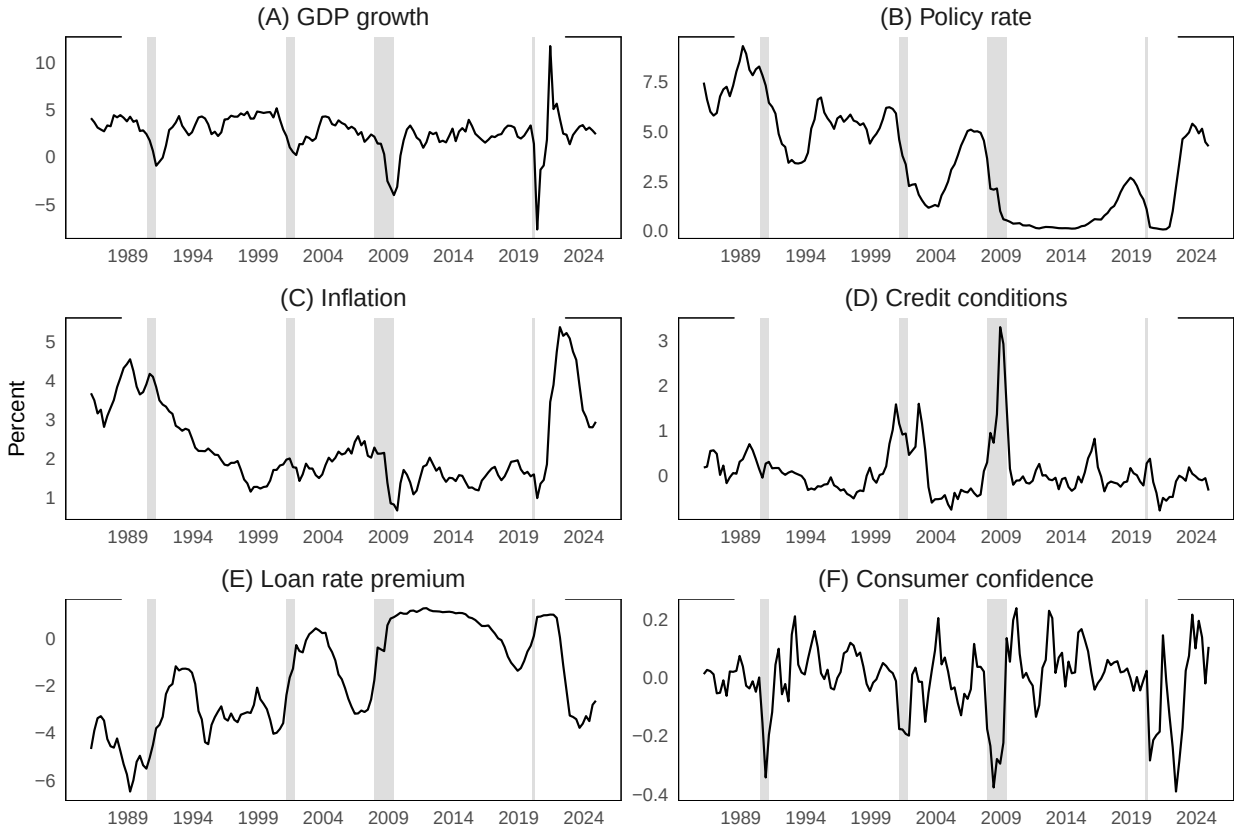


Notes: These plots present the number, entry, and exits of banks from the FFIEC Call Reports. A bank is born when its unique RSSD ID is observed for the first time in the sample, while a bank is considered to have exited when its RSSD ID has been observed for the last time. Bank exits are not necessarily failures, but can also represent acquisitions and mergers. Data are quarterly from 1984:Q1 through 2025:Q1.

A.2 Macroeconomic data

The endogenous variables used to represent the economy in the baseline (parsimonious) BVAR used in Section 4 and the medium-size BVAR used in Section 5 are standard. The endogenous variables used in the parsimonious BVAR are presented in Figure 16 and listed in Table 8. The endogenous variables used in the medium-size BVAR are listed in Tables 9, 10, and 11, as well as their data sources, stationary transformations, and historical availability. Monthly data are made quarterly by taking the quarterly average value.

Figure 16: Baseline BVAR endogenous variables



Notes: This plot presents the endogenous variables that define the parsimonious BVAR used in Section 4. Gray shaded regions mark NBER dated recessions. Data are quarterly from 1985:Q1 through 2024:Q4.

Table 8: Baseline BVAR endogenous variables

Variable	**Source**	**Transformation**
Loan rate	Call Reports	None
Consumer sentiment	FRED, *UMCSENT*	None
GDP	FRED, *GDPC1*	%YoY growth
Inflation (Core PCE)	FRED, *PCEPILFE*	%YoY growth
Policy rate	FRED, *DGS1*	None
Excess bond premium	Gilchrist and Zakrajšek 2012	None

Notes: This table lists the endogenous variables that define the baseline BVAR used in Section 4. Data sourced from FRED lists the mnemonics in italics. Loan rates are the loan-weighted average rates. The %YoY growth is calculated as $100 \cdot log(X_t/X_{t-4})$.

Table 9: Medium-size BVAR endogenous variables: banking sector

Variable	Source	Transformation	Start	End
Business loans	FRED, *TOTCI*	%YoY growth	1947:Q1	—
Real estate loans	FRED, *REALLN*	%YoY growth	1947:Q1	—
Consumer loans	FRED, *CLSACBW027SBOG*	%YoY growth	1973:Q1	—
30yr Mortgage rate	FRED, *MORTGAGE30US*	%YoY growth	1959:Q1	—
Consumer loan rate	FRED, *TERMCBPER24NS*	%YoY growth	1972:Q1	—
Business loan rate	FRED, *EEANQ*	%YoY growth	1997:Q2	2017:Q2
Total net worth	Call Reports	%YoY growth	—	—
Pct. banks tightening standards, large & mid-size firms	FRED, *DRTSCILM*	None	1990:Q2	—
Pct. banks tightening standards, small firms	FRED, *DRTSCIS*	None	1990:Q2	—
Pct. banks tightening loan covenants, large & mid-size firms	FRED, *SUBLPDCILTLNQ*	None	1990:Q2	—
Pct. banks tightening loan covenants, small firms	FRED, *SUBLPDCISTLNQ*	None	1990:Q2	—
Pct. banks increasing cost of credit lines, large & mid-size firms	FRED, *SUBLPDCILTCNQ*	None	1990:Q2	—
Pct. banks increasing cost of credit lines, small firms	FRED, *SUBLPDCISTCNQ*	None	1990:Q2	—

Notes: This table lists the endogenous variables that define the medium-size BVAR used in Section 5. Data sourced from FRED lists the mnemonics in italics. Loan rates are the loan-weighted average rates, while total loans and net worth are the simple sum of either object per period. The %YoY growth is calculated as $100 \cdot \log(X_t / X_{t-4})$. Data are quarterly from 1985:Q1 through 2023:Q3. "Start" and "End" report the full native date range of each FRED series; "—" in the End column denotes a series that is actively updated.

Table 10: Medium-size BVAR endogenous variables: non-bank lending financial markets

Variable	Source	Transformation	Start	End
Excess bond premium	Gilchrist and Zakrajšek 2012	None	1973:Q1	—
GZ spread	Gilchrist and Zakrajšek 2012	None	1973:Q1	—
Stock prices	Robert Shiller's website	None	1871:Q1	—
2-year Treasury rate	FRED, *DGS2*	None	1976:Q2	—
5-year Treasury rate	FRED, *DGS5*	None	1962:Q1	—
10-year Treasury rate	FRED, *DGS10*	None	1962:Q1	—
30-year Treasury rate	FRED, *DGS30*	None	1977:Q1	—

Notes: This table lists the endogenous variables that define the medium-size BVAR used in Section 5. Data sourced from FRED lists the mnemonics in italics. The %YoY growth is calculated as $100 \cdot \log(X_t/X_{t-4})$. Data are quarterly from 1985:Q1 through 2023:Q3. "Start" and "End" report the full native date range of each series; "—" in the End column denotes a series that is actively updated. Start and End dates for Gilchrist and Zakrajšek 2012 and Shiller series reflect the range of the underlying source data.

Table 11: Medium-size BVAR endogenous variables: activities, prices, and policy

Variable	Source	Transformation	Start	End
Policy rate	FRED, *DGS1*	None	1962:Q1	—
Consumer sentiment	FRED, *UMCSENT*	None	1952:Q4	—
Inflation (Core PCE)	FRED, *PCEPILFE*	%YoY growth	1960:Q1	—
Industrial production	FRED, *INDPRO*	%YoY growth	1919:Q1	—
Investment	FRED, *GPDIC1*	%YoY growth	1947:Q1	—
Government	FRED, *GCEC1*	%YoY growth	1947:Q1	—
Employment	FRED, *PAYEMS*	%YoY growth	1939:Q1	—
Durable consumption	FRED, *PCEDG*	%YoY growth	1959:Q1	—
Nondurable consumption	FRED, *PCEND*	%YoY growth	1959:Q1	—
Exports	FRED, *EXPGS*	%YoY growth	1947:Q1	—
Imports	FRED, *IMPGS*	%YoY growth	1947:Q1	—
Home prices	FRED, *USSTHPI*	%YoY growth	1975:Q1	—
Home starts	FRED, *HOUST*	%YoY growth	1959:Q1	—

Notes: This table lists the endogenous variables that define the medium-size BVAR used in Section 5. Data sourced from FRED lists the mnemonics in italics. The %YoY growth is calculated as $100 \cdot \log(X_t/X_{t-4})$. Data are quarterly from 1985:Q1 through 2023:Q3. "Start" and "End" report the full native date range of each FRED series; "—" in the End column denotes a series that is actively updated.

B Theoretical framework

This appendix defines the model equilibrium and market clearing conditions, shows how bank-level sentiment shocks may impact aggregate loan rate and lending outcomes, and finally proves the bank-specific loan rate equation.

B.1 Competitive Equilibrium

The loan market competitive equilibrium is characterized by the set of prices and allocations $\{R_{i,t}, C_{i,t}, L_{i,t}, N_{i,t}\}_{i\in\mathscr{B}}$, aggregate objects $\{R_t, L_t\}$, exogenous shocks and preferences $\{\psi_{i,t}, \varepsilon_{i,t}\}_{i\in\mathscr{B}}$, and stationary loan supply schedule such that:

- Borrowers demand loans $\{L_{i,t}\}_{i\in\mathscr{B}}$ according to Equation (1), taking $\{R_{i,t}, R_t, L_t\}$ as given
- Each bank i supplies $L_{i,t}$, given $N_{i,t}$, $C_{i,t}$, $\varepsilon_{i,t}$, and $\psi_{i,t}$, that solves its dynamic programming problem, Equation (2)
- The aggregate loan quantity satisfies $L_t = \sum_{i\in\mathscr{B}} L_{i,t}$
- The aggregate loan rate satisfies the CES price index

$$R_t = \left(\int_{i\in\mathscr{B}} \psi_{i,t}^{\theta_t} R_{i,t}^{1-\theta_t} di \right)^{\frac{1}{1-\theta_t}}$$

- Each bank-specific loan market clears, so that loan demand equals loan supply for all $i \in \mathscr{B}$
- Bank net worth evolves according to $N_{i,t+1} = N_{i,t} + \Pi_{i,t}$

While I refer to this as a competitive equilibrium for expositional simplicity, the loan market is characterized by Cournot competition among banks, implying that individual institutions have market power and internalize the effect of their lending decisions on equilibrium prices. As such, the equilibrium is more precisely interpreted as a Markov perfect equilibrium in a dynamic oligopolistic lending game, where each bank's policy functions are optimal given the state of the economy and the behavior of other banks. I retain the competitive equilibrium terminology to maintain consistency with the macroeconomic literature and to emphasize the tractability of the environment.

B.2 Bank sentiment shocks and aggregate outcomes

Bank-level sentiment shocks will increase aggregate loan rates. This conclusion follows simply from the CES preferences of the representative borrower implied effective loan rate:

$$R_t = \left(\int_{i \in \mathscr{B}} \psi_{i,t}^{\theta} R_{i,t}^{1-\theta} d_i \right)^{\frac{1}{1-\theta}}$$

which can be log-linearized to:

$$r_t = \sum_{i \in \mathscr{B}} \iota_{i,t} \, r_{i,t} \; + \; \theta \sum_{i \in \mathscr{B}} \iota_{i,t} \log \psi_{i,t} \qquad \iota_{i,t} = \frac{\psi_{i,t}^{\theta} R_{i,t}^{1-\theta}}{\sum_{j \in \mathscr{B}} \psi_{j,t}^{\theta} R_{j,t}^{1-\theta}}$$

and substituting in the log-linearized bank-level loan rate, Equation 8, yields the expanded form:

$$r_t = \sum_{i \in \mathscr{B}} \iota_{i,t} \Bigg[\log\left(\frac{1}{\beta_i}\right) + \log\left(\frac{\theta_t}{\theta_t - 1}\right) + s_{i,t} + c_{i,t} + \Gamma'(L_{i,t}/N_{i,t}) - \; ... \\ ... \; - \log \psi_{i,t} + \gamma_i + \rho_1 \lambda_{i,t-1} + \rho_2 \lambda_{t-1} + \varepsilon_{i,t} \Bigg] + \theta \sum_{i \in \mathscr{B}} \iota_{i,t} \log \psi_{i,t}$$

which reveals the derivative of aggregate log loan rates with respect to a sentiment shock to be positive.

Result 1 (Bank risk sentiment and the aggregate loan rate)
The aggregate loan rate is a weighted average of the bank-specific loan rates. Thus, a granular increase in a single bank's risk sentiments will increase the aggregate loan rate of the economy.

Bank-level sentiment shocks will decrease aggregate lending. The CES preferences of the representative borrower, yield the aggregate composite loan:

$$L_t = \left(\int_{i \in \mathscr{B}} \psi_{i,t} L_{i,t}^{\frac{\theta_t - 1}{\theta_t}} d_i \right)^{\frac{\theta_t}{\theta_t - 1}}$$

where $\psi_{i,t}$ is bank i appeal in time t. Thus, aggregate lending is a strictly increasing function of bank-specific lending. Simultaneously, bank-specific loan demand is:

$$L_{i,t} = \psi_{i,t} \left(\frac{R_t}{R_{i,t}} \right)^{\theta_t} L_t$$

which is a strictly decreasing function in aggregate loan rates. Therefore, by Result 1, a pessimistic

bank-level sentiment shock will decrease bank-specific and aggregate lending.

Result 2. (Bank risk sentiment and the aggregate loan supply)
An increase in bank-level sentiments will decrease the aggregate supply of loans in the economy.

B.3 Loan rate proof

Theorem 1 *A bank charges the loan portfolio average interest rate:*

$$R_{i,t} = \frac{1}{\beta_i}\frac{\theta_t}{\theta_t - 1}\frac{C_{i,t} + \Gamma'\left(\sigma^R_{i,t}\frac{L_{i,t}}{N_{i,t}}\right)}{1 - \mathbb{E}_t\lambda_{i,t}}\frac{1}{1 - s_{i,t}}$$

Proof. The bank's problem in series form follows:

$$\max_{\{R_{i,t}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty}\beta_i^t\Pi_{i,t} \quad \text{s.t.}$$

$$\Pi_{i,t} = R^*_{i,t}L_{i,t} - (L_{i,t} - N_{i,t})C_{i,t} - \Gamma\left(\sigma^R_{i,t}\frac{L_{i,t}}{N_{i,t}}\right) \qquad \text{(Profit function)}$$

$$R^*_{i,t} = (1 - \lambda_{i,t})R_{i,t} \qquad \text{(Realized gross return)}$$

$$N_{i,t} = N_{i,t-1} + \Pi_{i,t-1} \qquad \text{(Net worth LoM)}$$

After using the bank-specific demand curve $L_{i,t} = \psi_{i,t}\left(\frac{R_{i,t}}{R_t}\right)^{\theta_t}L_t$ to replace the loan rate $R_{i,t}$, the Lagrangian is:

$$\mathscr{L} = \sum_{t=0}^{\infty}\beta^t\left[(1 - \mathbb{E}_t\lambda_{i,t+1})R_{i,t}L_{i,t} - (L_{i,t} - N_{i,t})C_{i,t} - \Gamma\left(\sigma^R_{i,t}\frac{L_{i,t}}{N_{i,t}}\right)\right] \quad \text{s.t.}$$

$$N_{i,t} = N_{i,t-1} + \Pi_{i,t-1} \quad \text{with } N_{i,0} \text{ given}$$

Then, assuming the regulatory cost term takes a linear functional form: $\Gamma(\sigma^R_{i,t}L_{i,t}/N_{i,t}) = \zeta_t(\sigma^R_{i,t}L_{i,t}/N_{i,t})$ where $\zeta_t > 0$, the first order condition (FOC) is

$$\frac{\partial\mathscr{L}}{\partial L_{i,t}} = \beta_i L_i^{1/\theta_t}(1 - \mathbb{E}_t\lambda_i)\psi_i R^{1-1/\theta_t}L_i^{-1/\theta_t}\left(\frac{\theta_t - 1}{\theta_t}\right) - C_{i,t} - \zeta/N_i + \ldots$$
$$\ldots + \beta_i\left(\frac{L_t}{R_t}\right)^{1/\theta_t}\frac{\partial R}{\partial L_i}R_t^{1-1/\theta_t}\psi_i(1 - \mathbb{E}_t\lambda_i)\left(\frac{\theta_t - 1}{\theta_t}\right) - \left[\sum_{j=1}^{\infty}\beta_i^j o_{i,j}\right] - \left[\sum_{j=1}^{\infty}\beta_i^j\tau_{i,j}\right] = 0$$

where $o_{i,j}$ is the marginal increase in net worth due to relaxing regulatory costs in period j from the marginal increase in retained earnings in period t, and $\tau_{i,j}$ is the marginal increase in net worth due to decreasing capital funding costs in period j from the marginal increase in retained earnings in period t. Note that the condition

$$[\sum_{j=1}^{\infty} \beta_i^j o_{i,j}] = [\sum_{j=1}^{\infty} \beta_i^j \tau_{i,j}] = 0$$

must be true in the absence additional constraints (assuming a non-zero term implies a profit improving, thus net worth increasing, loan rate exists; formalizing a proof by contradiction using this logic is straightforward). Thus the FOC becomes a static equation:

$$\frac{\partial \mathscr{L}}{\partial L_i} = \beta_i L_i^{1/\theta}(1-\mathbb{E}_t\lambda_i)\psi_i R^{1-1/\theta} L_i^{-1/\theta}\left(\frac{\theta-1}{\theta}\right) - \ ...$$
$$... \ -C_i - \zeta\sigma_i^R/N_i + \beta_i\left(\frac{L}{R}\right)^{1/\theta}\frac{\partial R}{\partial L_i}R^{1-1/\theta}\psi_i(1-\mathbb{E}_t\lambda_i)\left(\frac{\theta-1}{\theta}\right) = 0$$

where I have dropped the t subscripts.

Solving the FOC for $R_{i,t}$ is then a simple matter of algebra:

$$\frac{\partial \mathscr{L}}{\partial L_i} = 0 \implies \frac{1}{\beta_i}\frac{\theta}{\theta-1}\frac{C_i+\zeta\sigma_i^R/N_i}{\psi_i(1-\mathbb{E}_t\lambda_i)} = \frac{L_t}{L_{i,t}}^{1/\theta} R^{1-1/\theta} + L^{1/\theta}R^{-1/\theta}\frac{\partial R}{\partial L_i}L_i^{1-1/\theta}$$

define $\chi_i = \frac{1}{\beta_i}\frac{\theta}{\theta-1}\frac{C_i+\zeta\sigma_i^R/N_i}{\psi_i(1-\mathbb{E}_t\lambda_i)}$ for convenience, then

$$\begin{aligned}\chi_i R^{1/\theta} &= L^{1/\theta}L_i^{-1/\theta}R_t + L_t^{1/\theta}\frac{\partial R}{\partial L_i}L_i L_i^{-1/\theta} \\ &= s_i^{-1/\theta} + s_i^{-1/\theta}\frac{L_i}{R}\frac{\partial R}{\partial L_i}\end{aligned}$$

where $s_i = L_i/L$ is the bank's loan market share, and using Shepard's Lemma, $-\frac{\partial R}{\partial L_i} = s_i$ we have:

$$\chi_i R^{-1/\theta} = s_i^{-1/\theta} - s_i s_i^{-1/\theta} = s_i^{-1/\theta}(1-s_i)$$

$$\implies \frac{1}{\beta_i}\frac{\theta}{\theta-1}\frac{C_i+\zeta\sigma_i^R/N_i}{\psi_i(1-\mathbb{E}_t\lambda_i)}\frac{1}{1-s_i} = \left(\frac{L}{L_t}\right)^{1/\theta} R^{1/\theta-1}$$

$$\implies \frac{1}{\beta_i}\frac{\theta}{\theta-1}\frac{C_i+-\zeta\sigma_i^R/N_i}{(1-\mathbb{E}_t\lambda_i)}\frac{1}{1-s_i} = \psi_i L^{1/\theta}L_i^{-1/\theta}R^{1/\theta-1} = L_i R_i$$

so the loan rate for a one unit loan is:

$$\frac{1}{\beta_i}\frac{\theta}{\theta-1}\frac{C_i+\Gamma'(\sigma_{i,t}^{R}\frac{L_i}{N_i})}{(1-\mathbb{E}_t\lambda_i)}\frac{1}{1-s_i}=R_i$$

where $\Gamma'(\frac{L_i}{N_i})$ is the marginal regulatory cost with respect to bank lending $L_{i,t}$. ■

C Measurement

C.1 Macro-approach and portfolio composition robustness

I will complement my baseline estimation of bank risk sentiment with an additional measurement strategy, a "macro-approach" that requires no additional assumptions beyond the theoretical model already introduced, but is restricted to only identifying sector-wide sentiments. The macro-approach will then be used to test the robustness of the bank sentiment measure against different postulated laws of motion for loan losses, as well as the effects of loan portfolio composition effects, namely loan maturity and borrower quality.

C.1.1 Macro-approach measurement strategy

A bank's risk sentiment is the bank's deviation from its rational expectations forecast of loan defaults. That is, taking $E_{RE}(\lambda_{i,t})$ as the rational expectations forecasting function of loan losses, sentiments can then be expressed as a wedge $\varepsilon_{i,t}$ between the bank's actual forecast of the default rate and the rational expectations forecast of the default rate:

$$\mathbb{E}\lambda_{i,t} = E_{RE}(\lambda_{i,t}) + \varepsilon_{i,t}$$

It then follows that sentiments directly impact loan rates through the forecasted losses, thus demanded compensation for risk:

$$R_{i,t} = \frac{1}{\beta_i} \cdot \frac{\theta_t}{\theta_t - 1} \frac{1}{1 - s_{i,t}} \cdot \frac{1}{1 - f(I_{i,t}) - \varepsilon_{i,t}} \cdot (C_{i,t} + \Gamma'(L_{i,t}/N_{i,t})) \tag{15}$$

so that positive sentiments (a belief that defaults will be higher than the rational expectations forecast would suggest) increase loan rates and decrease loan issuance.

While an agent's rational expectations forecast is unobservable, sentiments may be directly inferred from the bank's interest income. Given the structure of the loan market and resulting loan rates,

the log realized return is

$$R^*_{i,t} = \frac{1-\lambda_{i,t}}{1-\mathbb{E}\lambda_{i,t}} \cdot \frac{1}{\beta_i} \cdot \frac{\theta_t}{\theta_t - 1} \frac{1}{1-s_{i,t}} \cdot (C_{i,t} + \Gamma'(L_{i,t}/N_{i,t}))$$

$$= \frac{1-f(I_{i,t})-\nu_{i,t}}{1-f(I_{i,t})-\varepsilon_{i,t}} \cdot \frac{1}{\beta_i} \cdot \frac{\theta_t}{\theta_t - 1} \frac{1}{1-s_{i,t}} \cdot (C_{i,t} + \Gamma'(L_{i,t}/N_{i,t})) \implies$$

$$r^*_{i,t} = \varepsilon_{i,t} - \nu_{i,t} + \log\left(\frac{1}{\beta_i}\right) + \log\left(\frac{\theta_t}{\theta_t - 1}\right) + s_{i,t} + c_{i,t} + \Gamma'(L_{i,t}/N_{i,t})$$

for small enough default rates, marginal capital costs, and market shares. Note that the realized default rate $\lambda_{i,t}$ is rewritten as the rational expectations forecast and forecast error $\nu_{i,t}$. The rational expectations forecast ultimately falls out of the net return equation because the bank is compensated for the predictable risk, but bank's sentiment and forecast error remain.

By the properties of rational expectations with full information, we have that

$$\lambda_{i,t} = f(I_{i,t}) + \nu_{i,t}, \qquad \nu \sim \mathcal{N}(0, \Sigma)$$

for all banks, where $\nu_{i,t}$ is the bank-specific forecast error. Thus, the cross-sectional average forecast error is zero, and the sector-wide average realized return is:

$$\mathbb{E}r^*_{i,t} = \mathbb{E}\left[\varepsilon_{i,t} + \log\left(\frac{1}{\beta}\right) + \log\left(\frac{\theta_t}{\theta_t - 1}\right) + s_{i,t} + c_{i,t} + \Gamma'(L_{i,t}/N_{i,t})\right] \tag{16}$$

where $\mathbb{E}\nu_{i,t} = 0$ and consequentially falls out of the equation. Therefore, with observable proxies for the bank's markups, market share, marginal capital cost, and marginal regulatory costs, the log-linearized realized return equation can be estimated with standard econometric techniques, and the banking sector average sentiment may in turn be inferred from the cross-sectional average residual realized return.

C.1.2 Postulated law of motion for loan losses robustness test

The strength of the "macro-approach" is that it does not require an assumed law of motion for bank-level loan losses. In fact, the approach does not require any information on banks' rational expectations forecast of losses, which is advantageous given the unobservable nature of these forecasts. Since one is not required to postulate a law of motion for bank-level loan losses, the macro-approach is a useful robustness check on potential bias in the aggregate sentiment series introduced by a possibly misspecified law of motion.

However, there are two key limitations to the macro-approach. First, the implicit assumption that there is no aggregate risk. If this identifying assumption is violated, then the cross-sectional average $v_{i,t}$ is not necessarily zero, and one cannot separate BRS from aggregate shocks increasing default rates among borrowers. Second, the macro-approach does not allow for identifying bank-level sentiments.

C.1.3 Portfolio composition robustness tests

I will further use the macro approach to test the robustness of the aggregate bank risk sentiment to two portfolio composition considerations.

First, I will use the macro-approach to additionally test whether the composition of the loan portfolio influences the measure of aggregate bank risk sentiments. If it does, then the empirical measure does not successfully isolate animal spirit shocks. I first test whether bank risk sentiment may be capturing underlying features of the loan portfolio composition by loan type, for example, changes in the effective portfolio term premia. To do so, I include measures for the relative loan share of commercial, consumer, and real estate loans in the estimated log loan return equation that identifies bank sentiment in the macro-approach. These three types of loans vary in their average maturity, with real estate loans having the longest maturity on average. Thus, if BRS is capturing changes in term premia, then one would expect that an increase in the real estate share of the loan portfolio would be positively correlated with bank sentiment.

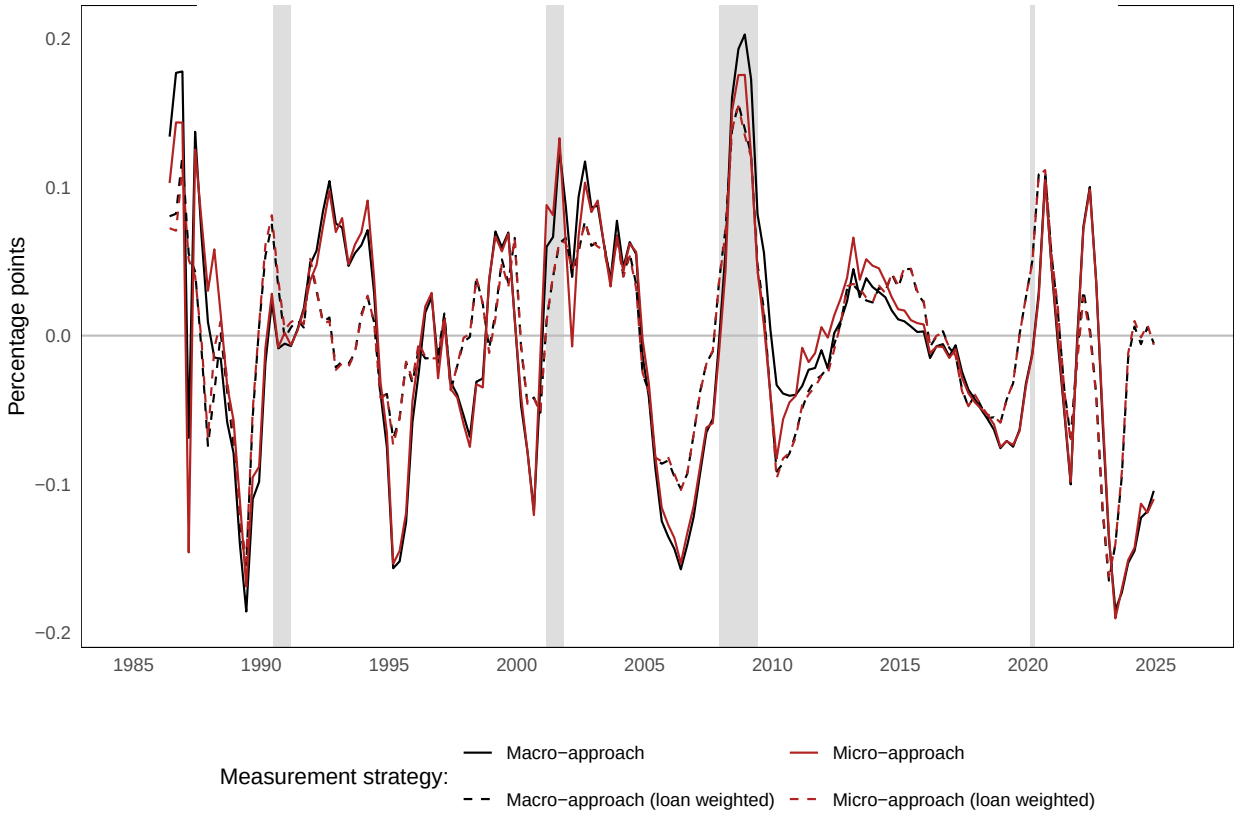
Second, I additionally test whether underlying portfolio borrower risk is influencing the bank risk sentiment. If so, then the BRS contains information about the riskiness of the portfolio, thus should be included in the rational expectations forecast of risk and cannot be reasonably described as noisy animal spirits. To do so, I include a measure of changes in the relative riskiness of borrowers in banks' loan portfolios. The relative risk proxy is the difference in the percent of banks responding to the Fed's Senior Loan Officer Opinion Survey (SLOOS) that they have observed an increase in demand for loans from small and medium size businesses and the percent of banks responding to the same survey that they have observed an increase in demand for loans from large businesses. This measure is meant to capture the changing allocation in the loan portfolio from safer large businesses to riskier small and medium-size businesses.

Table 12: Bank risk sentiment measurement equations

	Micro-approach (estimating $r_{i,t}$)		Macro-approach (estimating $r^*_{i,t}$)		
	(1)	(2)	(3)	(4)	(5)
capital cost: $c_{i,t}$	0.214^{***}	0.513^{***}	0.209^{***}	0.510^{***}	0.540^{***}
	(0.014)	(0.019)	(0.014)	(0.019)	(0.025)
regulatory cost: $\Gamma'_{i,t}$	-0.052^{***}	0.078^{***}	-0.040^{***}	0.095^{***}	0.029
	(0.009)	(0.030)	(0.010)	(0.033)	(0.027)
markups: $\log(\frac{\sigma_t}{\sigma_t-1}\frac{1}{1-s_{i,t}})$	-0.100^{***}	-0.014	-0.081^{**}	-0.015	-0.012
	(0.031)	(0.011)	(0.041)	(0.012)	(0.015)
idio. defaults: $\lambda_{i,t-1}$	-0.005^{***}	0.076			
	(0.002)	(0.054)			
agg. defaults: λ_{t-1}	-0.290^{***}	-0.121			
	(0.045)	(0.076)			
Relative risk					-0.000^{**}
					(0.000)
Consumer loan share					0.015
					(0.015)
Real estate loan share					-0.007
					(0.013)
Commercial loan share					-0.002
					(0.016)
Loan share weighted		✓		✓	✓
Time trend	✓	✓	✓	✓	✓
Bank FE	✓	✓	✓	✓	✓
Observations	781,738	781,738	781,738	781,738	581,433
Total R^2	0.898	0.913	0.895	0.912	0.880

Notes: This table presents the sentiment measurement equations. Columns (1) and (3) are unweighted; Columns (2), (4), and (5) are quarter-loan-share weighted. All models include a linear time trend and bank fixed effects. All models are estimated as within-group fixed effect linear panel models. The quarterly, unbalanced panel includes banks from 1986:Q1 through 2024:Q4. Parentheses report robust standard errors clustered at the bank level. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Figure 17: Bank risk sentiment measured by micro-approaches and macro-approaches



Notes: This plot shows the quarterly aggregate bank risk sentiment of U.S. commercial banks. The black lines present bank sentiment measured by the macro-approach. The red lines present bank sentiment measured by the micro-approach. Solid lines present the unweighted version of either measure, while the dashed lines present the loan-weighted version of either measure. Gray shaded regions mark NBER dated recessions. Data is from 1986:Q1 through 2024:Q4.

C.1.4 Measurement equations and robustness results

Following the measurement strategies laid out above, I estimate aggregate bank risk sentiments with the macro-approach. Table 12, columns (3) and (4), report the estimated loan pricing equations, estimated with and without loan-weighted observations. The macro-approach estimation accounts for a large percentage of bank-level realized income variation, which is quantitatively similar to the micro-approach (reproduced in Table 12 columns (1) and (2) for comparison). This is perhaps a reflection of the similarities in estimated pricing factor loadings across the two measurement equations. For example, the capital costs coefficients are almost identical, and the regulatory costs are only separated by approximately one basis point.

Postulated law of motion for losses. I also calculate aggregate bank risk sentiment —the quarter-by-quarter cross-sectional average residual realized income, which following Equation 16 is identifiable as the aggregate bank risk sentiment. Figure 17 shows a high degree of similarity between the aggregate bank risk sentiment estimated by micro- or macro-approach, whether weighted or unweighted by bank loans. The similarity across the two approaches is affirmative evidence that the postulated law of motion used in the micro-approach does not alter the measure of aggregate sentiments, thus does not pose a risk to the estimation of the macroeconomic importance and impacts of aggregate bank sentiment shocks.

Bank size. Lastly, bank size is not a source of heterogeneity in bank risk sentiments. This can be inferred from the qualitatively similar dynamics of the loan-weighted and unweighted measures of aggregate BRS presented in Figure 17. The loan-weighted measurement equation, thus the loan-weighted BRS, is dominated by the dynamics of the largest banks, while the unweighted measurement equation is dominated by the dynamics of small banks.[26] Therefore, the similarity across the two indexes reflects a similarity across the dynamics of small and large bank sentiments.[27]

Portfolio composition tests. First, bank sentiment is not systematically related to underlying portfolio risk. Table 12 column (5) shows that relative risk enters the measurement equation with a coefficient that is statistically significant at conventional levels, but economically negligible: a one percentage point increase in the relative demand for large versus small firm loans changes the predicted loan rate by less than one basis point. Thus, while precisely estimated, the effect is quantitatively immaterial for the measurement of bank sentiment.

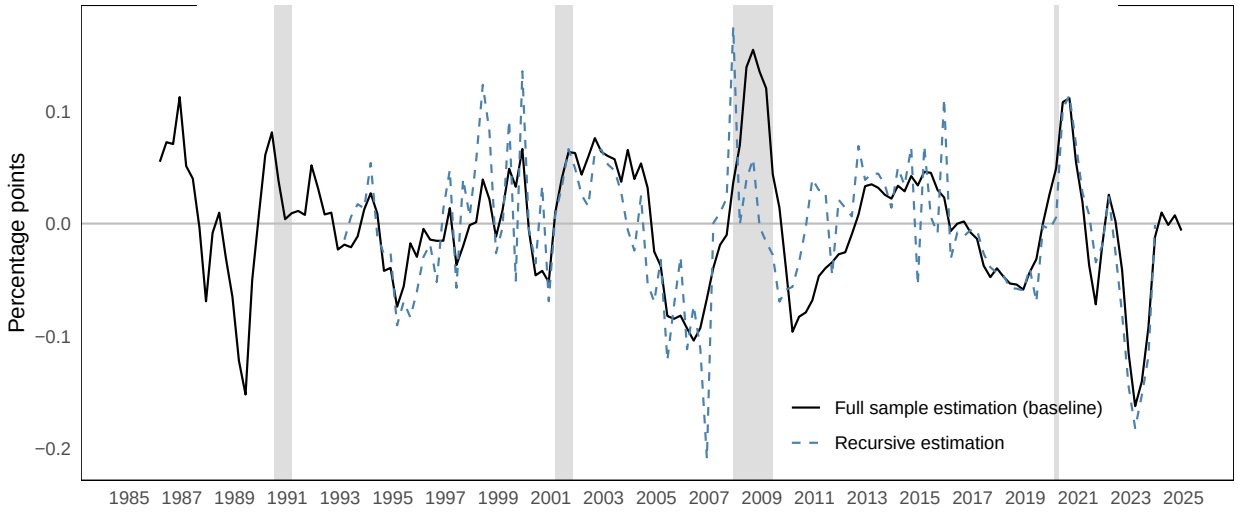
Second, bank sentiment is not meaningfully correlated with term premia or portfolio composition. Column (5) further shows that the shares of consumer, real estate, and commercial loans are neither statistically nor economically significant determinants of loan pricing. In turn, variation in the maturity or composition of the banking sector's loan portfolio does not load into the measurement equation.

Taken together, these results imply that the empirical measure of aggregate bank risk sentiment

[26] Loans are highly concentrated among a few large banks in the U.S. (Corbae and D'Erasmo 2021 show the top ten bank holding companies account for approximately 60 percent of bank assets), although there are several thousand banks actually operating in the United States (see Figure 15 in the Data Appendix A.1).

[27] A more direct test might be to compare sentiments among the largest banks to the aggregate BRS series, however, since there are fewer than 20 large banks that dominate U.S. lending markets, this instead would be a study of granular sentiment shocks to the largest banks, rather than the systematic influence of bank size on sentiment formations.

Figure 18: Recursive estimation of aggregate bank risk sentiment



Notes: This plot shows the quarterly aggregate bank risk sentiment of U.S. commercial banks, estimated with the full sample or with an expanding window starting in 1990:Q1. The solid black line presents the loan-weighted sentiment index and the dashed blue line presents aggregate sentiments estimated identically but with an expanding window information set starting in 1990:Q1. Gray shaded regions mark NBER dated recessions. Data is from 1986:Q1 through 2024:Q4.

does not reflect rational variation in either the underlying riskiness of borrowers or the composition of bank balance sheets. Rather, the residual variation captured by the measurement equation is orthogonal to these observable fundamentals, supporting its interpretation as a measure of sentiment rather than risk.

C.2 Recursive information set and historical robustness

Lastly, and related to the previous robustness check regarding the postulated law of motion for loan losses, I test how sensitive bank sentiments are to being estimated with the full sample. In particular, there may be a concern that estimating bank-level sentiments with the full sample will misrepresent the banks' rational expectations forecasts of loan default rates because it is contaminating their information sets with future data —this is a common problem in evaluating forecast performance without "real-time" information sets. To test this concern, I estimate the bank-level sentiments model one quarter at a time with an expanding information set, that is, attempting to preserve the pseudo-real-time forecasting information structure of the banks

I find there is a high degree of similarity in aggregate BRS between using the full sample to estimate

all bank-level risk sentiments with one estimated loan rate equation versus estimating bank-level sentiments one quarter at a time with an expanding window information set. Figure 18 shows similar qualitative dynamics of aggregate BRS estimated with the full sample compared to aggregate BRS estimated one quarter at a time. Additionally, the correlation coefficient between the two series is high, 0.694, and both series have a historical mean near zero.

The one key difference between the two series is sentiment's behavior during the GFC. The recursively estimated sentiment series is significantly more optimistic at the beginning of 2007 and then reaches a greater peak pessimism at the start of the crisis, compared to the baseline sentiment measure. The recursively estimated series is also quicker to reverse from optimism to pessimism at the start of the crisis, but then more quickly recovers as well. The two series only realign in 2012.

This exercise additionally shows that estimated bank-level sentiment is qualitatively robust to removing influential periods, such as the GFC and COVID-19 pandemic.

D Loan-level analysis

I estimate the causal effect of a change in a bank's risk sentiment on loan-level outcomes through a fixed effect panel model in the spirit of Khwaja and Mian 2008. The formal specification follows:

$$y_{l,f,i,t} = \gamma_{f,t} + \gamma_{f,i} + \delta \chi_{i,t} \varepsilon_{i,t} + \beta \Theta_t + \eta_{l,f,i,t} \tag{17}$$

where $y_{l,f,i,t}$ is the loan rate for loan l, firm f (i.e. borrower), bank i, and date t; $\gamma_{f,t}$ denotes a firm-quarter fixed effect, and $\gamma_{f,i}$ a borrower-lender fixed effect; $\varepsilon_{i,t}$ is the bank risk sentiment of bank i at date t; Θ_t collects the vector of loan and firm characteristics. My coefficient of interest when evaluating Equation 17 is δ, the response of loan outcome $y_{l,f,i,t}$ to a one percentage point change in a bank's risk sentiment.

Identification. I isolate the *within-firm* variation in loan outcomes attributable to variation in lenders' risk sentiment, and use this variation to estimate the *causal treatment effect* of bank-level risk sentiment shocks. To do so, I first narrow the sample of loans to those held by firms borrowing from multiple-syndicates in a given period, and in turn purge credit demand and other firm-specific factors with firm-quarter fixed effects. I then additionally control for individual borrower-lender relationships to ameliorate concerns of non-random matching in lending market, as well as loan specific characteristics, such as the presence of collateral or covenants, which may impact how lenders value loans after reassessments of risk.[28] These three steps isolate variation in outcomes attributable to lender specific factors (i.e. attributable to shifts in the supply of credit). Therefore, since all confounding sources of variation have been removed, the remaining variance explained by bank-level risk sentiment shocks can be interpreted as the causal response to structural shocks.

I do not incorporate additional controls for non-sentiment bank-specific factors that the literature typically includes to isolate the effect of credit supply shocks. This is because, by construction, bank-level risk sentiment shocks are orthogonal to these bank-specific controls. For example, BRS is orthogonal to the size of banks' balance sheets, profitability, and other variables utilized in works such as Di Giovanni et al. 2022. Therefore, adding further bank-level covariates is unnecessary to isolate variation due to bank-level risk sentiment shocks, if not detrimental in obtaining a precise measurement of the elasticity of interest.

Data. The loans studied in this analysis are in fact individual facilities, also known as tranches, of

[28] See Chodorow-Reich 2014 for a discussion of the stickiness of borrower-lender relationships and why they should be explicitly controlled for in the Khwaja-Mian research design.

Table 13: Summary statistics of matched bank-loan data

	Mean	SD	p(5)	p(25)	p(50)	p(75)	p(95)
Loan characteristics							
Loan amount	1921.2	2837.6	154.2	485.0	1100.0	2250.0	6066.1
Loan rate	245.6	133.6	67.5	160.0	225.0	300.0	475.0
Covenants present	64.4%						
Secured by collateral	79.9%						
Bank characteristics							
BRS	0.036	0.227	-0.351	-0.073	0.053	0.157	0.364
Bank net worth	1.699	3.060	0.002	0.073	0.616	1.948	5.687

Notes: This table reports the summary statistics for data used in estimating the loan-level impact of a change in BRS. The loan amount is reported in millions USD, and bank net worth in billions USD. The loan rate is the margin over reference (e.g. LIBOR), quoted in basis points. Covenant present and Secured by collateral are binary indicators. Loan characteristics are from DealScan. Bank characteristics are from U.S. Call Reports and author calculations. Dates range from 2000:Q1 through 2024:Q2.

syndicated loans from the LPC DealScan database.[29] The data covers nearly the universe of syndicated loans, which is in turn associated with borrowers (firms) that make up a majority of employment and production in the United States.[30] However, as I am interested in studying the impact of bank-level BRS on loan outcomes, I must narrow my study to tranches funded by lenders that can be matched to the U.S. Call Report records used to create my measure of bank-level BRS.[31] The matched bank-loan data set ultimately includes 180.5 thousand facility observations, ranging from 2000:Q1 through 2024:Q2, representing 287 unique lenders (banks) and 3569 borrowers (firms). However, my estimation sample will be a subsample of this matched-data set, with observations only being included if all model variables being present and the multi-lender multi-borrower requirements are met.

The DealScan data contains two types of observations, loan originations and loan refinancing agreements. I will focus on borrowers renegotiating the terms of a loan held on a bank's bal-

[29] A *syndicated loan* is a large or niche loan that requires a consortium, or syndicate, of lenders to fulfill. The loan can be broken up into discrete pieces, referred to as *tranches*. For all intents and purposes, tranches can act as independent, smaller, loans, with their own interest rates, payment schedules, covenants, and seniority.

[30] It should be noted that DealScan does not cover small business and household lending. One should consult Caglio et al. 2021 for a more comprehensive discussion of the limitations of DealScan's loan coverage.

[31] Lenders associated with a DealScan tranche are matched with FFIEC regulated banks by name and state. An additional fuzzy matching is attempted on remaining DealScan lenders, utilizing the routine put forth by Cohen et al. 2021, but no additional matches are made.

Table 14: Loan-level rate spread response to a bank sentiment shock

	(1)	(2)	(3)	(4)
Bank sentiment: $\varepsilon_{i,t-1}$	0.139	0.136	0.142	0.192***
	(0.158)	(0.159)	(0.159)	(0.066)
GFC only				✓
Bank characteristics			✓	✓
Loan characteristics		✓	✓	✓
Borrower-Quarter FE	✓	✓	✓	✓
Lender-Borrower FE	✓	✓	✓	✓
Observations	12,898	12,898	12,898	2,972
R^2	0.884	0.885	0.885	0.927
Adjusted R^2	0.804	0.804	0.804	0.880

Notes: This table reports the causal effects of a one percentage point increase in bank-level pessimism on bank-level loan rate margins (in percentage points relative to a reference rate such as LIBOR) offered in syndicated loans. Loan characteristics include an indicator for whether the loan is secured and an indicator for the presence of covenants. Bank characteristics include lender net worth. Column (4) restricts the sample to the Global Financial Crisis period (2007 through 2011). The loan rate has been winsorized at the 1st and 99th percentiles. Observations are weighted by bank size. Parentheses report robust standard errors, which are double clustered at bank and date levels. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

ance sheet to ensure that the loan is actually held by the bank. A majority of DealScan loans are originated by commercial banks, but in turn sold to non-bank lenders. Therefore, studying how changes in BRS impacts all syndicated loans would include studying how a bank's sentiment impacts loans that it will almost immediately sell off of its balance sheet. In the context of the analytical model used to identify BRS, there should be no relationship between a bank's risk sentiment and loans not held on its balance sheet, even if it is the entity that originates the loan. For a discussion of who participates in the syndicated loan market and the origination to distribution pipeline, see Fleckenstein et al. 2020, Buchak et al. 2024a, or Buchak et al. 2024b.

The sample covers a large range of loan, borrower, and lender sizes. Table 13 reports summary statistics for data in the matched bank-loan data set that will be used as covariates in the subsequent analysis. The loan (facility) amounts vary widely. The mean facility is for 1.9 billion dollars, but the majority are for less than one billion dollars, with the median facility being for 1.1 billion dollars and the 5th percentile worth only 154 million dollars. Banks and borrowers that participate in the syndicated loan market are likewise varied. The interquartile range of participating banks'

net worth (i.e. equity) is approximately 1.875 billion dollars, while the difference between the 95th and 5th percentiles is more than 5.685 billion dollars.

Results. A bank-level pessimistic sentiment shock increases loan rates. Table 14 shows that when a bank is hit with a one percentage point increase in risk sentiment (a pessimistic shock) then the loan rate margin increases by approximately 14 basis points on average across the pooled sample. However, while this point estimate is consistent across models with and without controls for loan- and borrower-level characteristics, the effect is too imprecisely estimated to be statistically significant at any standard level.

The effect of a pessimistic shock is larger and more precisely measured during crises, namely the Global Financial Crisis. Column (4) shows the impact of the bank-level sentiment shock when estimated between 2007 and 2011. The point estimate increases by 5 basis points (an approximately 36 percent increase relative to the pooled sample estimates) and becomes statistically significant at the five percent level.

The fact that the effect of sentiment shocks become larger and more precisely estimated during crises is notably the micro-level equivalent of the macroeconomic findings in Section 4.

E BRS as a valid instrument for sentiment shocks

For bank risk sentiment to be a valid instrument for bank sentiment shocks, then it must satisfy the standard relevance condition and exclusion restrictions. Put more formally, let ε_t^b be the structural bank sentiment shock and Z_t^b be the proposed instrumental variable. To be a valid instrument for the sentiment shock, Z_t^b must satisfy the conditions:

$$E[Z_t^b \varepsilon_t^{b'}] \neq 0$$

$$E[Z_t^b \varepsilon_t^{-b'}] = \mathbf{0}$$

or in words, Z_t^b must be correlated with ε_t^b but orthogonal to ε_t^{-b}, the vector of all other structural shocks. While BRS satisfies both theoretically, given the structural model presented in Section 2.1, I will also show both conditions are satisfied empirically as well.

E.1 Relevance condition

The relevance condition is easily shown to be satisfied. I will test this through the lens of an IV-local projection, as in Stock and Watson 2018. The econometric model is formally written as:

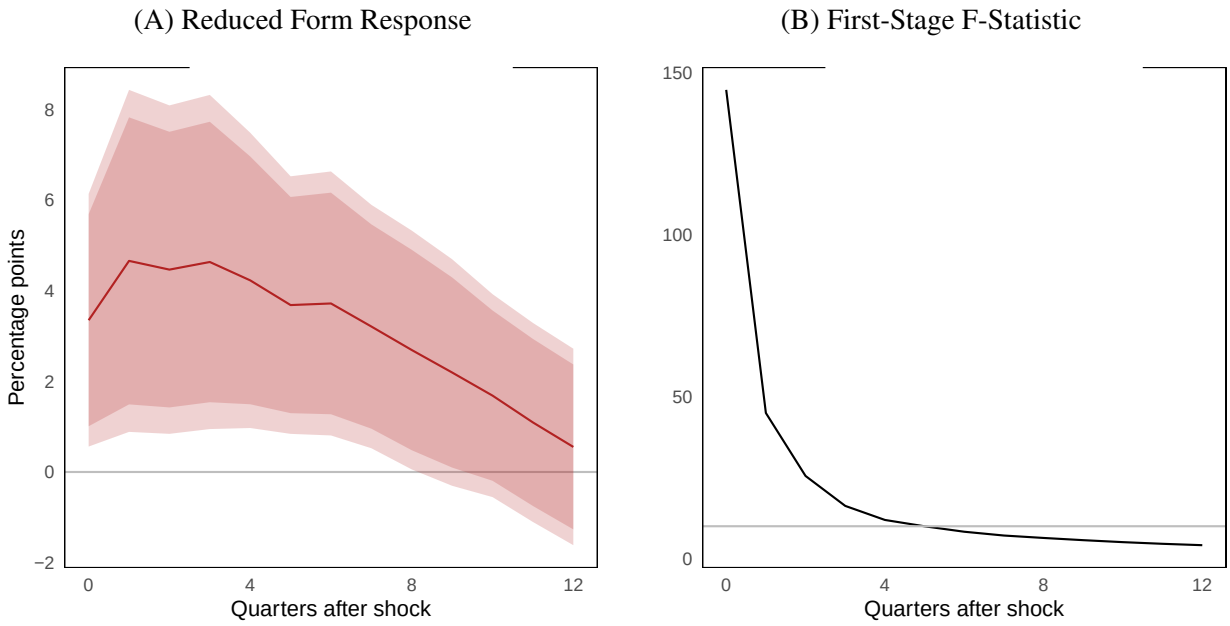
$$Y_{t+h} - Y_{t-1} = \alpha^h + \theta^h \hat{R}_t + \beta^h X_{t-1} + \varepsilon_{t+h} \tag{18}$$

$$R_t = \chi + \omega BRS_t + \zeta X_{t-1} + \eta_t, \quad \eta_t \perp \varepsilon_{t+h} \ \ \forall h \tag{19}$$

where Y_t is the economic outcome of interest in period t, α^h is the constant, $\hat{R}_t$ is the predicted aggregate loan rate, and X_{t-1} is the vector of six auto-regressive lags of real GDP growth, core PCE inflation, policy rate (proxied by the one year Treasury rate), excess bond premium, loan rate premium, and consumer sentiment, while the remaining unobserved determinants of Y_t are left to the residual ε_t. Note that in the local projection framework, the second stage, Equation 18, is estimated forecast horizon-by-horizon, thus coefficients depend on the horizon h. The first stage, Equation 19, in contrast is estimated once, projecting loan rate premium R_t onto bank risk sentiment, BRS_t, as a proxy for bank sentiment shocks at time t, while controlling for second stage regressors X_t and constant χ. Note that this local projection is the theoretically agnostic analog to the BVAR studied in Section 4. I use six lags in the following in the spirit of the saturated lags specification approach in Olea et al. 2024.

First, Figure 19 shows that the reduced form response of the loan rate premium is positive and statistically significant at the 95 percent confidence level. Following Angrist and Kolesár 2024, since bank sentiment shocks are just identified with one instrument, then given the first stage impact

Figure 19: Relevance of BRS as an instrument for bank sentiment shocks



Notes: This plot shows relevant instrument diagnostics for bank risk sentiment in an IV-LP model of the macroeconomy. Panel (A) shows the reduced from response of loan rate premiums to a one standard deviation bank sentiment shock, instrumented by fluctuations in bank risk sentiment. Panel (B) shows the first-stage F-statistics from the same model. The red band mark the Newey-West adjusted 95- and 90-percent confidence interval.

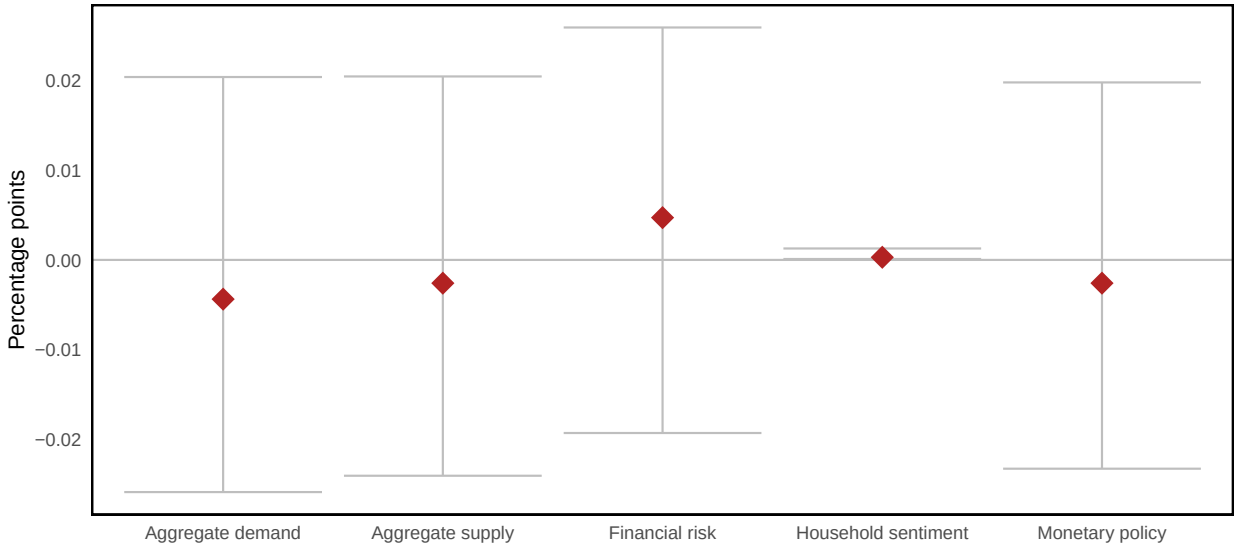
is positive as expected, the t-score of the second stage can be used for valid statistical inference. Therefore, the statistically significant impact of bank risk sentiment on the loan rate premium can be taken to indicate it is a relevant instrument.

Second, Figure 19 shows that the first-stage F-statistics is 145 on impact, indicating a very strong instrument. Moreover, the F-statistic stays above the heuristic cutoff value of 10 through 5 quarters after impact. This indicates that BRS is a relevant instrument for studying fluctuations in loan rate premiums both on impact, through one year after impact.

E.2 Exclusion restriction

I will use bank risk sentiment as an instrument for bank sentiment shocks in a parsimonious BVAR. Therefore, the bank risk sentiment must be statistically independent of the structural shocks in the VAR to satisfy the standard exclusion restriction. I will verify this to be the case by estimating an alternative specification of the BVAR studied in Section 4. The BVAR will be the same as specified in Section 4, but including BRS as an observable endogenous variable and excluding bank

Figure 20: Response of bank risk sentiment to aggregate structural shocks



Notes: This plot shows the response of bank risk sentiment to five aggregate shocks identified via IV and sign restrictions in a Bayesian VAR. Red diamonds mark the mean response in the BVAR posterior, while the gray bars mark the 68 percent credible set. The change in bank risk sentiment is measured in percentage points.

sentiment shocks. As a result, the remaining five estimated structural shocks of interest (aggregate demand, supply, monetary policy, financial risk, and household sentiment) will not be orthogonalized to bank risk sentiment by construction. I can then inspect the response of bank risk sentiment to these structural shocks, and if it does not respond, then it can be interpreted as statistically independent of the other structural shocks.

Bank risk sentiment satisfies the exclusion restriction. Figure 20 shows the response of bank risk sentiment to aggregate structural shocks on impact. BRS does not respond to any shock at any standard statistically significant level, Figure 20 shows very conservative 68 percent credible sets around the impacts. As a result, BRS may be interpreted as being statistically independent of the other structural shocks in the BVAR of interest, thus is a valid instrument for bank sentiment shocks in the macroeconometric model.

F Macroeconomic analysis

F.1 Posteriors and COVID-19 correction

This appendix shows the posterior distribution of the Giannone et al. 2015 hyperparameters and Lenza and Primiceri 2022 COVID-19 volatility scaling factor for the baseline BVAR model utilized in Section 4.

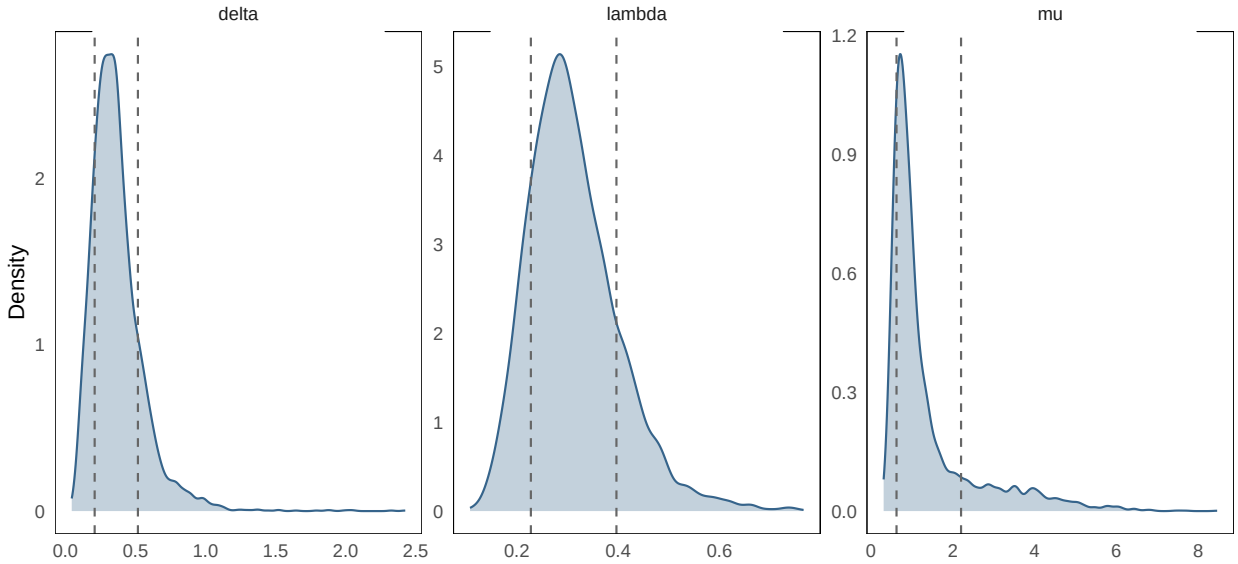
F.2 Historical and Forecast Error Variance Decompositions

The parsimonious BVAR presented in Section 4 produces both point estimates for the variance decomposition of endogenous variables in the system, as well as the credible set around such estimates. Figure 23 shows the credible sets around the forecast error variance decomposition of the contribution of bank sentiment shocks. Table 15 presents the variance decomposition for an extended set of horizons. Figure 24 lastly shows the historical decomposition of selected variables through time.

F.3 Impulse response functions

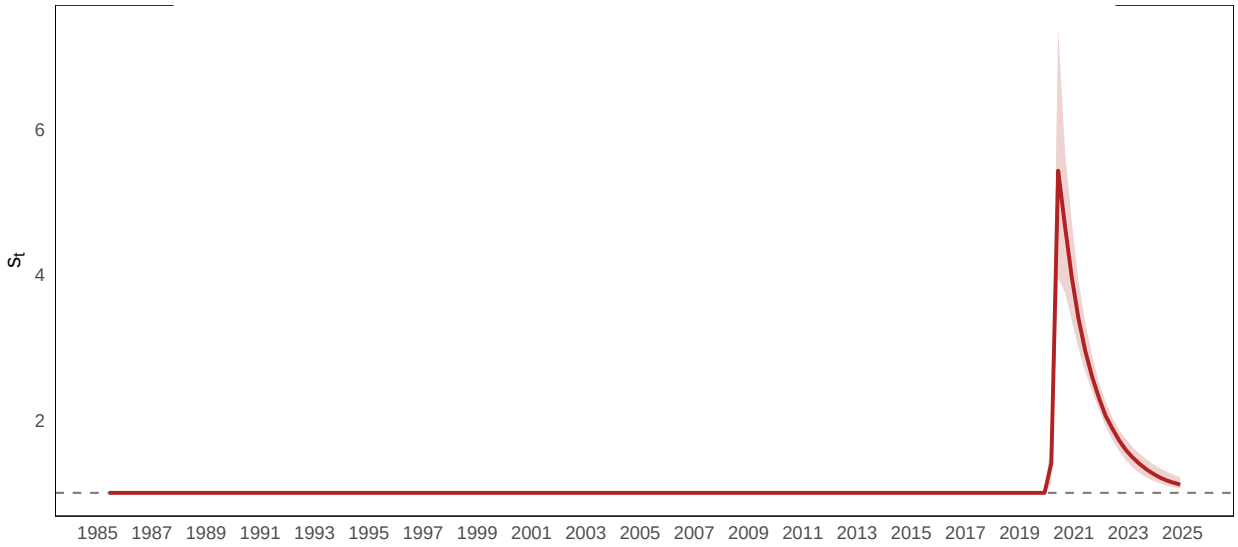
The parsimonious BVAR presented in Section 4 can also be used to estimate Impulse Response Functions (IRFs) to study the effect that one time, unanticipated, structural shocks have on the endogenous variables that make up the economy. Figure 25 presents the IRFs for a pessimistic bank sentiment shock and Figure 26 all other structural shocks.

Figure 21: Posterior densities of GLP hierarchical model parameters for the baseline BVAR



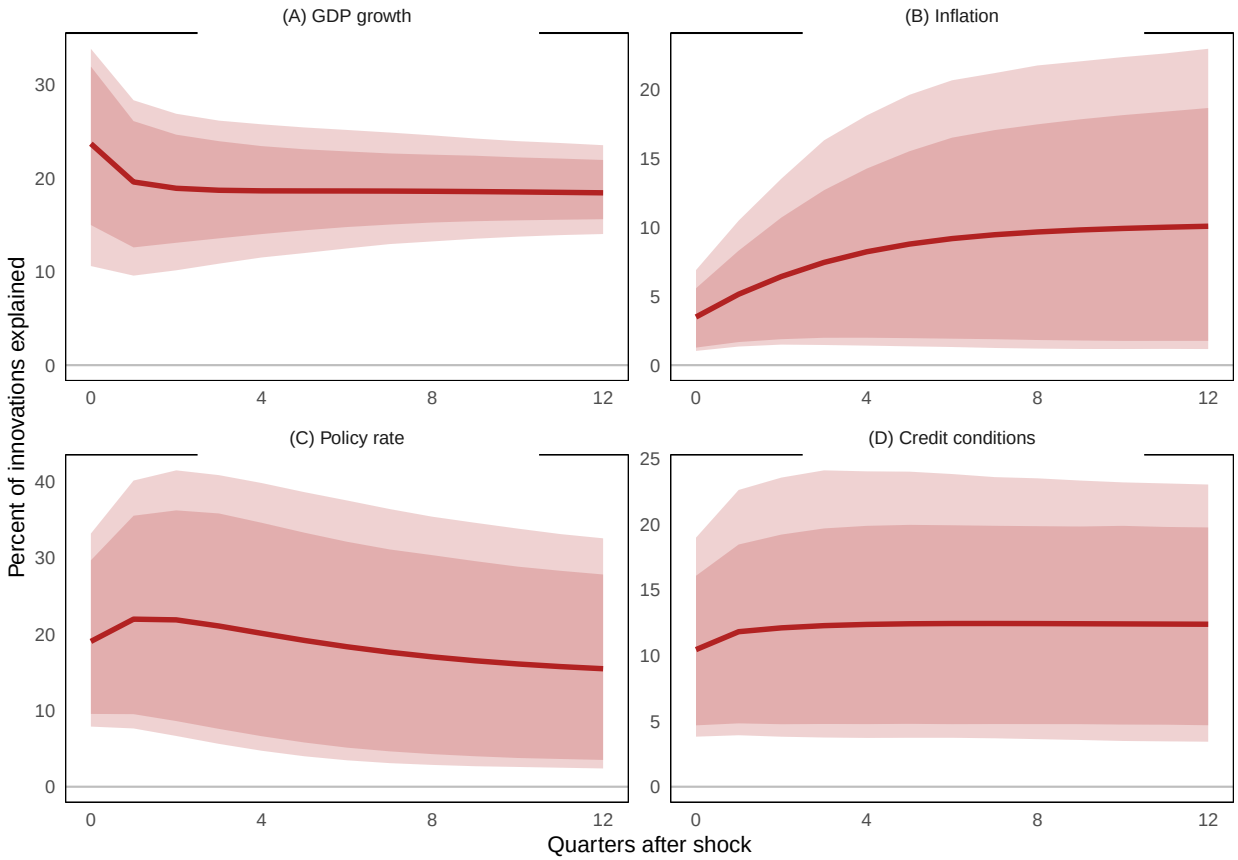
Notes: This plot shows the posterior distribution of the Giannone et al. 2015 hyperparameters in the small BVAR model. The dashed gray lines mark the 68 percent credible sets.

Figure 22: Estimated COVID-19 volatility correction factor in the baseline BVAR



Notes: This plot shows the Lenza and Primiceri 2022 estimated volatility scaling factor in the small BVAR model. The scaling is equal to one during pre-crisis volatility regimes, while a factor of four indicates volatility is four times greater than during the normal volatility regime.

Figure 23: Contribution of bank sentiment shocks to policy, activity, prices, and credit conditions



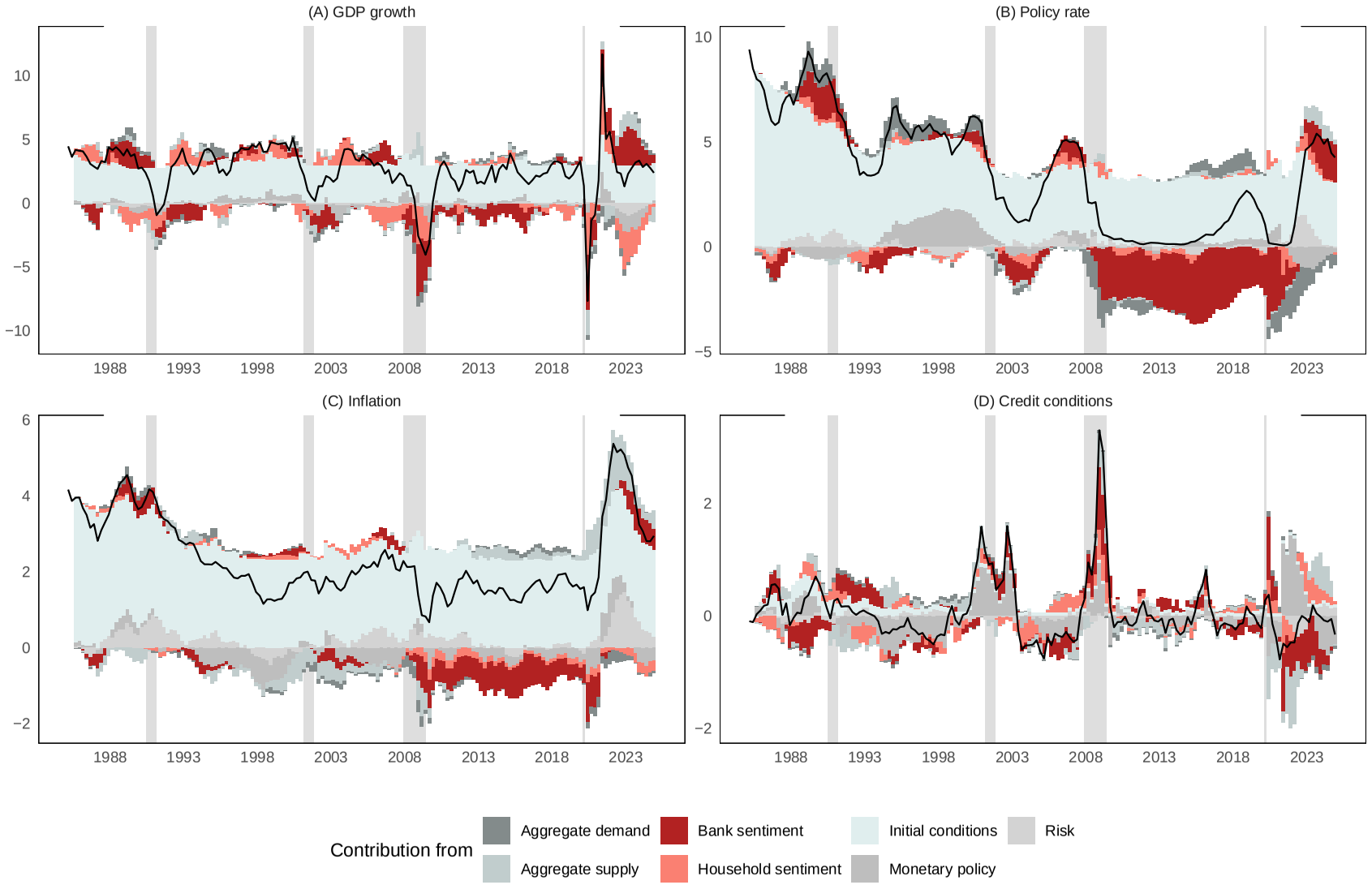
Notes: This plot presents the contribution to business cycle fluctuations in activity, prices, and credit conditions by bank sentiment shocks. Shaded bands mark the 68 and 90 percent credible sets. The solid red line marks the mean contribution to the variance decomposition.

Table 15: Long run variance decomposition of policy, activity, prices, and credit conditions

Endog. Variable	Horizon (quarters)	Percent of innovation explained by: Bank Sentiment	Household Sentiment	Aggregate Demand	Aggregate Supply	Monetary Policy	Financial Risk
Credit conditions	0	10.44	7.20	19.69	14.33	36.23	12.10
	4	12.36	6.65	10.96	17.67	36.31	16.05
	8	12.42	7.47	13.52	17.65	31.25	17.69
	12	12.37	8.12	14.75	17.37	28.90	18.49
	16	12.33	8.61	15.29	17.23	27.65	18.88
	20	12.32	8.99	15.57	17.18	26.83	19.11
	100	13.22	11.38	16.49	17.38	21.42	20.11
GDP growth	0	23.65	36.54	9.81	16.73	8.89	4.38
	4	18.62	21.82	20.00	13.52	12.61	13.44
	8	18.57	20.50	18.90	14.01	14.27	13.75
	12	18.41	20.01	18.54	14.30	14.72	14.02
	16	18.21	19.76	18.44	14.44	14.87	14.28
	20	18.05	19.60	18.41	14.52	14.96	14.46
	100	17.26	19.16	17.95	14.98	15.66	15.00
Inflation	0	3.50	1.75	22.01	30.34	20.77	21.63
	4	8.24	2.47	13.26	39.14	15.46	21.44
	8	9.68	3.07	13.79	37.91	12.32	23.23
	12	10.09	3.42	14.50	35.85	11.71	24.43
	16	10.34	3.62	15.02	34.25	12.00	24.78
	20	10.58	3.74	15.31	33.08	12.60	24.69
	100	13.43	4.21	15.43	27.52	17.61	21.79
Policy rate	0	19.03	6.56	19.16	13.44	13.26	28.55
	4	20.09	11.78	7.77	22.99	16.28	21.09
	8	17.01	14.03	8.67	24.31	14.05	21.94
	12	15.46	15.23	9.48	24.03	13.18	22.63
	16	14.78	15.97	9.95	23.58	12.99	22.72
	20	14.51	16.44	10.19	23.18	13.08	22.59
	100	14.75	17.73	11.09	20.59	14.73	21.12

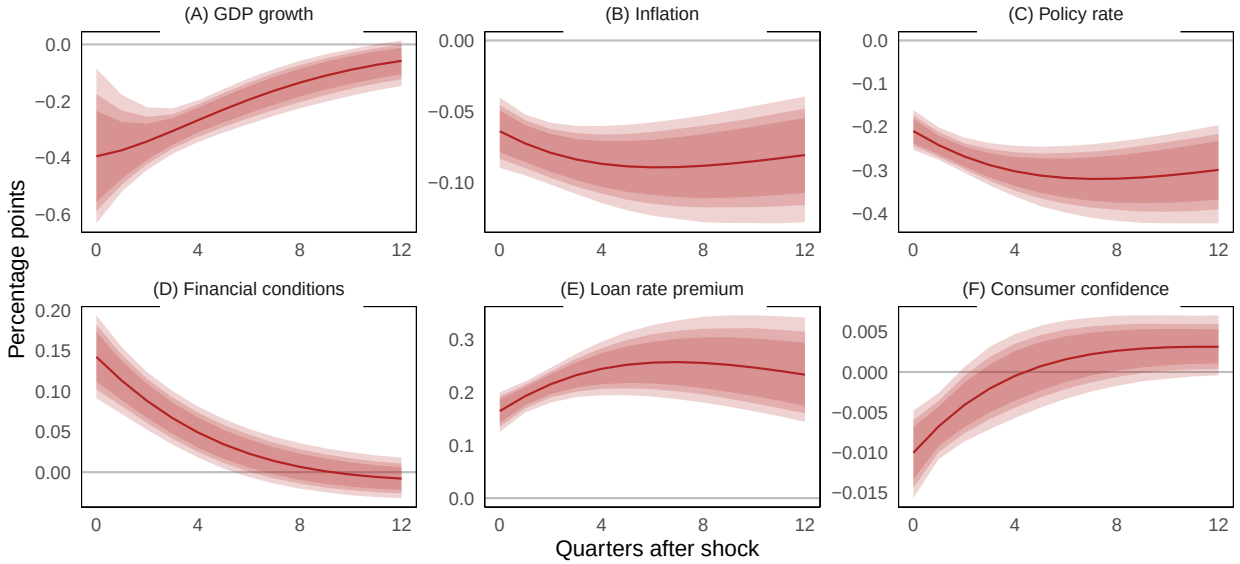
Notes: This table shows the variance decomposition of credit condition, policy, activity, and price fluctuations into contributions by structural shocks. The contribution of structural shocks account for 100 percent of the horizon-variable innovation. GDP growth is the annualized real quarterly growth rate, inflation is core PCE inflation, Financial conditions are proxied by the Gilchrist and Zakrajšek 2012 EBP, and the policy rate is proxied by the one year Treasury rate.

Figure 24: Historical decomposition of policy, activity, prices, and credit conditions



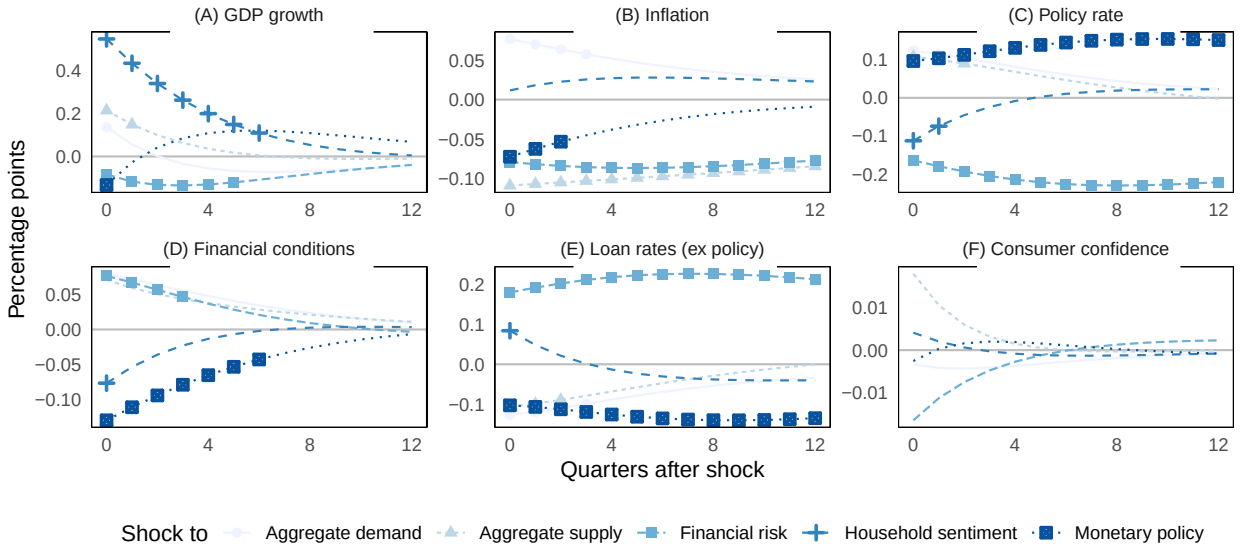
Notes: This plot presents the historical decomposition of business cycles and credit conditions into contributions by structural shocks. GDP growth is the annualized real quarterly growth rate, inflation is core PCE inflation, credit conditions are proxied by the Gilchrist and Zakrajšek 2012 EBP, and the policy rate is proxied by the one year Treasury rate. GDP growth, policy rate, and inflation data are in percent, and financial conditions are percentage points. Data are quarterly from 1986:Q1 through 2024:Q4.

Figure 25: Response of activity, prices, and credit conditions to a pessimistic bank sentiment shock



Notes: This plot presents the impulse response function of business cycle variables and credit conditions to a one standard deviation, pessimistic, bank sentiment shock. Bank sentiment shocks are instrumented by changes in BRS that act through the loan rate premium. Shaded bands mark the 68 and 90 percent credible sets.

Figure 26: Response of activity, prices, and credit conditions to a pessimistic other shocks



Notes: This plot presents the impulse response function of business cycle variables and credit conditions to a one standard deviation, pessimistic, bank sentiment shock. Markers indicate responses with 68 percent credible sets do not include zero.

F.4 Local projection robustness

I additionally estimate a theoretically agnostic empirical macroeconomic model, taking the form of an IV-local projection. The local projection, using a vector of lagged controls mirroring the endogenous variables in the BVAR presented in Section 4, asymptotically reproduces the same IRFs as the analogous VAR (Plagborg-Møller and Wolf 2021). However, in this small sample setting, such as the one presented here, Li et al. 2024 point out that the local projection framework is more robust to misspecification than a VAR, while Olea et al. 2024 prove that local projections is as accurate in its confidence intervals as a locally miss-specified VAR.

The econometric model is formally written as:

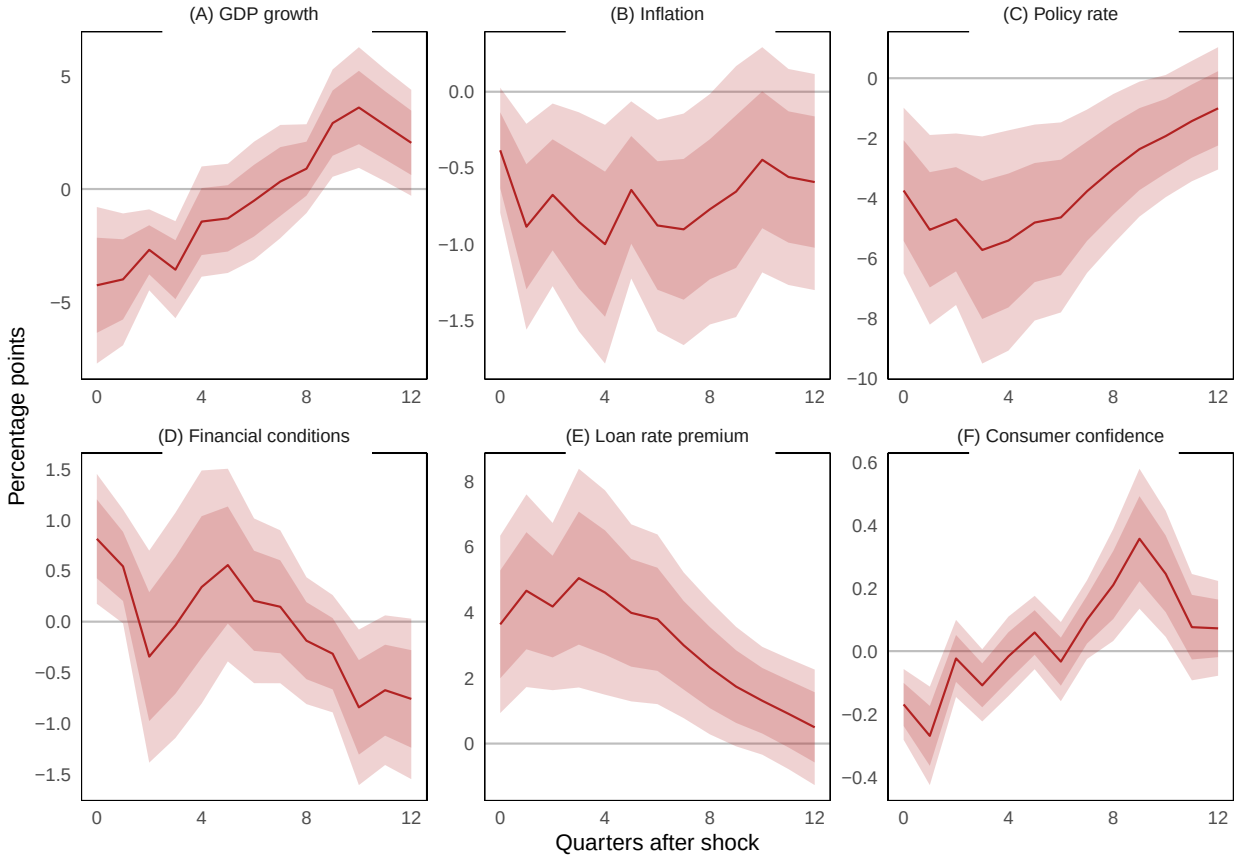
$$Y_{t+h} - Y_{t-1} = \alpha^h + \theta^h \hat{R}_t + \beta^h X_{t-1} + \varepsilon_{t+h} \tag{20}$$

$$R_t = \chi + \omega BRS_t + \zeta X_{t-1} + \eta_t, \quad \eta_t \perp \varepsilon_{t+h} \ \ \forall h \tag{21}$$

where Y_t is the economic outcome of interest in period t, α^h is the constant, $\hat{R}_t$ is the predicted aggregate loan rate, and X_{t-1} is the vector of four auto-regressive lags of real GDP growth, core PCE inflation, policy rate (proxied by the one year Treasury rate), excess bond premium, loan rate premium, and consumer sentiment, while the remaining unobserved determinants of Y_t are left to the residual ε_t. Note that in the local projection framework, the second stage, Equation 21, is estimated forecast horizon-by-horizon, thus coefficients depend on the horizon h. The first stage, Equation 20, in contrast is estimated once, projecting loan rate premium R_t onto bank risk sentiment, BRS_t, as a proxy for bank sentiment shocks at time t, while controlling for second stage regressors X_t and constant χ.

The dynamics characterized by the impulse response functions estimated via BVAR are robust to estimation via IV-LP. Figure 27 shows the response of activity, prices, policy, and credit conditions to a one standard deviation pessimistic bank sentiment shock, proxied by fluctuations in bank risk sentiment impacting the loan rate premium. While the sign of the impact of pessimistic sentiment shocks on different aspects of the macroeconomy is the same across LP and BVAR estimations, the magnitude of the impact is stronger in the local projection.

Figure 27: Macroeconomic response to a pessimistic bank sentiment shock, estimated via LP



Notes: This plot presents the impulse response function of business cycle variables and credit conditions to a one standard deviation, pessimistic, bank sentiment shock. The loan rate premium is the loan-weighted average loan rate less the policy rate, proxied by the one year Treasury yield. Bank sentiment shocks are instrumented by changes in BRS that act through the loan rate premium. Shaded bands mark the 68 and 90 percent Newey-West adjusted confidence intervals.

Appendix References

Angrist, Joshua and Michal Kolesár (2024). "One instrument to rule them all: The bias and coverage of just-ID IV". In: *Journal of Econometrics* 240.2, p. 105398.

Buchak, Greg et al. (2024a). "Aggregate Lending and Modern Financial Intermediation: Why Bank Balance Sheet Models Are Miscalibrated". In: *NBER Macroeconomics Annual* 38.1, pp. 239–287.

– (2024b). "Beyond the balance sheet model of banking: Implications for bank regulation and monetary policy". In: *Journal of Political Economy* 132.2, pp. 616–693.

Caglio, Cecilia R, R. Matthew Darst, and ebnem Kalemli-Özcan (Apr. 2021). *Collateral Heterogeneity and Monetary Policy Transmission: Evidence from Loans to SMEs and Large Firms*. Working Paper 28685. National Bureau of Economic Research. DOI: 10.3386/w28685.

Chodorow-Reich, Gabriel (2014). "The employment effects of credit market disruptions: Firm-level evidence from the 2008–9 financial crisis". In: *The Quarterly Journal of Economics* 129.1, pp. 1–59.

Cohen, Gregory J et al. (2021). "The US syndicated loan market: Matching data". In: *Journal of Financial Research* 44.4, pp. 695–723.

Corbae, Dean and Pablo D'Erasmo (2021). "Capital buffers in a quantitative model of banking industry dynamics". In: *Econometrica* 89.6, pp. 2975–3023.

De Loecker, Jan, Jan Eeckhout, and Gabriel Unger (2020). "The rise of market power and the macroeconomic implications". In: *The Quarterly Journal of Economics* 135.2, pp. 561–644.

Di Giovanni, Julian et al. (2022). "International spillovers and local credit cycles". In: *The Review of Economic Studies* 89.2, pp. 733–773.

Drechsler, Itamar, Alexi Savov, and Philipp Schnabl (2017). "The deposits channel of monetary policy". In: *The Quarterly Journal of Economics* 132.4, pp. 1819–1876.

Fleckenstein, Quirin et al. (2020). "Nonbank lending and credit cyclicality". In: *NYU Stern School of Business*.

Giannone, Domenico, Michele Lenza, and Giorgio E Primiceri (2015). "Prior selection for vector autoregressions". In: *Review of Economics and Statistics* 97.2, pp. 436–451.

Gilchrist, Simon and Egon Zakrajšek (2012). "Credit spreads and business cycle fluctuations". In: *American Economic Review* 102.4, pp. 1692–1720.

Jamilov, Rustam and Tommaso Monacelli (2025). "Bewley Banks". In: *Review of Economic Studies*.

Khwaja, Asim Ijaz and Atif Mian (2008). "Tracing the impact of bank liquidity shocks: Evidence from an emerging market". In: *American Economic Review* 98.4, pp. 1413–1442.

Lenza, Michele and Giorgio E Primiceri (2022). "How to estimate a vector autoregression after March 2020". In: *Journal of Applied Econometrics* 37.4, pp. 688–699.

Li, Dake, Mikkel Plagborg-Møller, and Christian K Wolf (2024). "Local projections vs. vars: Lessons from thousands of dgps". In: *Journal of Econometrics*, p. 105722.

Olea, José Luis Montiel et al. (2024). *Double Robustness of Local Projections and Some Unpleasant VARithmetic*. Tech. rep. National Bureau of Economic Research.

Plagborg-Møller, Mikkel and Christian K Wolf (2021). "Local projections and VARs estimate the same impulse responses". In: *Econometrica* 89.2, pp. 955–980.

Stock, James H and Mark W Watson (2018). "Identification and estimation of dynamic causal effects in macroeconomics using external instruments". In: *The Economic Journal* 128.610, pp. 917–948.